

Proceedings Of the
GLOBAL
SCHOLARLY
COLLOQUIUM - 2025

EDITOR

Dr. SUBATHRA CHELLADURAI

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Kanyakumari | Tamil Nadu | India
Email: capeforumyoutrust@gmail.com
<https://www.capeforumyoutrust.org>

Book Name : PROCEEDINGS OF THE GLOBAL SCHOLARLY COLLOQUIUM
Published by : OF BY AND FOR YOU PUBLICATION
Chief Editor : Dr. SUBATHRA CHELLADURAI
Copy Right : OF BY AND FOR YOU PUBLICATION®
Edition : First - Volume-I
Month & Year : June 2025
Pages : 1- 216

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ISBN : 9788198901637



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FROM THE EDITOR'S DESK



It is with great pleasure and academic pride that I present the Proceedings of the Global Scholarly Colloquium, a consolidated volume of the diverse and insightful papers presented during the event. This publication marks a significant milestone in our collective efforts to foster meaningful dialogue, interdisciplinary research, and global collaboration among scholars, educators, and practitioners.

The Colloquium brought together thought leaders and emerging researchers from various fields who shared their innovative perspectives, empirical studies, and conceptual frameworks. The contributions in this volume reflect a wide range of intellectual inquiries, highlighting contemporary issues, challenges, and future directions in their respective domains.

As the Editor of this volume, I had the privilege of reviewing and curating the papers, each of which underwent a process of academic scrutiny to ensure quality and relevance. I express my sincere appreciation to all the authors for their scholarly commitment and to the organizing committee for facilitating such an enriching platform for knowledge exchange.

I am confident that this edited book will serve as a valuable academic resource and a springboard for further research and collaboration. Let this publication stand as a testament to our shared vision for scholarly excellence and global academic engagement.

Warm regards,

Dr. SUBATHRA CHELLADURAI

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CO-EDITORS' MESSAGE

We are delighted to present this edited volume of the *Proceedings of the Global Scholarly Colloquium*, which compiles the rich array of academic papers presented during the event. It has been an honour to work alongside distinguished scholars and contributors whose work demonstrates the vibrancy and depth of contemporary academic inquiry.

This compilation reflects the colloquium's core objective: to bring together researchers and practitioners from diverse disciplines and geographies in a common intellectual space. Each paper included here has been selected for its originality, relevance, and scholarly value, and we believe the volume collectively captures the spirit of academic collaboration and innovation.

Working on this volume with the Editorial Team has been an inspiring journey. We express our heartfelt thanks to all the contributors for their valuable research, and to the Editor for the vision and leadership that guided this publication. We also extend our appreciation to the peer reviewers and organizing committee for their support in ensuring academic rigour and editorial excellence.

We hope this book will serve as a useful reference and a source of inspiration for researchers, educators, and students across disciplines.

Warm regards,

Dr. ABD GHOFUR, Indonesia
DURWIN A S, India

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IDENTITY POLITICS AS A TOOL FOR WINNING ELECTIONS: A CRITICAL REVIEW OF DEMOCRATIC PRACTICES

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ABSTRACT

This research focuses on the use of identity politics strategies in the electoral process in Indonesia and its influence on the quality of democracy and social solidarity. The approach used is qualitative with the method of literature study, examining various sources such as books, scientific journals, reports and mass media. The findings suggest that identity politics used in a systematic and manipulative manner can reinforce social divisions and increase polarization between groups of people. The use of this strategy has the potential to threaten democratic principles that uphold equality and inclusivity, while contributing to social conflicts in the life of the nation and state. Therefore, reform efforts are needed in the field of political strategy and election regulation so that more ethical and inclusive democratic practices can continue to develop and strengthen national cohesion.

Keywords : Identity Politics, Indonesian Democracy, Social Polarization, Political Reform.

INTRODUCTION

The impact of using identity politics as a strategic instrument in elections is very significant and cannot be ignored. The social polarization that arises from this practice has undermined the very foundations of deliberative democracy that should uphold dialogue and equal participation. On the other hand, social cohesion among citizens is also threatened, and in some cases it can even trigger horizontal conflicts between community groups. Democracy, which

is ideally inclusive and guarantees the fair and equal involvement of all elements of society, is actually distorted by manipulative, exclusive, and short-term political practices. When identity is used solely as a tool to gain political power, the substance of democracy that emphasizes the principles of equality, rationality and participation is at great risk of being shifted. Although many studies on identity politics have been conducted, most of the existing research tends to emphasize more on theoretical aspects or see it as part of socio-cultural reflection without specifically examining how identity politics strategies are carried out concretely in the practice of winning elections. This paper attempts to fill this gap by presenting a critical analysis of how political actors in Indonesia use identity politics as part of their electoral strategies, as well as how this has a direct impact on the quality of democracy and social integration of the nation. Through a qualitative approach with a literature study method, this paper aims to provide a more comprehensive understanding of the relationship between identity, power and democracy in the context of elections in Indonesia.

METHOD

This research uses a descriptive qualitative approach, namely the library research method, to analyze the phenomenon of identity politics in election winning strategies as a perspective of Indonesian democracy. The qualitative approach was chosen because it was considered relevant to thoroughly explore socio-political phenomena through the analysis of various relevant written sources without involving empirical data in the field (Moleong, 2017). The literature study method used includes a systematic review of relevant scientific literature, both primary and secondary sources that are suitable for the research title. Primary sources in this study include books on political theory, democracy and social identity. Secondary sources include national and international scientific journals, research reports, survey results, opinion articles and mass media documentation discussing elections and identity politics in Indonesia. These sources are searched through academic portals such as Google Scholar, Garuda Ristek, SINTA, JSTOR and Scopus, as

well as trusted online media such as Kompas, Tempo and The Conversation (Zed, 2004).

This research is descriptive-analytical with the aim of describing the socio-political reality of how identity politics is used as a tool to win electoral contests. Then, a critical analysis of the influence of these practices on the quality of democracy is used, especially in the perspective of social fragmentation, political exclusion and national cohesion. The data analysis techniques used are content analysis and discourse analysis. Content analysis was used to identify patterns of identity representation in the media, scientific literature, and political narratives that developed during elections. Meanwhile, discourse analysis is used to examine the construction of identity meaning in political statements, campaigns and community responses to identity issues. To maintain the validity and reliability of the data, this study only uses sources that have been verified and recognized academically. In addition, theoretical triangulation was carried out by comparing various theoretical perspectives on identity politics, democracy and power to enrich the analysis and avoid interpretation bias.

With this approach, it is hoped that this research can provide a conceptual solution in understanding the relationship between identity and power in a democratic system and also become a critical foundation for the practice of identity politics in the context of elections in Indonesia. The main question that wants to be answered in this study is: how is identity politics strategy applied as an instrument of victory in elections and what are the consequences for the quality of democracy in Indonesia? By exploring these questions, this study is expected to be able to make a meaningful scientific contribution in enriching the literature on identity politics. In addition, the results of this study are also expected to provide constructive input for policymakers and stakeholders in formulating political strategies and election regulations that are more ethical, inclusive and oriented towards strengthening substantial democracy.

The impact of using identity politics as a strategic tool for elections is very important and cannot be ignored. The social polarization that arises from this practice undermines the democratic soil of advisors, which are supposed to maintain equal participation in dialogue. Meanwhile, social cohesion among citizens is also threatened, but in some cases it can lead to horizontal conflicts between groups of people, democracy, inclusive ideals, guaranteeing that all elements of society are included in a just and equal way, in fact distorted by manipulative, exclusive and short-term political practices. When identity is used solely as a means of acquiring political power, the democratic date entity emphasizes the principles of equality, rationality and participation. Identity Policy Studies have been conducted, but most of the existing research focuses more on theoretical aspects or as part of sociocultural reflection.

This paper seeks to occupy this position by representing a key analysis of how Indonesian political actors use identity politics as part of their electoral strategies and how this directly affects the quality of democracy and the country's social integration. Through a qualitative approach using literature research methods, this paper aims to convey a more comprehensive understanding of the relationship between identity, power and democracy in the context of Indonesian elections. The main questions that must be answered in this study are: how identity politics strategies are used as profit instruments for elections and what are the consequences of Indonesia's democratic quality. By investigating these questions, it is hoped that this research will be able to contribute significantly to the enrichment of the identity policy literature. In addition, the findings of this study are expected to provide input constructive to political decision-makers and stakeholders in the development of political strategies and electoral rules to strengthen ethical, comprehensive and essential democracy.

RESULTS AND DISCUSSION

Result

The findings of this study suggest that identity politics has transformed into a method that is not only widely used, but also very

effective in gaining electoral support in Indonesia. In a diverse society, politicians deliberately exploit identity sentiments based on religion, ethnicity, and culture to build strong political loyalties. This phenomenon does not only occur by chance, but is systematically constructed through political campaigns, social media, and support from elites and figures in society.

From the many real examples of this practice, it can be observed in the 2017 Jakarta Regional Elections, that every religious symbol and identity narrative is used massively to influence the viewpoint of voters. Candidates who can identify themselves with a particular identity group are likely to receive significant support. This support is not based on a person's work program or experience, but rather on emotional attachment to a certain group of people. This marked a shift in the political orientation of society from a rational approach towards identity-based affection.

Although it has proven to be effective in attracting votes, the use of identity politics has a negative impact on the quality of democracy. One of the main consequences is increasing social polarization. In the last few elections, society has been fragmented into identity groups that are suspicious of each other and show a decline in tolerance. Differences in political choices that should be commonplace in democracy and even turn into rigid social identities. Healthy public discourse is replaced by attacks on each other based on their identity names, such as the use of the terms "cebong" and "kampret" that show extreme divisions.

More deeply, identity politics also maintains the practice of exclusion in the political realm. Minority and opposition groups are often marginalized, even to the point of weakening their political legitimacy in the public sphere. Identities that do not conform to the dominant current are often perceived as a threat to national stability. In fact, the essence of democracy lies in the recognition of diversity and equality. Thus, identity politics is not only a tool for winning elections, but also an instrument of control over access to power and space for public participation.

In addition, the institutionalization of identity politics in democratic practice shows a shift in the political paradigm. When voters consider the similarity of identity more than leadership skills, the meaning of democracy is at risk of erosion. Democracy, which was ideally based on rational and deliberative participation, now tends to be trapped in the symbolism of exclusive representation. This is a danger signal that future political dynamics could become increasingly trapped in identity narratives, ignoring substantive issues such as social justice, welfare and governance.

Finally, this study emphasizes the importance of ethical awareness in political practice. If identity politics is not accompanied by a commitment to the principles of justice and inclusion, then social exclusion and political inequality will continue. Therefore, the active role of various elements such as educational institutions, mass media and civil society organizations is needed in encouraging critical political education, building democratic literacy and creating spaces for dialogue between identities. In this way, Indonesian democracy can develop not only procedurally, but also substantially and civilized

DISCUSSION

Identity Politics Becomes a Strategic Instrument in Election Contestation, Identity politics is now a tactic that is often used by political actors to gain electoral support, especially in countries with diverse social characters such as Indonesia (Aspinal, 2017). In this case, identity is no longer just a means of cultural representation, but has turned into a bargaining tool in the struggle for power. Candidates often position themselves as representatives of certain identity groups, whether based on religion, ethnicity, or social class, in order to gain legitimacy from the community (Mietzner, 2018). The most striking example is seen in the Jakarta Regional Elections. Jakarta in 2017, where religious issues were intensely used to influence voters' perceptions of the candidates who were fighting. Religious narratives, which were previously moral and spiritual, have turned into an excellent instrument of political mobilization, supported by religious leaders, community organizations, and the

use of social media. This phenomenon shows that identity can make political loyalty more maximal emotionally and ideologically.

Technically, identity politics is implemented through a number of approaches: first, the use of identity symbols in campaigns; second, exclusive rhetoric that creates a dichotomy of "us" versus "them", and third, community-based political patronage that offers rewards if a particular candidate wins an election. These strategies are effective because they touch on the emotional and collectivist sides, instead of relying solely on voter rationality. Then Social Polarization and Political Fragmentation, The impact of the use of identity politics in general elections often causes polemics of differences in the social realm that are significant. Society is divided into several groups that will be suspicious of each other, even openly opposed, according to political preferences that stem from certain identities. This kind of polarization affects the space for dialogue, lowers social solidarity and gives birth to an exclusive attitude among groups. In the long run, this has the potential to reduce the quality of democracy and weaken social integration. Polarization does not only exist at the level of the political elite, but even in the realm of the lower society (Mouffe, 2005). On social media, for example, identity feuds are clearly seen through terms such as "cebong" and "kampret" that are highlighted in the context of the 2019 Presidential Election. This reflects that political inequality is already part of the social identity inherent in society.

According to Chantal Mouffe, this condition reflects a destructive form of political antagonism, which should be diverted into a form of agonism, namely a healthy competition of ideas in a democratic space. However, when identity is used as an exclusive political tool, the space of difference turns into a completely unproductive arena of conflict. Then in other cases the threat to democracy and participatory exclusion. One of the very dangerous things of identity politics is the emergence of attempts to delegitimize democratic institutions and processes. Minority groups or opposition parties are very often considered illegitimate or even portrayed as a threat, simply because of different identities. This gives rise to political

exclusion, where only the dominant group is recognized and given space in decision-making.

In extreme cases, this exclusion can give rise to discriminatory policies, repressive actions, or violence against groups considered "other". After the 2017 Regional Elections and even the 2019 Presidential Elections, in Indonesia there were various restrictions on political expression from certain groups, both in the form of regulations and social norms. This situation indicates a deterioration of democracy that is supposed to ensure inclusivity and diversity. Fukuyama (2018), emphasizes that when identity becomes the main center in politics, solidarity as citizens will continue to be depleted. Group identity, namely in terms of religion or ethnicity, will take precedence over citizenship or nationality identity. In the long run, this condition can disrupt national unity and weaken social fragmentation. From this observation also normalizes Identity Politics and Ethical Challenges, the danger that continues to loom over identity politics not only in its use that eliminates other groups, but also if such things begin to be considered common in politics. If this kind of thing is not given attention, what will happen is that the political process will tend to change from a contestation of ideas and programs to a contest of identity. Leaders are chosen not based on capacity or integrity, but simply because they are considered to "represent our identity".

From the point of view of political ethics, this kind of thing will raise serious problems. Politics should uphold the public interest and values such as justice, equality, and solidarity. However, with the increasingly dominant identity politics, these values are at risk of being marginalized because politicians focus more on their own groups (Nurhasim, 2018).

In the future, the fundamental challenge is to create a rational, inclusive and value-based political climate. Political education is the main key to increasing public awareness so that it is not easily provoked and provoked by identity rhetoric. and also, the media and civil society organizations must also play an important role in the

dissemination of discourses that position diversity as the strength of the nation, not only as a source of division (Rawls, 1999).

CONCLUSION

This study convinces that it has become an important component of electoral dynamics in Indonesia, especially since the reform era. This phenomenon is no longer the current campaign strategy, but has been converted into structured and systematic electrical equipment. It happens at the national and regional levels. From the results of the literature study conducted, the exploitation of identity in politics seems to be rooted and is part of the main strategy for profitability. Identity, including elements of religion, ethnicity and cultural attribution, is very accustomed to mobilizing political support. Without realizing it, it strengthens the sense of pluralism towards democracy, the use of a strong identity promotes exclusivity and disguises political entities based on program and rationality. The risk is a decline in the quality of advisory democracy and a decline in the same discussion in health in society. The use of Identity Policy also encourages deeper polarization of society. Social tensions and fragmentation not only occur in the political elite, but also extend to the broader level of society, creating sharp divisions that are difficult to bridge. This situation worsens the relationship between social groups, increasing the risk of horizontal conflict and marginalized minority groups for comparable political participation.

Overall, identity politics, which were originally interpreted only as expressions of democratic inequality, are used to confirm the dominance of certain groups. Then identity is used as a tool of exclusive difference, democracy until it loses its integration power and national values threaten democracy. If this trend remains in society, it is not impossible for the country to be affected by the impact of very serious social disturbances.

To face challenges like these, Indonesia must announce a political transformation from symbolic and sectarian to rational, inclusive and grounded. Reforming political education, strengthening the relationship between democratic institutions and the active role and

discipline of civil society is key to maintaining the values of diversity, maintaining health and maintaining the values of diversity.

Acknowledgments

With great respect and enthusiasm, the author expresses his great appreciation to all parties who have contributed to the process of compiling this article. In particular, the author expresses his deep gratitude to the superiors who have provided consistent and sincere direction, support, and guidance during the research and writing process. His advice and insights have been an important foothold in shaping the framework and flow of thought in this article.

The author also wishes to express his gratitude to academics, experts and researchers whose work is the main reference and source of intellectual inspiration in exploring issues surrounding the application of technology in governance. The insights they offer through various scientific publications have provided depth and breadth in understanding the topics discussed.

Not to forget, the author would like to thank all credible and relevant sources of information, which made it possible for this article to be compiled in its entirety and comprehensively. Thanks are also extended to our colleagues, the drafting team, and early readers who have provided critical feedback and constructive suggestions for the improvement of this article. Finally, I hope that this article can make a real contribution to the development of academic literature, expand understanding of innovation in digital governance, and encourage the creation of more transparent, efficient and accountable governance in the future.

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ENHANCING AUTONOMY AND ACCURACY: AI TOOLS IN LANGUAGE LEARNING AND ASSESSMENT

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ABSTRACT

The rapid evolution of artificial intelligence (AI) has transformed language teaching and assessment. This study investigates the integration of AI tools to prepare students for the Cambridge Linguaskill examination. It explores the dual impact of AI on learner engagement and performance and evaluates the effectiveness of AI-driven assessment in comparison with traditional methods. A group of 66 learners aged 16–20 years underwent diagnostic testing followed by 30 hours of targeted training using AI tools. The study found significant improvement in language proficiency and learner autonomy, with AI tools providing rapid adaptive feedback and consistent evaluation. The findings support the growing role of AI in language education and its potential to complement human instruction.

Keywords: Artificial Intelligence (AI), Cambridge Linguaskill, Adaptive Feedback, AI Tools in Education

1. INTRODUCTION:

The integration of Artificial Intelligence (AI) into language education is transforming the landscape of English language teaching and assessment. AI-driven tools are increasingly recognized for their ability to deliver personalized, adaptive learning experiences that cater to individual learner needs. By offering real-time feedback,

data-informed instruction, and consistent evaluation, these tools not only enhance student engagement but also significantly improve teaching efficiency and learning outcomes. Akbarani (2024) underscores the usefulness of AI in alleviating teacher workload while simultaneously elevating the quality of instruction. The application of AI in English language education, particularly in the domain of assessment, has shown promising results. Notably, AI technologies have been instrumental in fostering learner autonomy and supporting self-directed learning. With the capability to provide rapid, targeted feedback and streamline assessment procedures, AI tools empower educators to deliver more frequent and precise evaluations. A key example of AI's impact in language assessment is the Cambridge Linguaskill examination—a computer-based English proficiency test developed by Cambridge Assessment English. Incorporating AI to assess skills across listening, speaking, reading, and writing, Linguaskill exemplifies how advanced technologies can ensure accuracy, objectivity, and scalability in language testing. This study explores the implementation of AI tools in preparation for the Linguaskill exam, examining their influence on learner performance and comparing the effectiveness of AI-based assessment with traditional human evaluation methods. Through this investigation, the study aims to contribute to the growing body of research supporting the integration of AI in language education as a means to enhance both teaching quality and learner outcomes.

2. LITERATURE REVIEW

The integration of Artificial Intelligence (AI) into education has significantly influenced English language teaching and assessment, promoting personalized, adaptive, and learner-centered instruction. AI tools offer real-time feedback and data-driven support, aligning with modern pedagogical trends.

AI's capacity for personalization is a major advantage. Heift and Schulze (2015) highlight how AI systems adjust content based on learner responses, enhancing engagement and fostering autonomy (Little, 2007). Intelligent Tutoring Systems (ITS) and Natural Language Processing (NLP) play key roles in language instruction by

enabling interactive grammar and vocabulary practice and delivering accurate feedback (Wang et al., 2020).

In assessment, AI addresses issues like bias and scalability. Automated essay scoring tools such as e-rater and IntelliMetric ensure objective evaluation (Burststein et al., 2013), while Automated Speech Recognition (ASR) reliably assesses spoken responses (Zechner et al., 2009). The Cambridge Linguaskill exam exemplifies AI application, offering adaptive, fast, and skill-specific feedback (Cambridge Assessment English, 2021).

While AI enhances efficiency, scholars note its limitations in emotional intelligence and contextual judgment (Popenici & Kerr, 2017), and caution against ethical risks such as data privacy concerns (Selwyn, 2019). Thus, AI is best seen as a complement to human teaching.

In conclusion, research supports AI's transformative potential in language education. However, more empirical studies are needed to evaluate long-term impact and guide effective integration.

3. RESEARCH OBJECTIVES

This study was guided by two core research questions: First, it aimed to explore the ways in which the integration of AI tools into language teaching affects learners' performance and engagement, particularly during preparation for the Cambridge Linguaskill examination. Second, it sought to examine the effectiveness of AI-driven assessments in evaluating language proficiency and how such evaluations compare with traditional human-conducted assessments.

4. METHODOLOGY

The study involved 66 students aged between 16 and 20, whose English proficiency levels ranged from A2 to B2 on the CEFR scale. A diagnostic test was initially administered to assess learners' baseline abilities. Subsequently, the participants underwent a structured 30-hour training programme integrated into their coursework, focusing

on listening, speaking, reading, and writing. AI tools were embedded into the curriculum to support each language skill area of the course. Upon completion of the training, the students took the Cambridge Linguaskill examination, and their performance was analyzed to assess improvements and the utility of the tools used.

The AI tools integrated into the learning process included Cambridge's Write & Improve and Speak & Improve platforms, which provide real-time, adaptive feedback on writing and speaking. LanguageTool.org was employed for grammar and style correction, allowing learners to self-edit and revise their work with minimal teacher intervention. Additional resources such as Elllo.org and Rewordify.com were used to enhance listening comprehension and simplify complex vocabulary, thereby supporting learners' overall language development.

5. RESULTS AND FINDINGS

The findings of this study provide strong evidence of the pedagogical value of AI tools in language education. In terms of writing, Write & Improve encourages students to revise their work iteratively, resulting in better grammar, structure, and coherence. For speaking, the use of Speak & Improve led to notable gains in fluency and self-confidence among students. LanguageTool supports learners in independently identifying and correcting errors, fostering a sense of autonomy. Tools such as Elllo.org and Rewordify.com played key roles in vocabulary acquisition and improved reading and listening comprehension.

The learners' feedback reflected high levels of motivation and engagement. The students appreciated the personalized and interactive feedback provided by the AI tools, which made their learning experience more dynamic and responsive to their individual needs.

In terms of assessment, AI-generated scores from the Linguaskill exam showed close alignment with those provided by human assessors. The speed and specificity of the feedback provided by AI

tools often exceeded what was possible through traditional grading. Furthermore, the tools highlighted subtle language issues that might be overlooked in manual evaluations. A comparative analysis of the diagnostic and post-training test scores revealed a clear upward shift in performance. For instance, the number of students in the basic proficiency band decreased from 27 to zero, whereas those in the highest proficiency band increased from 14 to 28.

Performance Band	Diagnostic Test	Linguaskill Exam
Red	27	0
Amber	14	11
Yellow	11	22
Green	14	28

6. DISCUSSION

This study demonstrates that AI tools can significantly enhance language learning effectiveness and efficiency. By offering instant adaptive feedback and enabling learners to identify and correct errors autonomously, these tools support deeper and more sustained engagement with learning materials. Importantly, while AI provides a scalable and consistent means of evaluation, the role of human teachers remains crucial in providing contextual understanding, emotional support, and tailored instructions. Thus, AI should not be viewed as a replacement, but as a powerful complement to human educators.

7. CONCLUSION

The integration of AI-driven tools into language teaching and assessment frameworks offers considerable promise to educators. In the context of the Cambridge Linguaskill exam, platforms such as Write & Improve, Speak & Improve, and LanguageTool have proven instrumental in improving learner performance, particularly in

writing and speaking. These tools not only made the preparation process more interactive and responsive, but also enabled learners to take greater control of their progress. The study concludes that AI-enhanced learning environments can foster improved outcomes and support personalized, engaging, and effective language education.

8. REFERENCES

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ECONOMIC STATUS OF FOLK ARTISTS

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ABSTRACT

Folk arts are the internal autobiographies of folks. These are the creative rhetoric's or critical investigations of traditional, regional and prevailing life of working class. They are the stable development concepts of concerned region representing cultural diversity and plurality. The purpose, action, technique, style and skills are different. Such valuable and different skills are created by folk artists. They are the most intelligent and creative. They have dedicated their life to preserve the stable culture of this land, cultural distinctness of communities, to teach lessons of making movement against exploitation and to propound the morality. They have taught the lesson of happiness, values and ideal education. They have dedicated their life for public development. Due to this reason, today they are suffering various pains struggles and challenges. Their economic condition is highly pathetic. Their income cannot be compared to any of the workers of this land. They are tired of economic instability. They have become weak losing self confidence. The reason for this is many. The artists have lost the 'the traditional exhibiting place' to exhibit their art. The traditional art nurturing systems have become silent and oblivion. These arts are confiscated by the modern communication medias, authoritarians, individualism proponents and capitalists. Modern machine based system given scope for monopoly. This has resulted in the failure of folk artists to catch the attention of the consumers. Mass are turning away from this traditional art. Belief in god is lessened. This has crushed the traditional folk exhibition art. Due to such innumerable reasons the earnings of folk artists turned pathetic. Due to this their economy is collapsed. So, day by day, they are becoming psychologically weak. The sections turning away from this field to another is in a brisk process.

Moreover ours is a superior culture. The formal or the cultured artists are diplomatically suppressing the folk artists and are abiding them from development. Individualism has substituted the mass representation. The havoc of art agents is maximum. Such various reasons are suppressing the economic ways of these artists. The video's conquered the place of direct exhibition of folk arts. The spectators of such videos are cores. Such issues have multiplied the art market of folk artists. Over above the UNO, UNESCO and Development institutions are giving prominence to such folk arts. By using folk arts as applied art in education, stable development concept plans are implemented. Various associations' societies have come into existence to develop these arts and they are executing its work. The activities of such institutions and their assurances given a life to folk artists but they have failed completely in bringing economic confidence. Various reasons for this are theoretically and scientifically considered for discussion.

Keywords: Identifying the economic status of folk artists and searching for problems and challenges. To analyze theoretically the scientific reasons to find the solution for these problems.

INTRODUCTION

The number of the strength of folk artists in India is quite high. The account of their static is till now not precise. No one tried to take a brave step to know the statistics. The reason for not knowing may be due to fear or they might have considered them ignorant. They are separated from the administration system of India. It indicates they are separated from the development perspectives of this land. They have just a point of privilege in the development. So they are invisible. So the folk artists participation is lost in great number in development role. Moreover these artists are insulted and discarded with abuses. They are controlled from reaching the mainstream. So the number of exploiters, suppressers and pained are countless. For example they are discriminated as separativeness from formal or polite artists. So the folk artists are losing self confidence. Moreover their exhibiting places are conquered by the authoritarians and capitalists. They are making

folk art as part of their economic earnings and popularity sources. They are trying to make these artists as their slaves. The authoritarians, capitalists, wealthy people and educated are utilizing these artists especially by registering associations in the name of these illiterate artists. There are various examples of the capitalists registering more than ten associations in their names. He applies for a tender of government and private programs and earns money by taking the artists to exhibit their skills. 'There was a program in Bhopal and it was a quite big program. Thirteen teams of Karnataka represented the program with 192 artists. They suggested to bring ATM card for the leader of the team. They submitted the train ticket only after took the ATM and pan-card. Excellent art was exhibited in the program. The amount was credited on the same day to our account. It was in our account for only fifteen minutes. The Agent has drawn the complete amount suddenly the art exhibition intoxication was lost. We reached Bengaluru with sorrow. ATM's returned to us and we were given one thousand rupees as daily wage such is the horrid economic life of us'¹ So the folk artists are suffering from such economic exploitation. By questing them, the study will be followed.

OBJECTIVES

The present article have various objectives. Few are the following:

1. The present study aims to identify economic status of folk artists. It means to understand different perspectives of the birth of folk art, contexts, creation and economic structure theoretically, what was the economic status of folk artists in the past? Why? What are the reasons for such status? What is the status of today? Why? What are the effects on art and artists due to such status? How it is affecting the cultural system? Is it effecting on whose cultural identify? How the cultural identifies are responding? Which are the plans made? Or not? If plans are made, how is it? How it if effecting on economy of artists? Such questions are searched?

2. Various folk arts are coming to existence in the terminology of 'folk artists' expanse. They are noted as god centred, folk arts, clan centred folk arts, practice centred folk arts, person centred folk arts, sect centred folk arts, worshipping based folk arts, entertainment centred folk art. The structure, worshipping section, audience section, artists sections of these folk art are varied. So the economic status of artists are different. So the study aims to know which of these sections has gained the success and which section is weak. Such issue is theoretically or philosophically discussed.

3. These artists are classified as professional singers, semi professional singers, passion centred artists, nomadic artists, semi-nomadic artists, modern artists, male artists, female artists, transgender artists, youth artists, child artists. These artists exhibiting the art of superior community, and there are entertainment artists. The identity and values of there art are different. They have kept lively the economic, social, religious, cultural status of folk artists lively. So the intent is to discuss with various examples which artists are weak economically, why? How? If economically development, for what reason, how and what are the factors behind? Through there few remedies will be found.

4. World is growing fast. Digital field is to the peak. This is taking control over every field. Due to this online forum is extending day by day folk forum is also in progress. This is a healthy development. This is a way to step with the period. But there is no account of folk artists in this digital account. The non-artists are opening the digital account and exporting the artists by cheating them. This is the digital financial crime. Few artists are becoming the victims of such scam. Even here they suffer the economic crisis. Few folk artists have opened account. In this regard information from both will be taken and inquisition will be made. Because this is the new exhibiting place for folk artists. So this concept will be analyzed philosophically.

5. Government has made various plans for the development of folk artists. The objective in this context is to analyze and discuss. The

philosophical aspects related to, how these plans are developing economically these folk artists? Why? How is the economic structure? How for is to reaching the folk artists? What is the ratio of economic profit by them? What are the problems and challenges? What are the remedies for these problems?

6. Rarely few of the folk arts are becoming the incoming generating arts at the present period for example the drum music or dance or "Dollu Music", Lavani Haradeshi-Nageshi, Sri Krishna Parijita, Moharam Songs[Muharram Rubaiyats]. Observing the income from these art, the capitalists, educates, entrepreneurs, especially the highly qualified, authoritarians, the wealthy persons and entertaining artists entered this field. There is a competition between traditional artists and capitalists. This has resulted in the birth of creative artists and non-creative artists, literate and illiterate artists, Amogasiddha sect and Malingaraya set, vibhuti sect and Bhandara sect, lingadhari sect and non-lingadhari sect resulting in debate and they are singing songs of revolution. They are composing songs with entertaining form for earning money. This has effected traditional artists. Philosophical analysis will be made on what way these are effecting? How the lives of these artists are crushed? How their music turned obscene? What are the challenges to the artists wishing to continue in this field? How will be the financial perspective? What will be the mindset of the audience? What are the challenges to the artists to prepare the useless to convince the audience? Many more factors will be analysed on the north Karnataka folk art Lavani, Bhajan, Folk songs. The experienced artists narration will be considered for the analysis.

7. In few folk artists impurity concepts, caste discriminations, art discrimination and class discriminations persist. These discriminations has resulted in the economic crisis and art exhibitions of co-artists. There is a structure for economic distribution or financial distribution. By this, few artists are facing the economic crisis. These factors also will be considered for the study. The objective is to know what are the intents behind the

discrimination? Which section is working here? By this, which section and caste artists are facing economic crisis? What are the effects on the life of artists? The study aims to search suitable remedy for this.

Philosophical review of economic status of folk artists

The folk art exhibiting structure is distinct. Nurturing of exhibition of art and artists is found here. For example 'Tagara Jogi's. The shepherd after listening to the narrative exhibitions has to offer a sheep for a herd. This is the mandatory rule of this art. This is called the religious norm. Especially this is the delicious norm of Halumata religion [community]. In the research terminology it is called as 'Religious alms'. There is a lot of differences between beggary and 'religious alms'. This is alms taken by the devotees from folk artists. Such kind of alms is followed by folk artists in Karnataka. For example, this system exists in Kamsale, Ganeyavru and Matangi artists. This system had made the economic life of folk artists stable. Our work is Tagarajogi mean the shepherd's work. Since generation, this occupation is followed by us. For this shepherds work various families of our generation dedicated. Each shepherd's were offering lamb. Then there were thousands of sheep. Now the numbers of herds are lessened. The population of us has increased. There are now nuclear families. Herding sheep families or the live stocks have decreased. This has been distributed between families. Now each families have 30 or 40 sheep's and our children may get the share of five or six. Now we don't offer sheep's every year. One lamb will be offered once in three years. Some will never offer. We get thirty maximum lambs for three years. The lambs will have the illness. Moreover the young shepherds lost trust on us.² So offering sheep's is reduced such reasons have lessened their income. It has reduced to three thousand from fifteen thousand. So they suffer the instability. At earlier nomadic life was followed and moved to different places on horseback. But now motor bikes substituted horse. Even this has increased expenditure. The expenditure on bike created problems. Such changes are brought from these religious alms. So the artists are facing economic crisis. Especially religious

occupational singers like Gorava's, Gane's, Gondaliga's, Jogati's, Kamsale artists and Neelagaras and others are facing economic crisis.

2. Oblivion of well trust or goodness and economic instability of folk artists.

There is a sacred feeling on few of folk arts of Karnataka. This feeling has lessened today. It was believed that through puppet shows and chowdike traditions, something good sign for the family and it will be a blessing for continuation of progeny. It was believed, the drought will vanish, there will be rain and the cattle's will be cured from diseases. So the folk believed in continuing the tradition of few art exhibitions. At the earlier days, when we went to our native place, we watched performance of various arts by the artists. We exhibited our art for months and even the rural folk demanded it. They loved us and we stayed in the temples. All the neighbours fed us early morning with tea, breakfast and lunch. We were seen with delicious dishes with enriched protein foods of multigrain's, curd, ghee, milk etc., we never suffered from illness. We used to have sumptuous meal. My grandmother lived for hundred years. The life made us so strong and never faced financial crisis. But today no soul turns towards our art. We are waiting just for one show.³ These words indicates economic system and its oblivion picture. Due to the decrease in the ratio of belief, the economic problems can be seen.

3. Distortedness of capitalists: Economic life of folk artists suffering from crisis.

Only few folk arts are progressing with good earning indicating healthy development. For example, Yakshagana, Hulikunita (Tigerdance) Drum music or Dollu Kunita, Lavani, Haradesi-Nageshi, Sri Krishna Parijata etc., Drum music is earning more than Yakshagana. There are such thousands of drum music teams are in north Karnataka. The uniqueness of this team is there are people of all age and all sections. The teams of girls is more than the boys. More than hundreds of girls entered this field in two decades. This is a great movement in the folk artists field of India. A

senior and a creative artist will lead and constructed ten to twenty teams. For all these teams, he will give the training. Composing songs and the incharge of programme will be his. The songs will be changed or altered for each exhibition will displaying the dialogues. So he has to write the lyrics always. Remuneration and amount will be fixed for this. Giving remuneration is considered as disrespect. There is a competition, pride and honor in offer. They believe offering in healthy sheep for Guru. 12 to 14 sheeps will be offered from each team. Each master teaches songs for fifteen to twenty teams. There are Gurus who have taught for more than fifty teams for example Sri Kulkarni master of Vijayapura, Sri. Aratal Kalmesh master and Kulkarni master has given up job and joined to drum music. He earned crores of money by this art. He purchased fourteen acres of land from this music art.”⁴ This is the earning from Drum musicians. Their economic status is good. Due to this there is a competition in learning this art. This is the competition in earning. This competition has brought various changes.

3.1. Drum musician capitalists, Authoritarian attack

After knowing by the capitalists, authoritarians, highly qualified about the good life of drum musician artists entered their professional life and took the lead by making the artists to sing for their companies and used them for their selfish greed. They are utilizing the poor artist’s talent and making the profit. They are selling music car sets, CD in the digital market. Little remuneration is given for the artists only for once. But the capitalists are gaining monthly profit for this. This is a cheating stand of capitalists on folk artists. Moreover this is the tactics of snatching the income of artists of ‘outdoor exhibition’ and the audience. Artists without audience are like finance less artists. Moreover the capitalists encourages for the trend music in place of Halumata culture representing music and songs imparting public education. Due to this observe drum music have occupied the foremost place. The songs contains the theme of abusing women with observe words and the songs indicate the women as inferior. Due to such songs, music world losing the culture. Due to this the traditional artists of elder generation are

neglected. They are discarded as not versatile singers, non-creative artists and lacks the knowledge to please the audience of modern audience. The elder singers have disregard for such songs and reject the privileges' to sing for such audience. They respect the culture and songs. This feeling have created economic crisis for them. This is the economic chaos and instability. They are limited to customs, fairs and profitless exhibitions. So such artists are in critical economic crisis.

3.2. Folk songs converting as the songs of disorderly people: Economic backwardness of artists

The world of folk music is becoming the world of disorderly people. Example : Drum music and the trend of argument is in practice. The songs of Bhandara sect, Lingayat sect, Halumata sect, Sheelamata sect, Malingaraya sect, Amoghasiddha sect are in practice. The songs include the theme of men-women as superior. The songs try to justify Malingaraya as superior sect and Amogha Siddha's try to justify Amoghasiddha as superior. The winners of the argument are given prizes, bracelets and are honored. To win the prizes they follow any inferior ways. This is the economic slavery victimization, selfishness and madness. Similarly the singing argument highlights the superiority of their deities. These arguments sometimes may lead to fights and sometimes the hot arguments may result in clashes between art teams. This is the economic war. Even if the musicians abscond after the incident but are not given up and are attacked severely by searching them. A singer called Sri Umadi Manasiddha sings in the stage by tying gun in his waist belt. Such new developments have brought the meaningless revolutionary or conflicting songs to the forefront. So the artist's life is floating meaninglessly in the exhibiting world. In few years, these songs may vanish. So the real traditional singers are staying away from Drum music field as unable to adjust to such system. This is the technique to keep the traditional singers at the backstage. Due to these techniques, thousands of artists become silent staying away from music. The arguments in Lavani, Bhajan and art made the traditional artists to keep still. This is a politics of attacking this field.

4. Caste discrimination and economic backwardness of folk artists.

Folk art has the ability to maintain the identity by harmony, communal unity and interest of public. But today few folk artists are involved in caste discrimination. For example, Smt. Shantamma Maharudrappa Kalmani of Haveri is a Sobhane singer. Her age is 73 and she says, "I visit every house indiscriminate to caste of invited, but my request to them is I am Panchamashali and I will not consume food from other castes. They have to search for the people of our caste and they should cook for us and in absence of our caste people, they must go for neighboring village and our caste people should be provided with ration to prepare food"⁵ Such stand is found in many folk artists. Especially such stand is found in upper section. They need opportunity to conduct programs in the house of Dalit's and lower community. They need the money, honor from them. They need the grains by their hour. But they don't need the shelter and food prepared by them. Tired by this, the youth artists will not incite this grandmother and her companions. If they feel their caste, house and food as superior, we need not to suffer in this regard. So the people stopped inviting her and so the opportunities lessened to her. This has resulted in lowering of income. So the team is facing the economic problem. Similarly various artists are facing caste discrimination. So the opportunities are decreasing. This is the mass economic backwardness.

5. Prohibiting art teaching for other castes: economic confinement of artists.

The origin and development of folk artist is distinct. Few arts are oriented on clan, few are centered on god, worship centered, occupation centered with this focus they have come into existence and developed. Now the art is in inevitable situation 50 survive in a healthy environment. Knowing this, few of the folk artists are extending beyond traditional exhibition. Unable to bear this, few of the caste based persons are exploiting the artists 'Veeregase is the art of Lingayats. It is the art of male and customary art. Today it has exceeded the limit of caste, gender and custom. To teach this art in schools and colleges, no caste or creed discrimination is followed.

But the lingayat community elders of village and political leaders are pressurizing not to teach Veeragase for other castes and lower community folk. This troubles me a lot. So I stopped teaching this art for children.⁷ This is the backwardness of art development and economic progress of artists. So due to the pressure of casteism politics the folk artists are facing crisis. This is dangerous to change such attitudes education should be given. This transformation should be initiated by Karnataka folklore artists.

6. Economic victimization of folk women artists due to Men economic politics.

Women were restricted to exhibit few folk arts crossing the restrictions, women are exhibiting art. This has not been accepted by male folk artists. Moreover they are composing lyrics by inferioring the women. For example;

“Bull has reached the Hubris
It is offered with ground durum wheat
It lacks the consciousness of life
Never approach its side
Women came to catch the bull
But tangled to its horn
Plucked your buckle
And rubbed the ground”⁸

With such obscene composition, insulted women on the public stage. This is an insult for women folk artists. Few of the folk women artists are in disturbance. Such economic agitation can be seen in the experienced narration of Haradeshe-Nageshi singers. The women suffering the insults stayed away from such art. So the male artists are doing such exploitation by insulting women in order to keep them away from such art. This is one of the reasons for economic backwardness of women. This must not continue. This is the rejection of women cultural identity and backwardness.

7. Remuneration distribution based on caste: Lower community artists under economic agitation.

Exhibition form of folk art is unique. There is more politics in folk theatre. The role of heroes should be given and enacted by the leader or Gowda of the village and the role of attendant should be performed by the Dalit. The harmonium master will belong to the strong or higher community and drum players will be from lower community.

The economy of exhibitions depends on the remuneration distribution the drum players get merely 400 to 500 amount from 10,000 rupees. The stronger community artists get more remuneration than lower community. This has created chaos in lower community. The folk artists of lower community are facing discrimination of finance.

So the folk artists are facing severe economic crisis due to various problems. Different criteria's are needed in this regard. The artists from Dalit, tribes, neglected community are more in number. Due to this they are suffering backwardness. The histories of neglected communities are deviating. So due to their development, emergency plans should be implemented.

MAJOR FINDINGS

1.The folk artists of Karnataka are in pathetic economic condition. They suffer financial crisis. The art exhibitions are lessening day by day. The artists are demanding for a place to exhibit their art. They suffer the lack of food grain. They believed art is their property but it is now a mirage so for their livelihood, they depend on daily wages and work as laborers. The seniors and elder artists are waiting for opportunities and struggling for food and clothes. The aged are becoming psychologically imbalanced. Their children and grand children are shifting their art. They suffer the lack of food grain. They believed art is their property but it is now a mirage so for their livelihood, they depend on daily wages and work as laborers. The seniors and elder artists are waiting for opportunities and struggling for food and clothes. The aged are becoming psychologically imbalanced. Their children and grand children are shifting to cities as street vendors. The aged are living in tents and wait for opportunities. They lead life by such opportunities.

2. There are different types of artists in folk artists. They are professional singers, semi professional singers, hobby singers, nomads, semi nomads etc., even these artists have lost exhibiting places and are suffering anxiety. For the daily life, they have structured economic plans. They are regular worships, weekly worships, full moon day, new moon day worships, fairs, festival, opening ceremony, marriage ect., In all these ceremonies they exhibited folk art and they were getting grains and money to lead life. Example Mailaralinga Goravas and Matangi's, ritual of Paddalagi or the sacred bamboo basket held by the choudike artists and such practices are vanished and the other practices are confiscated by the priestly classes. Due to this folk artists are in financial crisis and struggling for economic stability.

3. Few of the folk art especially profit based art are confiscated by the local leaders, educated, graduate unemployed, capitalists, higher section and politicians. They made the folk artists as labourers. The artists suffer the inferiority complex.

4. There is competition between senior and junior artists. This is not merely the exhibiting competition but economic competition. The elderly original artists are insulted. This is the economic rejection and insult folk artists are binded in such whirlpool. They are awaiting for training of such traditional artist.

5. Caste politics is found in folk artists. Higher caste artists get more remuneration than the lower section artists. So the lower sections are facing the lower economic status. Moreover it is making them slaves and is exceeding day by day.

6. Folk art is sacrificed due to caste politics. Due to this, they are utilizing its significance and uniqueness. There are certain sections that trouble the artists in educational field, applied field and development field. This stand is curtailing the economic development of folk artists. They are facing such economic problems.

Folk artists are in such economic problems. The list is big and such list need to be erased. Then only the folk artists can progress economically. For this there are certain suggestions and guidelines.

SUGGESTIONS

1. Folk artists should be involved in creative economy. There is a need to enter in to the global art business and in this regard finance management training and techniques to be taught to exhibit art. Among them, Market skills, social media management, digital exhibition technique should be trained. The training should include life skills of present period and development methods, money management, innovative entrepreneurship etc.,

2. Government should create employment for providing economic security to folk artists. This measure is inevitable for government. It is true that all artists cannot be given the privilege of government jobs. But eligible artists should be appointed according to rules. “During the appoint of 2009, appoints of music, dance, painting, teachers were made similar consideration should be made for folk artists.”⁸ Folk art educations should be started and art teachers should be appointed.

3. Folk artists eligible for teaching and practical training should be appointed in Karnataka Folk University and Gangubai Music University.

4. Cultural conferences, art fairs should be organized. Folk festivals should be conducted folk artists should be given opportunities to exhibit their art in fairs, festivals, marriage, tourism, festivals and temples. For this rules should be formed.

5. For women folk artists separate training workshop should be conducted because the artists need orientation. Folk artists training workshop should be conducted under tribal women development plans, nomadic development plans, SC and ST development plans

and women development plans. Through this vocational education, folk instruments and costumes should be arranged credit policies should be implemented. Market policies should be framed.

6. The pension of artists should be increased. The pension eligibility of artists should be reduced to 48 to 50 age. Karnataka government should give 1000 rupee pension to the artists from 18 to 60 years aged. Similar to West Bengal government.

7. Various plans are implemented in Karnataka for the development of folk artists. Folk fair is one among them. This fair program should be increased.

8. Tamilnadu government “Tamilnadu folk artists welfare Board have organized different program for the development of folk artists. There are 50,882 members in this board. It has appointed for 1,98,72,917 rupees. Similarly in Karnataka such folk artists development board should be started.

9. The development plans of Yakshagana artists are complimentary to the development of artists. Example, Yakshnidhi. In this fund, medical checkup, health services for artists are made. These service schemes should be given for the other folk art artists.

10. The children and youth should be given training on folk art and exhibiting training should be implemented. Example as Nali-Kuni plans of Yakshagana, other plans should be implemented. These artists should be made teachers.

11. Survey of folk artists should be made. Survey should be conducted as caste wise and occupation wise. Then their economic status can be understood.

CONCLUSION

Economic status of folk artists is heinous. Attention should be made regarding this. It is true as there is a saying that like

Saraswati river, folk artists are vanishing. The reason for this is cultural system. This system should be eradicated. Then the folk artists can come to the mainstream. It is possible to improve their economic status. This economy is not the economic development of artists. It is the development of this land. This is cultural trade and development. So few development plans should be formed. If these plans are brought to action then the folk artists can dream development and the identity of the community rejected from education can remain lively and cultural history of this land will have life.

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CHALLENGES FOR THE GOVERNMENT IN REALIZING CHILD FRIENDLY SCHOOL

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ABSTRACT

Child Friendly School (SRA) is a program that aims to create a comfortable school environment and become the second home for children. This program ensures that children's rights are met and provide protection for them. The Indonesian government through the Ministry of Women's Empowerment and Child Protection (KemenPPPA) has actively encouraged the implementation of Child Friendly Schools (SRA) in various regions. This aims to ensure that children's rights are guaranteed as long as they are in the school environment, so that the safety and security of children is always maintained. This study aims to find out the challenges faced by the government in realizing child friendly schools. This study uses qualitative research methods and data is collected through relevant written sources, such as scientific journals, articles, and official government documents. The results showed that the government in realizing this child friendly school was still faced with various challenges. The challenges faced include socialization and training that are not sustainable, limited resources, budgets and facilities and infrastructure, lack of active participation from schools, and weak cooperation between stakeholders and policy implementers.

Keywords: Implementation, Government, School, Child Friendly

INTRODUCTION

Education is the main foundation in the development of superior and competitive human resources. Children as students play a central role in the educational process and therefore need to receive

adequate attention and protection during the learning process. School as a formal environment for children to receive education must be able to create an environment that supports physical, mental, emotional, and social development. For this reason, an educational approach that pays attention to the rights and needs of children is very important to apply.

The concept of Child Friendly School (SRA) is present in response to various problems in the education sector which often ignores the rights of children. Child friendly school is an approach that places children as the center of attention in every aspect of education. This includes efforts to ensure that their basic rights are fulfilled, such as the right to be free from violence, discrimination, mental pressure, and unfair treatment. Child friendly school's also emphasize the importance of children's active participation in the learning process, decision making, and the development of an inclusive, safe and supportive school environment. Thus, schools not only function as a place of learning, but also as a protection and empowerment room for every child.

The Indonesian government through the Ministry of Women's Empowerment and Child Protection (KemenPPPA) has actively encouraged the implementation of Child Friendly Schools (SRA) in various regions. One of the concrete steps in this effort is the issuance of Minister of Women's Empowerment and Child Protection Regulation Number 8 of 2014, which functions as a basis for implementing Child Friendly School Programs (SRA) in various educational institutions. This is part of the national and international commitment in implementing the child rights convention and achieving the Sustainable Development Goals (SDGs), especially the fourth goal that emphasizes the importance of inclusive and quality education. In addition, cross ministry and institutional cooperation continue to be established so that the application of Child Friendly Schools (SRA) can be done thoroughly and sustainably throughout Indonesia.

SD Lab School FIP UMJ is one of the schools that is committed as a Child Friendly School (SRA). The principal as a policy maker has taken various steps to realize this vision. These steps include the preparation of implementation time, training for teachers and staff, identifying the needs of children, the preparation of the action plan, to the implementation and evaluation of the action plan. With the implementation of the concept of child friendly school, the quality of learning at SD Lab School FIP UMJ has experienced a significant increase. The teachers become more sensitive to the needs and feelings of students, so they are able to adjust more effective teaching methods. They also pay attention to the balance between students' cognitive and emotional aspects. Implementation of Child Friendly Schools at SD Lab School FIP UMJ succeeded in creating a safer and more comfortable environment for all students. (<https://kspstendik.dikdasmen.go.id/artikel/detail/implementasi-sekolah-ramah-anak-di-sd-lab-school-fip-umj-antara-peluang-dan-tantangan>)

The Implementation of Child Friendly Schools (SRA) cannot run optimally without the active involvement of various elements of society. Although the government plays an important role in designing regulations and implementation guidelines, the participation of educators, parents, and the surrounding environment is a determinant of success. The teacher is responsible for creating a safe, inclusive learning atmosphere, and respecting diversity, while parents contribute through supportive parenting and constructive communication with the school. On the other hand, society plays a role in creating a social environment that supports the growth and development of children holistically. This cross sector collaboration is an important foundation in realizing a comprehensive and sustainable child friendly school.

However, the implementation of this program still faces various challenges. One of them is the limitations of human resources who understand the principles of child friendly. There are still many educators and education personnel who continue to use conventional teaching approaches, which are often accompanied by

verbal and physical violence as a form of discipline. On the other hand, educational infrastructure in various locations, especially in remote areas, does not yet support the creation of a truly child-friendly school. Lack of training, low budget, and uneven understanding of this concept also complicated the implementation process.

Social and cultural factors also play a role in the success of this program. In some communities, the practice of violence in educating children is still considered normal and is part of a disciplined culture. This indicates that the challenges faced are not only technical, but also related to the values and habits of people who have been rooted. Therefore, creating shared awareness among educators, parents, and society becomes very important in encouraging changes towards an education system that pays more attention to children.

METHOD

This research was compiled using a qualitative approach through literature studies to analyze the challenges faced by the government in realizing child friendly schools. The data used comes from various relevant written sources, such as scientific journals, articles, and documents official government that discusses the topic of child friendly schools.

The data collection process is carried out by examining and summarizing information from these sources. Furthermore, data was analyzed descriptively to identify various challenges faced by the government in implementing child friendly school policies.

Results and Discussion Result

Children are the next generation of the nation who are undergoing an important phase in the formation of identity. But unfortunately cases of violence against children still often occur, including in the school environment. This can be seen from the many reports to the Indonesian Child Protection Commission (KPAI) related to violence that befell children. Schools, families, and the environment should

be a safe place for children to grow, develop, and express their talents and interests.

Responding to cases of violence against children, especially in schools, the government through the Ministry of Women's Empowerment and Child Protection has launched a Child Friendly School Program (SRA). This aims to ensure that children's rights are guaranteed as long as they are in the school environment, so that the safety and security of children is always maintained. Child Friendly School (SRA) is defined as a formal, non-formal, and informal educational institution, which provides a safe, clean and healthy environment. This program is expected to make schools a safe place for children to grow and develop themselves, as well as being an arena for them to channel interest and talent, in accordance with the principles carried by Child Friendly Schools (SRA). (Nurfarida 2021) in Dindin Alawi et al (2022).

In addition, child friendly schools also support children's participation in various aspects such as planning, policies, learning, supervision, and complaints mechanisms related to the fulfillment of children's rights and protection in the field of education. Thus, it can be concluded that Child Friendly Schools (SRA) are educational institutions formed to ensure the fulfillment of children's rights and protection in the school environment.

DISCUSSION

Child Friendly School (SRA) is a program that aims to create a comfortable school environment and become the second home for children. This program ensures that children's rights are met and provide protection for them. Therefore, the administration of child friendly schools is very important for all education units. The principles underlying the formation of child friendly schools stated in the Regulation of the Minister of Women's Empowerment and Child Protection Number 8 of 2014 are as follows.

1. Non-discrimination, that is, every child has the right to get their rights without discrimination, so that all children can participate equally
2. The best interests for children, meaning that all policies and decisions taken must really prioritize the best interests in children's education.
3. Life, survival, and development, namely the school environment is committed to respecting the dignity of every child and giving a guarantee of optimal development for them.
4. Respect for children's views, meaning that every child's view must be appreciated, because it has an important impact on their development process.
5. Good management, schools must carry out the principles of transparency, accounting, participation, information disclosure, and uphold the rule of law to create a supportive environment.

Child Friendly Schools have been agreed to be a forum for education units in the Ministry of National Education, as well as related programs at the Ministry of Women's Empowerment and Child Protection. The formation of child-friendly schools is a concrete step from the ministry to fulfill the rights of children in the educational environment. The education unit, which functions as an educational service provider that directly interacts with children and the community is under the coordination of the Ministry of National Education and Regional Government. Ministries and institutions related to responsibility to ensure security, safety and health in order to support the growth and survival of children in the educational environment.

In the midst of efforts to create a more inclusive education system, the government is faced with various challenges in realizing child friendly schools. The challenges intended include:

1. Socialization and training that is not sustainable.
The government faces challenges in maintaining the consistency of the implementation of socialization and

training related to child friendly schools. Introduction and limitations in providing debriefing to educators and related parties have resulted in understanding and application of this concept to be less optimal. In addition, training that is one or limited is not enough to form sustainable competencies among implementing in the field

2. Limited resources, budget, and facilities and infrastructure. Implementation of Child Friendly Schools (SRA) requires adequate support of resources, including adequate budgets, competent professionals, and adequate physical facilities. Unfortunately, many schools are still facing limited education budget constraints, as well as lack of child friendly facilities, such as playrooms, decent toilets, counseling rooms, and safe and clean environment. This limitation is a barrier in an effort to create a school environment that meets child friendly standards.
3. Lack of active participation from school.
The level of school involvement in initiating Child Friendly School Programs (SRA) is still relatively low. Some schools have not yet made this program a priority, caused by various factors, including the limitations of information, the absence of internal policies that support, and the assumption that the application of Child Friendly Schools (SRA) will increase administrative burdens or require large costs. In fact, the success of this program is very dependent on the initiative and commitment of all members of the school community itself.
4. Weak stakeholder cooperation and policy implementers.
The involvement of various parties including local governments, schools, parents, communities, is very important to support the success of a program. Unfortunately, the weak coordination and synergy between all parties involved often makes the implementation of the policy run separately without a clear direction.

To realize a comprehensive child friendly school (SRA), a long term commitment is needed from the government, increasing the capacity of human resources in the field of education, adequate budget support, and strong cross sectoral collaboration.

CONCLUSION

Education is the main foundation in the development of superior and competitive human resources. Children as students play a central role in the educational process and therefore need to receive adequate attention and protection during the learning process. The concept of Child Friendly School (SRA) is present in response to various problems in the education sector which often ignores the rights of children. Child Friendly School (SRA) is defined as a formal, non-formal, and informal educational institution, which provides a safe, clean and healthy environment. The Indonesian government through the Ministry of Women's Empowerment and Child Protection has actively encourages the implementation of child friendly schools in various regions, with the issuance of the Minister of Women's Empowerment and Child Protection Regulation Number 8 of 2014.

SD Lab School FIP UMJ is one of the schools that is committed as a child friendly school. With the implementation of child friendly schools, the quality of learning at SD Lab School FIP UMJ has experienced a significant increase.

In the midst of efforts to create a more inclusive education system, the government is faced with various challenges in realizing child friendly schools, including socialization and training that is not sustainable, limited resources, budget and facilities and infrastructure, lackactive participation from schools, and weak cooperation of stakeholders and policy implementers.

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A STUDY ON THE IMPACT OF ONLINE ADVERTISEMENT TOWARDS COSMETIC PRODUCTS WITH SPECIAL REFERENCE TO CHENNAI CITY

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ABSTRACT

Online marketing is a new form of marketing media. It plays a vital role in the recent global era and is also inseparable from the younger generation. There are different kinds of communication platforms, including social media to reach internet buyers. The researcher has collected primary data through a structured questionnaire method and the convenience sampling method were used. 100 samples were collected for the present study. There are different in age, gender, educational background and occupation. The datas were collected from people who buy cosmetics in online. Online platforms will result in time savings, economic friendliness and a decrease in stress. According to this survey the majority of consumers prefer to purchase products through online and also expect new offers. Mobile phones, Laptops, PCs and Tablets are easy ways to access internet platforms. Most of the consumers are receiving damaged plastic and glass products in online purchasing. Moreover, while the consumers purchasing the products through online it seems entirely vary from online advertising by way of size, colour and quality.

Keywords : Online Advertisement, Cosmetics, Online shopping platforms and Personal Computer.

INTRODUCTION

Online marketing has been the digital aspect in the current period. Online advertising is also known as digital or internet marketing. It is

a way of promoting goods and services to reach potential consumers. It covers several kinds of marketing communications that are delivered via internet-based platforms such as social media, websites, email, and various display advertising (including web banners), as well as mobile and search engine advertising. The new modern media is having a huge possible impact on the appearance of a new division of consumers. The elders are enjoying in the online shopping because, it reduces travel tension and saves time. Modern consumers believes that they are comfortable to buy branded products than non-branded products through online. The online advertisement attracting consumers to catch their minds and also increasing their purchasing power. This research emphasizes the importance of understanding the buying process and consumers opinion about in online marketing. The respondents between the age group of 18 to 45 years spend more money for costumes across the family income. Both genders are purchasing products like children's clothing, jewellery, watches and different varieties of goods in the international market. Consumers are buying foreign branded goods even the price is higher than the domestic market. The most popular online shopping platforms in India are Amazon, Flipkart, Myntra, Nykaa, Meesho, Shopsy, Purple and JioMart.

STATEMENT OF THE PROBLEM

Advertising is one of the key factors in impressing on consumer attitude, or desire to buy in advertising. The researcher has carried out the study topic entitled "**A STUDY ON THE IMPACT OF ONLINE ADVERTISEMENT TOWARDS COSMETIC PRODUCTS WITH SPECIAL REFERENCE TO CHENNAI CITY.**" The purpose of the research is to identify the awareness of online advertisement, consumers perception, buying behaviour and pros and cons. In the present scenario, people are more conscious about their personal care. In common parlance, the high income group people are ready to spend more money on cosmetic items. Middle-class and lower-middle-class people are also willing to pay more for cosmetics and they are considered it a part and parcel of their family budget. Cosmetics

were used by only luxurious people. But now it is considered as essential product for middle- and lower-class people. Based on the age group, consumers taste and preferences will be different. Unfortunately, most of the consumers prefer to buy foreign branded products, which is not good for our national economic growth and development.

OBJECTIVES OF THE STUDY

1. To study the impact of online advertisement of cosmetic products.
2. To analyse the awareness of online advertisement of cosmetic products.
3. To examine the perception of online consumers towards online advertisement of cosmetic products.
4. To find out the pros and cons of online advertisement of cosmetic products.

LIMITATION OF THE STUDY

1. Frequently used cosmetic products only selected for the present study.
2. The study is confined only in Chennai city.
3. The researcher has collected only 35 questions due to time constraints.

REVIEW OF LITERATURE

Kwang Yeun Chun et al., (2014)¹ According to them, the internet environment has changed the format of internet advertising. One emerging form of online marketing communication is 'contextual advertising', in which marketers strive to develop customized images or texts more relevant to customers based on the content of web pages. This study investigates the effectiveness of internet contextual ads. In particular, they examine the effect of internet contextual ads on brand memory (i.e. recall and recognition) and attitudes towards the advertisement and/or brand using the theory

of priming effect. They found that the complexity of banner ads moderates the relationship between a contextual advertisement and its effectiveness, and generates two distinct priming effects (i.e. assimilation and contrast effects). The results demonstrate that the internet contextual advertisement enhances brand recognition and induces favorable attitudes towards the ad. In addition, consumers have higher recall rates and attitudes towards the brand when they are exposed to a less complex contextual advertisement or when they are exposed to a complex, non-contextual advertisement

Patil, Hema (2012)² Firstly this research identifies the behavioral factors: based on which consumers evaluate any cosmetic product. Secondly the research has identified the behavioral factors which are sensitive to the demography. Thirdly the research has identified separate set of factors which influence the marketing strategy of chemical and herbal cosmetic companies. Fourthly the research has identified, major discriminating factors between chemical and herbal cosmetic products.

Sivaneshgvaran S (2016)³ This objectives are categorized into educational products, career / job opportunities, commercial products, online advertisement with social importance and safety and security measures of online advertisements. This will ultimately raise a question whether the online advertisement creates any impacts. Though there many people using internet for information gathering, entertainment, chatting, exchanging information and ideas, buying and selling of product and services, teenagers are mostly attracted by internet sophistications. Hence the report of the study conducted signifies the constructive and caustic impact of advertisement among the teenagers.

Punyatoya, Plavini (2011)⁴ **Internet** being the fastest-growing medium of all time provides a lot of opportunity to almost every industry in the 21st century. As consumers move online, so do advertisers. The purpose of their research was to study the features that lead to an effective internet banner advertising and to compare and contrast the pattern of banner ad that is being used by advertisers in two large categories (products and services). Content

analysis was conducted for comparing product and service banner ads posted on Indian websites. It was found that the banners ads were effective due to absence of promotional incentives, presence of information and absence of animation. But ineffectiveness is present in the banner ads due to presence of company logo and action phrase on them. Firms also need to increase size of the banner and length of the message in the ad, to make them more effective. Product and service banner ads were also found to be different in terms of effectiveness. This research will help advertiser and marketers to have an effective internet banner ad, so that consumer will form a positive attitude for the product or service banner ad.

Puneet Arora (2011)⁵ has conveyed in his article stated that due to technology accessibility and increased advertising clutter commercial avoidance has more in the last decade and has become one of the main worries of advertisers and media firms. He has suggested that the message should be conveyed to the target group by means “right message, right group, right medium and right time will reduce the zapping of advertisement”.

RESEARCH METHODOLOGY

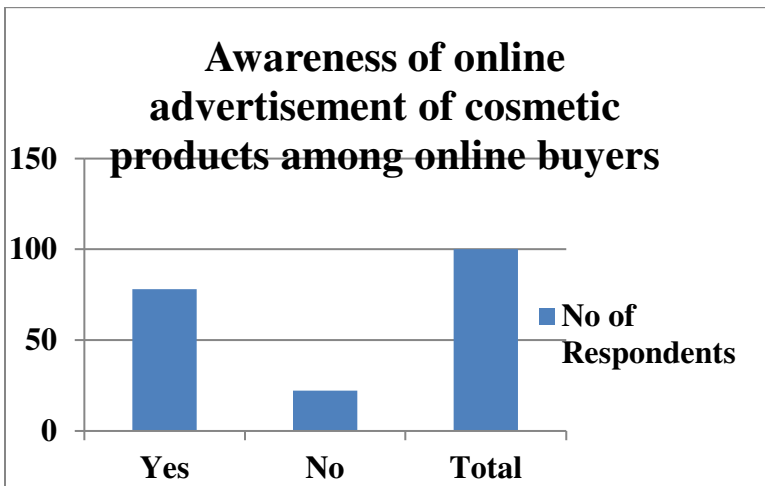
The researcher has used the primary data method. The researcher has selected 100 respondents to collecting questionnaire on the impact of online advertisement of cosmetic products. The convenient sampling technique was used. Data were collected using a well-structured questionnaire method and it involves two parts. Part I determines the questions related to demographic factors, and Part II determines the online buying consumers behaviours in Chennai city. The collected data were analyzed using a simple bar diagram.

DATA ANALYSIS

TABLE 1 SHOWING THAT THE AWARENESS OF ONLINE ADVERTISEMENT OF COSMETIC PRODUCTS

Awareness of online advertisement of cosmetic products among online buyers	No of Respondents
Yes	78
No	22
Total	100

Figure 1



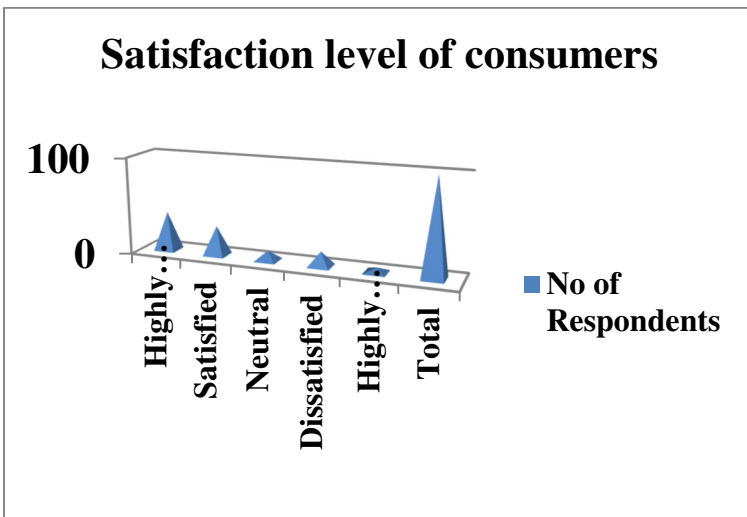
INTERPRETATION

The above table shows that the majority of the respondents are aware about online advertisement of cosmetic products.

TABLE 2 SHOWING THAT THE SATISFACTION LEVEL OF CONSUMERS.

Satisfaction level of consumers	No of Respondents
Highly satisfied	40
Satisfied	30
Neutral	10
Dissatisfied	15
Highly dissatisfied	5
Total	100

Figure 2



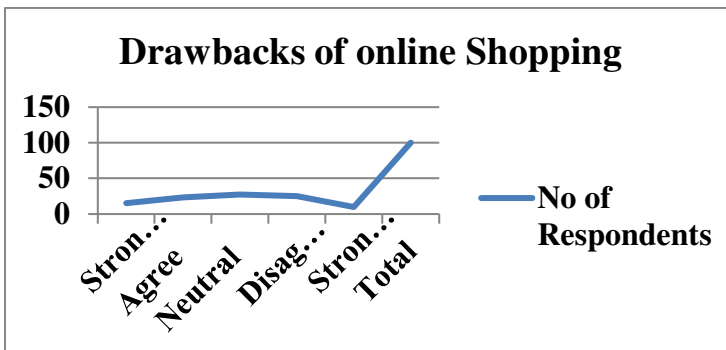
INTERPRETATION

As per the above table, the majority of respondents are highly satisfied with online purchasing.

TABLE 3 SHOWING THAT DRAWBACKS OF ONLINE SHOPPING.

Drawbacks of online Shopping	No of Respondents
Strongly Agree	15
Agree	23
Neutral	27
Disagree	25
Strongly Disagree	10
Total	100

Figure 3



INTERPRETATION

It is clear from the above table that some of the respondents feel that there are given drawbacks in online shopping.

FINDINGS

- The majority of the respondents are females.
- There are different age groups of the respondents using online advertisement: 15-20 years 10.8%, 20-30 years 67.6% and 31-40 years 16.2%.

- The majority 40.5% of the respondents are postgraduate, 32.4% are undergraduate and 10.8% are professional degree holders.
- The majority 43.2% of the respondents are students, 29.7% of the respondents who earn a monthly income of Rs.10,000 to 25,000 and 24.3% of the respondents who earn a monthly income of Rs.25,000 to 50,000.
- The majority of the respondents buy cosmetic products for their own needs.
- The majority of the respondents are using the cash on delivery mode of payment.
- The majority 54.1% of the respondents purchase only the branded products, 24.3% of the respondents purchase for the price and 21.6% of the respondents purchase the features.
- The majority of the respondents preferred the Amazon online platform.

SUGGESTION AND CONCLUSION

This study was conducted to find out the impact of advertisements on the buying behaviour of people in cosmetic items. Online advertising has had a great impact on people for the past few years. Most of the consumers have good knowledge of online advertisement. The majority of the respondents are educated and salaried employees and they have good knowledge about online shopping. The consumers are satisfied with the lowest prices, sales and services, Easy to buy, less travelling expenses, reduced stress, more variety, and product quality in online shopping. Some of the respondents expecting chemical free products in online shopping. Few consumers request to reduce shipping cost while purchase products through online and purchase also expecting a quick refund policy.

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TREND ANALYSIS ON HUMAN DEVELOPMENT INDEX IN TAMIL NADU FROM 1993-2021

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ABSTRACT

The Human Development Index (HDI) is a composite statistical measure developed by the United Nations Development Programme (UNDP) to assess the overall development and well-being of a country's population. It focuses on three key dimensions of human development: health (measured by life expectancy at birth), education (measured by mean years of schooling and expected years of schooling), and standard of living (measured by Gross National Income per capita adjusted for purchasing power parity). By combining these indicators, the HDI offers a broader perspective on development beyond mere economic growth, emphasizing people-centered progress. HDI values range from 0 to 1, where higher values indicate greater human development. The index serves as a powerful tool for comparing development levels across countries and regions, identifying inequalities, and informing policy decisions aimed at improving quality of life. Hence this study focuses on to assess HDI indicators and the composite growth rate in Tamil Nadu from 1993 to 2021.

Keywords: HDI, UDNP, wealth, standard of living etc..

INTRODUCTION

Human development is a process of enlarging people's choices and enhancing human capabilities—the range of things people can do and the freedom they enjoy in life. As defined by the United Nations Development Programme (UNDP), human development involves improving health, expanding access to education, and ensuring a decent standard of living. To quantify and compare development across regions, the UNDP introduced the Human Development Index (HDI)—a composite measure based on three key dimensions: life expectancy at birth (health), education level (years of schooling), and per capita income (standard of living). In a country as diverse as India, human development outcomes vary widely across states and districts. Tamil Nadu, located in the southern part of India, is often recognized as one of the better-performing states in terms of social and economic development. With a history of progressive policies in healthcare, education, and welfare, Tamil Nadu has achieved relatively high HDI scores compared to the national average. It is home to a robust public distribution system, pioneering social security schemes, and a well-established public health infrastructure that have significantly contributed to its development journey.

However, despite these achievements, the intra-state disparity in human development remains a critical concern. While metropolitan areas such as Chennai, Coimbatore, and Kanchipuram demonstrate high levels of human development, several districts in the interior and rural parts of the state—such as Ariyalur, Perambalur, and Viluppuram—lag behind in one or more HDI dimensions. These disparities reflect structural inequalities rooted in access to resources, socio-economic status, regional infrastructure, gender, caste, and urban-rural divides.

This study is undertaken with the aim of examining the status, patterns, and trends of human development in Tamil Nadu, with particular focus on understanding the causes of spatial and socio-economic inequalities across districts. It evaluates the components

of HDI—health, education, and income—through available data over a period of time to identify both improvements and stagnation points. The study will also assess the impact of public policies, such as free education schemes, public health missions, and poverty alleviation programs, on human development outcomes in the state.

Furthermore, this research attempts to explore how effectively Tamil Nadu has translated economic growth into human development, and whether this development is inclusive and equitable. It aims to highlight the district-level differentials, gender disparities, and the role of governance and institutional mechanisms in shaping human development trajectories.

By drawing on official data, previous human development reports, and secondary research, this study aspires to offer a nuanced understanding of Tamil Nadu's development model. The insights derived can help inform policy decisions, direct resources to lagging regions, and contribute to the discourse on achieving balanced and sustainable human development across all sections of society.

A comprehensive assessment of HDI in Tamil Nadu is crucial not only for recognizing the progress made but also for identifying the persistent gaps that must be addressed in the pursuit of equitable development. The findings of this study will thus provide valuable input for policymakers, development planners, and academic researchers committed to building a just and inclusive society.

REVIEW OF LITERATURE

The concept of human development has been the subject of extensive scholarly research over the past few decades, particularly in the Indian context where regional disparities are pronounced. This review examines the major studies and academic contributions that have informed the understanding of HDI and its components—health, education, and standard of living—with special reference to Tamil Nadu.

1. Theoretical Foundations of Human Development

The Human Development Index was first introduced in the UNDP Human Development Report (1990), conceptualized by Mahbub ul Haq and supported by Amartya Sen's capabilities approach. Sen emphasized that development should be assessed by the expansion of human freedoms and opportunities, not merely by economic growth.

Numerous scholars such as Anand and Sen (1994) have argued that HDI offers a broader view of development by focusing on people-centric indicators. While HDI has its limitations—such as ignoring inequality and environmental sustainability—it remains a globally accepted tool for comparing regional and national progress.

2. HDI in the Indian Context

India has published periodic reports at national and sub-national levels. The India Human Development Report (2011) by the Institute of Applied Manpower Research (IAMR) highlighted stark disparities across states and emphasized the role of public policy in influencing HDI outcomes.

Studies like Desai et al. (2015) using the India Human Development Survey (IHDS) showed that improvements in HDI indicators are closely linked to public spending in education and health, especially in states like Kerala and Tamil Nadu. Dreze and Sen (2013) in their work "An Uncertain Glory" argued that Tamil Nadu's development model—similar to Kerala—combines economic growth with strong social sector investments.

3. Human Development in Tamil Nadu

Tamil Nadu has been widely studied for its unique approach to development:

- Tamil Nadu Human Development Report (2003): The first state-level HDR in India, this report highlighted high achievements in education and health compared to national averages. It credited welfare schemes such as midday meals, free school uniforms, and public health investments for improving HDI outcomes.
- Tamil Nadu HDR (2017): This updated report focused on inter-district disparities, gender development indices, and inclusive growth. It found that while districts like Chennai and Kanyakumari ranked high in HDI, backward districts such as Ariyalur and Perambalur continued to lag, especially in income and educational attainment.
- MIDS (Madras Institute of Development Studies) research has shown that HDI improvements in Tamil Nadu are correlated with political commitment to social justice and targeted welfare measures.

4. District-Level Disparities

Several micro-level studies and doctoral theses have explored the intra-state inequalities in Tamil Nadu:

- Ramakrishnan (2018), in a study on "District-Level Human Development in Tamil Nadu", analyzed HDI components and showed that access to basic amenities, gender empowerment, and literacy were unevenly distributed.
- Annamalai and Rajan (2020) focused on gender-based disparities in HDI using the Gender Development Index (GDI) and Gender Inequality Index (GII), noting the need for gender-responsive planning at the district level.
- A study by Sundaram and Mukherjee (2016) emphasized that while health and education indicators have improved steadily in Tamil Nadu, income inequality and urban-rural divide remain persistent challenges.

5. Role of Public Policy

Scholars have consistently highlighted the impact of Tamil Nadu's welfare-centric governance on human development. Schemes like:

- Amma Canteens, Amma Baby Kits, and Universal PDS
- Free bus passes for students, cradle baby schemes, and urban health missions

It has significantly contributed to human development outcomes, especially among marginalized groups. Harriss (2000) described Tamil Nadu's model as a "pragmatic populism" where welfare and growth go hand-in-hand.

6. Link with Sustainable Development Goals (SDGs)

Recent literature also aligns HDI with SDGs, especially SDG 3 (Good Health), SDG 4 (Quality Education), and SDG 10 (Reduced Inequalities). Studies by NITI Aayog (2020) and UNDP India show that Tamil Nadu ranks among the top five performers in SDG achievement, reflecting its consistent human development trajectory.

OBJECTIVES OF THE STUDY

The main objective of this study is to analyze and evaluate the Human Development Index (HDI) of Tamil Nadu with a focus on regional disparities, progress over time, and contributing socio-economic factors. The specific objectives include:

- To assess the overall HDI status of Tamil Nadu
- To examine district-wise disparities in human development
- To analyze the trends and progress in HDI components since 1993

- To provide policy recommendations for inclusive human development

RESEARCH METHODOLOGY

This research aims to find the trend of HDI in Tamil Nadu since 1993 to 2021. This research uses secondary data which is obtained from, database and reports by the Sub-national Human Development Index (SHDI) - Tamil Nadu, India (1993–2021). Further, this research paper aims to study the major components of HDI indicators like education, income and health and their growth rate over the period of years and to find the compound rate of HDI from 1993 to 2021. This paper is based on descriptive and analytical study of State-wise analysis of Tamil Nadu with a focus on district-level variations.

RESULTS AND DISCUSSIONS

1. Tamil Nadu's Overall HDI

- **State HDI (2023–24):** According to the UNDP/Global Data Lab, Tamil Nadu's HDI was around **0.692**—well above India's national average (~0.644) and placing the state in the '**high human development**' category
- **National Rank:** In 2023, among Indian states and union territories, Tamil Nadu ranked **14th**, following states like Goa, Kerala, Chandigarh, and Puducherry.

2. Progress of HDI from 1993 to 2021

Here are the **education, income, and health indices** for **Tamil Nadu** from **1993 to 2021** (latest year with full data available), according to the Global Data Lab (subnational HDI):

Table No:1.1: Progress of HDI from 1993 to 2021

Year	Education Index	Income Index	Health Index	Composite HDI
1993	0.443	0.445	0.635	0.476
1994	0.448	0.448	0.641	0.478
1995	0.454	0.462	0.647	0.484
1996	0.462	0.471	0.658	0.488
1997	0.471	0.481	0.668	0.494
1998	0.481	0.485	0.679	0.500
1999	0.493	0.493	0.689	0.507
2000	0.504	0.504	0.700	0.513
2001	0.508	0.508	0.711	0.519
2002	0.513	0.513	0.722	0.528
2003	0.517	0.517	0.731	0.536
2004	0.526	0.526	0.740	0.543
2005	0.536	0.536	0.750	0.552
2006	0.546	0.546	0.760	0.569
2007	0.556	0.556	0.769	0.581
2008	0.573	0.573	0.778	0.593
2009	0.582	0.582	0.789	0.606
2010	0.599	0.599	0.790	0.615
2011	0.616	0.616	0.794	0.622
2012	0.630	0.630	0.797	0.628

2013	0.643	0.643	0.801	0.638
2014	0.645	0.645	0.806	0.653
2015	0.649	0.649	0.812	0.662
2016	0.654	0.654	0.818	0.667
2017	0.658	0.658	0.826	0.676
2018	0.664	0.664	0.832	0.682
2019	0.669	0.669	0.838	0.691
2020	0.670	0.670	0.836	0.693
2021	— (0.608)	— (0.670)	0.770	0.687*

www.Global Data Lab. (2024). Sub-national Human Development Index (SHDI) - Tamil Nadu, India (1993–2021). Institute for Management Research, Radboud University. Retrieved from <https://globaldatalab.org/shdi>

Figure No:1.1: Trend of HDI in Tamil Nadu

**Source: Computed from secondary data
Education Index (1993–2021)**

- 1993–2005: Slow and steady improvement due to expanding school enrollment and basic literacy.
- 2005–2015: Accelerated improvement, likely due to SSA, RTE Act, and mid-day meal schemes.
- 2016–2020: Plateauing near 0.670, showing maturing of basic education metrics.
- 2021: Data not directly available, but pandemic may have reduced effective learning (est. value: 0.608).

Income Index Trend (1993–2021)

- Steady rise reflecting economic growth from industrialization, IT services, and infrastructure.

- Stable trend post-2015, indicating income improvements reaching a saturation-like stage.
- 2021 estimated at 0.670 — stable, but inflation and employment shocks may have affected real income.

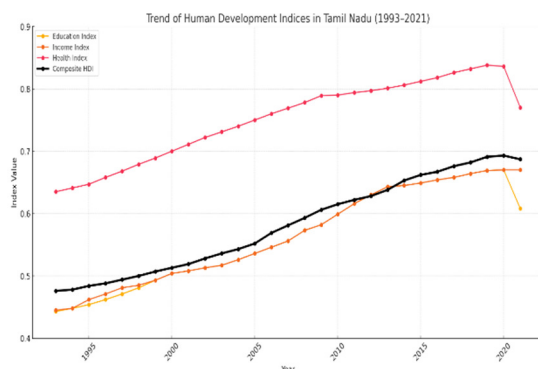
Health Index Trend (1993–2021)

- Tamil Nadu started with a strong health base even in 1993.
- Gradual improvement due to health infrastructure, maternal care, immunization, and nutrition programs.
- Highest index of the three, showing Tamil Nadu's continued strength in health outcomes.
- Only index that maintained growth even during COVID-19 (2021).

3. Inter-District Variation

District-level disparities are significant:

- **Top HDI districts (2024 estimates):**
 - Chennai: 0.841
 - Kanchipuram: 0.824
 - Erode: 0.823



- Coimbatore: 0.822
- Thiruvallur: 0.817

- **Lowest HDI districts:**
 - Nagapattinam: 0.699
 - Viluppuram: 0.688
 - Earlier (2017 report) data showed other extremes:
- **Highest (2017):** Kanyakumari (0.944), Virudhunagar (0.855), Thoothukudi (0.852), Chennai (0.847), Kancheepuram (0.845).
- **Lowest (2017):** Thiruvavarur (0.568), Villupuram (0.561), Theni (0.539), Perambalur (0.447), Ariyalur (0.282)

Testing of Hypothesis:

H_0 : There is no significant disparity in Human Development Index among the districts of Tamil Nadu.

H_1 : There exists significant disparity in Human Development Index among the districts of Tamil Nadu.

Based on the provided HDI data from both 2024 and 2017:

- **Mean HDI** across selected districts: **0.713**
- **Standard Deviation (SD): 0.184**
- **Coefficient of Variation (CV): 25.75%**

Reject the null hypothesis (H_0): There is no significant disparity in HDI among the districts of Tamil Nadu. **There is statistically significant variation in human development across Tamil Nadu's districts**—confirming developmental imbalance that requires policy attention.

4. Underlying Indices and Social Indicators

- **Education:** Expected schooling ~11.9 years; mean years ~6.7 years for India (~similar for TN).
- **Health:** Life expectancy ~74 years (TN, 2019).
- **Income:** Tamil Nadu's per capita GDP (PPP 2025 est.) is around **US \$22,944**—higher than national average

5. Drivers & Challenges

- **Social policies:** Public support systems like midday meals, public health infrastructure, social welfare have been pivotal in Tamil Nadu's high health and education outcomes
- **District disparities:** Despite strong state-level performance, backward districts like Ariyalur and Perambalur lag significantly.
- **Emerging concerns:** Growth of non-communicable diseases (obesity, diabetes) means newer health challenges are emerging

6. Key Milestones

- **2003:** The first Tamil Nadu Human Development Report was published by the State Planning Commission with UNDP support, based on Baseline district HDI, policy focus
- **2017:** A second, comprehensive report (supported by NITI Aayog, UNDP, and MIDS Chennai) highlighted inter-district disparities and gender-inequality dimensions, highlighted to Inequality, gender, inter-district gaps.
- **2021:** Third HDI report prepared by State Planning Commission focus on COVID-19 impact, SDG indicators.

POLICY IMPLICATIONS

1.Strengthen Equitable Development Across Districts

- Launch district-focused human development missions (like Aspirational Districts Programme).
- Increase budgetary allocations and special packages for backward districts.
- Enhance infrastructure (schools, hospitals, internet access) in low-HDI areas.

2. Post-COVID Recovery in Education

- Invest in bridging learning gaps through remedial education, digital learning access, and teacher training.
- Strengthen school resilience systems (e.g., digital infrastructure for hybrid learning).
- Implement early warning systems for dropout risks in rural areas.

3. Address Health Transitions and NCDs

- Shift focus to preventive healthcare, lifestyle education, and screening for NCDs.
- Expand urban wellness centers and rural health sub-centers to include NCD care.
- Integrate mental health services within primary health infrastructure.

4. Boost Quality and Skills in Education

- Focus on quality of learning outcomes, not just access.
- Promote vocational education, soft skills, and digital literacy.
- Align curriculum with future job market demands (AI, automation, green skills).

5. Tackle Income Growth Saturation

- Incentivize diversification in rural and semi-urban economies (e.g., agro-processing, MSMEs).
- Encourage green and digital economy sectors (e.g., solar energy, IT-enabled services in tier-2 towns).
- Create inclusive job opportunities for women, youth, and informal workers.

• 6. Continue Gender and Social Equity Focus

- Strengthen gender budgeting and women-focused welfare schemes (education, employment).
- Promote inclusive policies for SC/ST communities, especially in education and skill development.
- Improve data collection on marginalized communities for targeted planning.

7. Align HDI Goals with SDG Targets

- Localize Sustainable Development Goals (SDGs) into state and district development plans.
- Use data-driven dashboards to track health, education, and income targets.
- Institutionalize inter-departmental coordination for integrated HDI–SDG planning.

8. Enhance Data Systems and Monitoring

- Invest in real-time district-level data collection systems.
- Collaborate with academic institutions (e.g., MIDS) and civil society for data validation and transparency.
- Publish regular HDI updates and open-access dashboards for public and academic use.

CONCLUSION

Tamil Nadu has made noteworthy progress in human development—rising from an HDI of 0.457 in 1993 to ~0.687 in 2021, and currently ranked amongst the top-performing Indian states. Strong public infrastructure in education and health have driven this growth. Yet, intra-state inequalities across districts and rising lifestyle health challenges call for focused policy interventions. Tamil Nadu’s HDI trajectory is a success story in many respects but needs fine-tuned, inclusive, and forward-looking policies to sustain and distribute progress equitably. Focusing on backward districts,

education quality, health transitions, and income diversification will be key to the next stage of development.

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PRINCIPAL'S POLICY INNOVATION IN INTEGRATING DOUBLE- TRACK PROGRAMS WITH THE REGULAR CURRICULUM

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Abstract

This qualitative descriptive study examines the principal's policy innovation in integrating the double track program with the regular curriculum at State Senior High School 1 Galis, Pamekasan Regency, East Java, Indonesia. The research employed observation, interviews, and documentation analysis using Miles and Huberman's data analysis model to investigate how educational leadership transforms traditional academic frameworks through innovative policy implementation. The study reveals five key dimensions of successful program integration: strategic policy innovation through talent-based skill selection and competent trainer recruitment; systematic implementation via Training of Trainers programs with East Java Provincial Office and Sepuluh Nopember Institute of Technology collaboration; measurable outputs producing graduates with dual competencies for immediate workforce entry; comprehensive supporting factors including institutional commitment, qualified trainers, modern infrastructure, and industry partnerships; and adaptive solutions addressing implementation challenges through flexible scheduling and collaborative learning systems. The findings demonstrate that the principal's transformational leadership successfully bridges academic and vocational education, producing graduates who secure employment in business and industry sectors while meeting local economic development needs. The research contributes a replicable model for educational institutions seeking to enhance program relevance through strategic policy innovation, establishing a framework where institutional partnerships, human resource development, and adaptive leadership create sustainable educational transformation

that serves diverse student pathways and community economic requirements.

Keywords: principal leadership, policy innovation, double track program, curriculum integration, vocational education

INTRODUCTION

The Indonesian education system faces challenges in preparing graduates with both academic and technical skills for the workforce. Vocational education has a long history in Indonesia but struggles with inadequate facilities, teacher quality, and industry support (Suharno et al., 2020). Higher education curricula often lack relevance to industry needs, leading to a mismatch between graduate competencies and job requirements (Rianto et al., 2022). Implementing a dual vocational education system presents difficulties in public-private cooperation and contextual adaptation (Wibowo et al., 2022). To address these issues, Indonesia is focusing on improving access, quality, and industry linkages in education (Indrawati & Kuncoro, 2021). Developing a competency framework for "k-workers" and enhancing work-based learning through industrial internships are proposed solutions (Santoso & Hassan, 2018; Sutiman et al., 2022). Additionally, establishing occupational-specific competency standards for vocational teachers is crucial for improving the system (Estriyanto et al., 2017).

The double track program is an innovative educational approach that combines general and vocational education to provide students with more flexible learning paths (Cholidah et al., 2024; Saputra, 2022). Implemented in various regions of Indonesia, including East Java and Gorontalo, this program aims to develop students' entrepreneurial skills and reduce unemployment among high school graduates (Alfiyah et al., 2022; Yulikah et al., 2021). The program has shown positive impacts on students' practical and academic skills, as well as enhancing school reputation (Cholidah et al., 2024). It includes various entrepreneurship education forms such as culinary, make-up, fashion, and computer skills (Mayudho et al., 2023). Despite challenges like limited funding and equipment, the

program has been successful in fostering students' entrepreneurial spirit and independence (Cahyono & Subiyantoro, 2022; Kholis et al., 2022). The implementation process involves planning, organizing, executing, monitoring, and evaluating stages, with collaboration between schools and external partners (Kholis et al., 2022; Mayudho et al., 2023).

The principal has a strategic role in improving the quality of education through effective leadership. As a visionary leader, the principal is responsible for managing resources, formulating innovative policies, and motivating teachers to improve performance (Moh, 2024). Effective leadership strategies include improving the quality of human resources, developing ICT infrastructure, open leadership, and collaboration with various stakeholders (Aryawan, 2019; Juharyanto et al., 2020). The principal also needs to function as an educator, manager, administrator, supervisor, and innovator (Agustin & Firmansyah, 2023). A dialogical approach and open communication with staff can build good relationships and increase work effectiveness (Rohman, 2020). Although an authoritarian leadership style is sometimes necessary, it is important to balance it with openness to input and self-evaluation (Anisa, 2020).

The integration of double-track programs with regular curricula requires specific strategies due to differences in orientation and learning methods. Integration models should balance religious and general education, emphasizing character development and aligning with the institution's vision and mission. Double-track programs aim to enhance students' entrepreneurial skills and provide additional competencies (Cahyono & Subiyantoro, 2022; Cholidah et al., 2024). Strategic management is crucial for effective implementation, ensuring systematic, planned, and controlled learning processes (Setiawan & Hermansyah, 2023). Successful integration requires a strong commitment from teachers, effective collaboration with parents, and addressing challenges such as limited funding and equipment (Alfiyah et al., 2022). SWOT analysis can help formulate aggressive growth strategies for double-track programs (Lutfi & Churiyah, 2023).

Secondary schools in Indonesia face challenges in balancing academic education with technical skills. Several vocational high schools have difficulty preparing graduates for the workforce because the Teaching Factory program has not fully met industry needs (Sariwulan et al., 2020). To overcome this, several schools have implemented robotics training (Hozairi & Kurdianto, 2019) and computer skills (Montolalu & Langi, 2018) to improve student competency. Evaluation of competency certification programs is also carried out to increase effectiveness (Priyanto et al., 2024). The principal plays an important role in improving the quality of learning through professional supervision and implementation of innovative programs (Marhawati & Sulkify, 2021).

The implementation of educational programs, including dual vocational training and multicultural education, faces several challenges in Indonesia. Limited human resources, facilities, and time are major constraints (Anwar, 2018; Rahim, 2022). Schools struggle with inadequate budgets, weak management skills, and low teacher compensation (Rahim, 2022). The mismatch between industry requirements and available skills hinders the successful implementation of dual vocational education (Wibowo et al., 2022). Other challenges include low parental interest in Islamic schools, inefficient financial planning, and lack of coordination between school committees and education authorities (Anwar, 2018; Wahyudi et al., 2021).

The success of the double-track program in enhancing students' skills is significantly influenced by the school principal's ability to create synergy between academic and vocational education through effective policies. Research indicates that principals play a crucial role in implementing these programs, as they coordinate resources and foster collaboration among stakeholders (Lita et al., 2021; Wahyudin, 2018). Evaluations of various double-track implementations show positive outcomes in developing students' entrepreneurial skills and interests, despite challenges such as limited facilities (Alfiyah et al., 2022; Cholidah et al., 2024). Additionally, the integration of entrepreneurship education within these programs has proven effective in promoting student

independence and creativity (Cahyono & Subiyantoro, 2022; Kholis et al., 2022). Overall, the principal's leadership and strategic policy-making are essential for the successful integration of academic and vocational education (Angga & Iskandar, 2022; Saputra, 2022).

The "Double Track" program has been evaluated across various high schools in Indonesia, demonstrating its effectiveness in enhancing students' entrepreneurial skills and reducing unemployment rates. Cholidah et al. (2024) found that the program positively impacts students' practical and academic skills, despite facing funding and resource challenges. Similarly, Alfiah et al. (2022) reported successful implementation in Banyuwangi, highlighting student enthusiasm and technological support as key factors. Cahyono & Subiyantoro (2022) noted a significant positive influence on students' entrepreneurial interest, while Dinaloni emphasized their role in addressing high unemployment among high school graduates (Dinaloni & Indrawati, 2021). Nur Cholis et al. (2022) illustrated improvements in creative economic skills through partnerships and local curriculum integration. Lutfi & Churiyah (2023) proposed a SWOT-based strategy for enhancing the program, and Asrori highlighted its potential to increase job opportunities and economic growth in East Java (Asrori et al., 2022). Alhamidi affirmed the program's viability in expanding secondary education (Alhamidi et al., 2021).

There is a need for in-depth research on the innovation of principal policies in overcoming the challenges of integration to produce a model that can be adapted by other schools. This research is important to identify innovative strategies that have proven successful in overcoming the challenges of integrating double-track programs with the regular curriculum. Through in-depth research, practical solutions can be found related to scheduling, resource allocation, and effective evaluation methods, so that the results can be used as a guide for other principals in designing similar policies.

METHOD

This study uses a descriptive qualitative approach to examine the principal's policy innovation in integrating the double-track

program with the regular curriculum. The location of study was conducted at State Senior High School 1 Galis located on Konang Street, Konang, Galis District, Pamekasan Regency, East Java, Indonesia. The selection of this location was based on the fact that the school has implemented a double-track program that aims to build student skills by providing entrepreneurial knowledge, especially for students who cannot continue to college. This program is the result of collaboration between the school and the East Java Provincial Government in collaboration with the Sepuluh Nopember Institute of Technology (ITS), thus providing a rich context for research.

The types of data collected in this study are grouped into primary data and secondary data, using three data collection techniques, namely observation, interviews, and documentation to obtain comprehensive information about the implementation of the policy. Data analysis refers to the Miles and Huberman model which consists of three stages, namely data condensation, data presentation, and drawing conclusions to interpret the findings systematically.

To ensure the validity of the data, this study applies a credibility test through observation persistence, observation extension, triangulation (sources and methods), negative case analysis, use of reference materials, and member checking to verify findings with informants. In addition, this study also applies a validity test that includes transferability to ensure that research results can be applied to other contexts, dependability to maintain data consistency and stability, and confirmability to ensure research objectivity and avoid bias, so that research results have high credibility and validity.

RESULT

Based on the results of qualitative research conducted at State Senior High School 1 Galis, Pamekasan Regency, comprehensive findings were obtained regarding the principal's policy innovation in integrating the double track program with the regular curriculum. These findings indicate that the principal has

implemented various innovative strategies ranging from strategic planning, integrated implementation, to continuous evaluation to ensure the success of the program. Through in-depth observation, interviews with various stakeholders, and documentation analysis, this study identified five main aspects that describe the complexity and success of the policy innovation, as described in the following findings.

Principal's Strategic Policy Innovation

In implementing strategic policy innovation, the principal demonstrates comprehensive leadership through three key initiatives. 1) The principal implements a skill selection policy based on an analysis of student talents and interests and the needs of the local community to ensure the relevance of the program to student potential. 2) The principal sets standards for recruiting competent trainers as a key policy to ensure the quality of student guidance in developing an entrepreneurial spirit and practical skills. 3) The principal makes a special targeting policy for Private College students with an interest selection system, such as a make-up program that is of interest to students with a makeup hobby. These interconnected policy innovations reflect the principal's holistic approach to integrating double track programs with regular curriculum while maintaining focus on student-centered learning and community relevance.

Implementation of Program Integration Policy

The implementation of the program integration policy demonstrates the principal's systematic approach to operationalizing the double-track program within the existing educational framework. 1) The principal provides full support through the trainer human resources development policy via the TOT (Training of Trainers) program in collaboration with the East Java Provincial Office and Sepuluh Nopember Institute of Technology (ITS), including the provision of an official travel order letter for weekly workshops. 2) The principal establishes a flexible scheduling policy between the double-track program and the regular curriculum to ensure harmonious integration without disrupting the

main learning. 3) The principal initiates a partnership policy with the Business and Industry Sector as a place for students' skills application and a bridge to the world of work. These implementation strategies collectively ensure that the integration process maintains educational quality while providing practical pathways for student development and career preparation.

Integration Policy Output

The integration policy demonstrates tangible outcomes that validate the effectiveness of the principal's innovative approach to combining academic and vocational education. 1) Students gain dual competency (academic and vocational) that prepares them to directly enter the workforce for those who do not continue to college. 2) The program produces graduates who are absorbed in the business and the industry sector, with culinary alumni working in the bakery industry and makeup alumni opening independent salons. These measurable outputs illustrate the successful transformation of educational policy into practical career opportunities, providing students with viable alternatives to higher education while meeting local economic needs.

Policy Supporting Factors

The success of the integration policy is underpinned by multiple reinforcing factors that create a conducive environment for program implementation. 1) Full support from school management and teachers as ongoing motivation for trainers through participation in competency development webinars. 2) The availability of highly qualified trainers as the backbone of the success of program integration with the regular curriculum. 3) Modern and updated facilities and infrastructure that support the effectiveness of integrated learning practices. 4) Positive synergy with business and industry sector and community support that utilizes the services of the double track program. These interconnected supporting elements form a comprehensive ecosystem that enables the principal's innovative policies to flourish and deliver sustained educational outcomes.

Challenges and Solutions for Policy Innovation

The implementation of the integration policy encounters specific challenges that require innovative solutions to maintain program effectiveness. The principal addresses the lack of student discipline and consistency, along with complaints of fatigue due to a busy 5-day schedule, through strategic policy adaptations. As innovative solutions, the principal implements a flexible schedule policy and a group/post system per program to increase team responsibility, motivation, and collaboration in integrated learning. This adaptive approach demonstrates the principal's capacity to transform operational challenges into opportunities for enhanced student engagement and program sustainability.

DISCUSSION

Principal's Strategic Policy Innovation

The findings on the principal's strategic policy innovations demonstrate a transformative leadership approach that prioritizes holistic student development. The three policy initiatives implemented reflect the principal's deep understanding of the complexity of modern education, which must be able to address the challenges of individual student needs as well as community demands. There is a growing emphasis on regional approaches to skills policy, integrating place-based perspectives and enhancing cross-sector opportunities (Corradini et al., 2023). The concept of 21st-century skills has gained traction globally, influencing education policies to focus on creativity, responsibility, and self-regulation (Hilt et al., 2019).

The skills selection policy, based on an analysis of student talents and interests, as well as the needs of the local community, demonstrates a strategic vision that is not only academically oriented but also focused on developing practical competencies relevant to the socio-economic context of the surrounding environment. Progressive and reconstructionist pedagogies, such as project-based learning and youth participatory action research, have shown promise in developing these competencies (Dobson, 2023). However, challenges remain in translating broad policy goals into

practical implementation, with practitioners often struggling to make sense of vague policy contexts (Muñiz, 2020).

The policy of recruiting competent trainers as a foundation for the quality of guidance indicates the principal's awareness that the success of the program depends not only on the curriculum design but also on the quality of the human resources who implement it. Research indicates that effective entrepreneurship education requires competent trainers and standardized competencies to develop students' entrepreneurial skills (Huang et al., 2020; Miço & Cungu, 2023). Successful programs integrate experiential learning opportunities, including guest speakers, pitches, and industry events (Bliemel, 2013).

Standardization of trainer competencies in developing students' entrepreneurial spirit and practical skills is key to ensuring effective transfer of knowledge and experience. This strengthens the argument that educational innovation requires continuous investment in developing the capacity of educators who not only master technical aspects but are also able to inspire and guide students in exploring their potential. Entrepreneurship education should be incorporated across curricula and extracurricular activities (Isrososiawan, 2013), with collaborative and practical methods like hackathons and role-play enhancing entrepreneurial competencies (Iglesias-Sánchez et al., 2019). Studies emphasize the importance of developing resilience, innovation, and opportunity recognition in students to prepare them for uncertain futures (Hardie et al., 2020). Vocational education plays a crucial role in developing entrepreneurial competencies, particularly through extracurricular activities and entrepreneurial environments (Perić et al., 2020).

The policy of targeting students specifically for private colleges with an interest selection system, such as the make-up program, demonstrates the principal's flexibility and responsiveness to the diversity of student backgrounds and aspirations. This program not only provides space for students to develop hobbies into commercially viable skills but also creates an alternative path to education that is more personal and meaningful. The integration of the double-track program with the regular curriculum through a

student-centered learning approach demonstrates the principal's ability to create an inclusive and adaptive educational ecosystem, where each student can find a development path that suits their potential and interests while maintaining the required academic standards.

This strategic policy innovation strengthens the position of the principal as an agent of change who is able to transform the educational paradigm from being uniform to being more differentiated and contextual. The holistic approach applied not only has an impact on increasing the relevance of education to the needs of students and society but also creates a model of educational leadership that can be adapted by other institutions in facing the challenges of diversifying learning needs in the contemporary era.

Implementation of Program Integration Policy

The implementation of the program integration policy carried out by the principal demonstrates a strategic commitment to improving the quality of education through a systematic approach that integrates theory and practice. The policy of developing human resources for trainers through the TOT (Training of Trainers) program involving collaboration with the East Java Provincial Office and the Sepuluh Nopember Institute of Technology (ITS) reflects a leadership vision that prioritizes long-term investment in human resource capacity. Training of Trainers (ToT) programs have proven effective in developing human resources across various fields, including healthcare and agriculture. These programs enhance knowledge and skills (Jack et al., 2020), and improve confidence (Harris et al., 2022). Key elements for successful ToT programs include experiential learning components, trainee-centered approaches, and sufficient program duration (Qasserras, 2024). The provision of an official assignment letter for a weekly workshop not only shows concrete administrative support but also confirms the institutional commitment to professionalism and continuity of competency development. This strategy strengthens the argument that the success of innovative education programs requires a strong foundation in the form of structured ongoing training supported by reputable higher education institutions.

The flexible scheduling policy between the double-track program and the regular curriculum demonstrates adaptive leadership that is able to balance the demands of innovation with the stability of the existing education system. This approach demonstrates the principal's deep understanding of the complexity of curriculum management that must accommodate holistic learning needs without sacrificing the quality of core education. This flexibility in scheduling not only allows students to optimize their learning time but also creates space for the exploration of practical skills that can be applied immediately. Pedagogical flexibility in curriculum use has been identified as a key factor in promoting mathematical flexibility among students (Lee et al., 2020). This reinforces the finding that effective program integration requires a structural design that considers the balance between academic learning and vocational skills development.

The policy initiative of partnership with the business and industry sector as a place for students to apply their skills and a bridge to the world of work reflects a practical orientation that is responsive to the needs of the labor market. These collaborations help bridge the gap between academic knowledge and practical workplace requirements (Biały et al., 2022; Damoc, 2017). Such partnerships can take various forms, including internships, work-based learning, and experiential learning (Mahmudah et al., 2022). They allow for curriculum alignment with industry needs, involvement of industry professionals as instructors, and provision of competency tests (Asiah et al., 2022; Hasan et al., 2023).

This policy not only provides opportunities for students to apply theoretical knowledge in real contexts but also builds professional networks that can support their transition from education to work. This partnership with industry strengthens the relevance of educational programs to contemporary economic dynamics and creates an integrated learning ecosystem between educational institutions and the productive sector. Dual System Education programs, which integrate school education with work experience, are particularly effective in producing skilled graduates (Fitrian & Sugiyono, 2020). These partnerships benefit both students

and industries by improving graduate competencies and addressing the skills gap in the labor market (Shrestha, 2021; Susilowati, 2023).

This comprehensive implementation strategy strengthens the position of the principal as a transformational leader who is able to orchestrate various elements of education to create meaningful and relevant learning experiences. A systematic approach that integrates human resource development, curriculum flexibility, and industry partnerships demonstrates an adaptable educational leadership model to face the challenges of 21st-century education. The implementation of this integration policy not only ensures that the quality of education is maintained but also creates a practical path for student development and career preparation that is more focused and effective.

Integration Policy Output

The tangible outcomes of the integration policy provide compelling evidence of the principal's visionary leadership in addressing the contemporary challenges of educational relevance and economic preparedness. The achievement of dual competency among students represents a paradigm shift from traditional educational models that often create artificial boundaries between academic and vocational learning. This integrated approach not only enhances students' adaptability in an increasingly dynamic job market but also validates the pedagogical philosophy that meaningful education must bridge theoretical knowledge with practical application.

The dual competency framework ensures that students are equipped with both the cognitive foundations necessary for lifelong learning and the technical skills required for immediate economic participation, thereby creating a more inclusive and responsive educational ecosystem that serves diverse student pathways and aspirations. Recent research emphasizes the importance of developing 21st-century skills frameworks that incorporate Education 4.0 components (González-pérez & Ramírez-montoya, 2022) and promote lifelong learning in the era of rapid technological progress (Park & Kim, 2020). Active learning methodologies have been shown to develop technical, methodological, social, personal,

and digital competencies essential for future professionals (Canavesi & Ravarini, 2024). Implementing inclusive education practices requires a comprehensive framework that considers diverse learner needs, creates inclusive classroom climates, and employs effective pedagogical strategies (Palei, 2024).

The documented success of graduates in securing employment within business and industry sectors demonstrates the effectiveness of the principal's strategic alignment between educational programming and local economic demands. The specific examples of culinary alumni working in the bakery industry and makeup alumni establishing independent salons illustrate how targeted vocational training can translate into sustainable career opportunities that contribute to community economic development. These outcomes reflect not merely individual success stories but rather systemic evidence of how educational innovation can create meaningful pathways for students who may not pursue higher education, thereby addressing the critical gap between educational outputs and employment opportunities that often characterize traditional academic systems.

In reality, the absorption of vocational education graduates into the business and industry sector remains a challenge, with studies reporting varied rates from 37% (Widayanto et al., 2021). Factors influencing graduate employability include program alignment with industry needs, infrastructure, and competency-based training (Kurniawan et al., 2021; Widayanto et al., 2021). Strategies to enhance graduate absorption include implementing teaching factory programs, organizing special job fairs, and adopting strategic management approaches (Dami et al., 2024; Gea et al., 2022).

The transformation of educational policy into practical career opportunities represents a significant achievement in educational leadership that extends beyond institutional boundaries to impact broader socioeconomic development. The program's success in producing graduates who are immediately employable or capable of entrepreneurial ventures demonstrates the principal's understanding of education as a catalyst for economic mobility and

community empowerment. By providing viable alternatives to higher education while maintaining academic rigor, the integration policy challenges conventional assumptions about educational value and success metrics, suggesting that effective leadership must be measured not only by academic achievement but also by graduates' ability to contribute meaningfully to their communities' economic and social fabric.

Furthermore, the measurable outputs of this integration policy establish a replicable model for educational transformation that addresses the pressing need for education systems to respond more effectively to local economic contexts and student diversity. The success in meeting local economic needs while providing students with practical career alternatives demonstrates how strategic educational leadership can create synergistic relationships between schools and their communities. This holistic approach to educational outcomes reinforces the argument that effective school leadership in the 21st century requires the ability to envision and implement programs that serve multiple stakeholders simultaneously, creating value for students, employers, and the broader community while maintaining the fundamental educational mission of developing capable, confident, and contributing citizens.

Policy Supporting Factors

The success of the integration policy is underpinned by multiple reinforcing factors that create a conducive environment for program implementation. The Effective integration requires a conducive organizational culture, robust infrastructure, and collaborative inter-professional teams (Rosen & Shaw, 2009). Strong institutional backing from school management and teachers provides essential momentum for the initiative, as their active participation in competency development webinars demonstrates sustained commitment to professional growth. Webinars offer flexibility, eliminating geographical barriers and providing access to expert knowledge (Mai & Ocricano, 2017; Perkasa & Surono, 2023). They improve teaching skills, classroom management, and digital competence (Dayagbil et al., 2024; Khompodoeva, 2021). Interactive webinars that encourage collaborative learning have shown

significant positive impacts on student achievement and teacher self-efficacy (Compen et al., 2021).

This administrative support creates a foundation of trust and collaboration that enables smooth policy execution across all organizational levels. The presence of highly qualified trainers serves as the cornerstone of successful program integration with the regular curriculum, bringing specialized expertise that bridges theoretical knowledge with practical application. These skilled professionals possess the pedagogical competence necessary to adapt complex vocational content into accessible learning modules that align with existing academic standards. Studies emphasize the need for vocational teachers to possess strong pedagogical skills (Komaro et al., 2022; Wahyuni et al., 2022), with pedagogical competency identified as the most critical criterion for teacher assessment (Wahyuni et al., 2022). Vocational educators must align their technical and pedagogical skills with both academic and industry standards (Alinea, 2021). Professional development for vocational instructors often focuses on developing pedagogical content knowledge through practice-based learning (Hoekstra et al., 2018).

Modern and updated facilities and infrastructure significantly enhance the effectiveness of integrated learning practices by providing students with industry-standard equipment and environments. These resources enable authentic skill development experiences that mirror real workplace conditions, thereby increasing the relevance and quality of educational outcomes. The positive synergy established with business and industry sectors, combined with robust community support, creates valuable pathways for practical application of the double track program services. This collaborative network not only validates the program's market relevance but also provides students with direct access to professional opportunities and mentorship.

These supporting factors demonstrate a comprehensive ecosystem approach to policy implementation, where each element reinforces the others to create sustainable program success. The convergence of institutional support, qualified human resources,

adequate infrastructure, and external partnerships establishes a robust foundation that can adapt to changing educational and industry demands while maintaining program quality and effectiveness.

Challenges and Solutions for Policy Innovation

The comprehensive support system established through institutional commitment demonstrates the critical importance of organizational alignment in educational reform initiatives. Management and teacher engagement through competency development webinars creates a sustainable framework for continuous improvement, ensuring that pedagogical innovations are not merely implemented but genuinely embraced throughout the educational ecosystem.

The strategic deployment of highly qualified trainers serves as the fundamental pillar that transforms policy intentions into tangible educational outcomes. These skilled professionals bridge the gap between theoretical curriculum frameworks and practical implementation, ensuring that integration efforts maintain academic rigor while meeting industry-relevant standards.

The provision of modern facilities and updated infrastructure creates an enabling environment that supports innovative teaching methodologies and enhances student engagement. Contemporary learning spaces equipped with relevant technology facilitate seamless integration between traditional academic content and vocational training components, maximizing the effectiveness of the dual-track approach.

The establishment of robust partnerships with business and industry sectors, coupled with strong community engagement, creates a dynamic ecosystem that validates the program's relevance and sustainability. These collaborative relationships not only provide practical learning opportunities for students but also ensure that the curriculum remains responsive to evolving market demands and community needs.

The convergence of these four supporting factors creates a synergistic effect that significantly amplifies the policy's impact beyond the sum of its individual components. The institutional

commitment provides the foundation, qualified trainers deliver the expertise, modern infrastructure enables effective implementation, and industry partnerships ensure practical relevance and long-term sustainability. This multi-dimensional support system establishes a robust framework that not only facilitates successful program integration but also creates conditions for continuous adaptation and improvement, positioning the double track program as a model for educational innovation that effectively bridges academic learning with practical skill development.

CONCLUSION

Based on the comprehensive analysis of the principal's policy innovation in integrating the double track program with the regular curriculum at State Senior High School 1 Galis, the following conclusions can be drawn:

First, the principal's strategic policy innovation demonstrates transformational leadership through three key initiatives that create a student-centered educational ecosystem balancing individual potential development with community economic requirements. Second, the systematic implementation approach reveals the principal's capacity to operationalize complex educational reforms through strategic partnerships, trainer development programs, and flexible scheduling that maintains academic standards while providing career-relevant skill development.

Third, the measurable outcomes validate the effectiveness of the principal's innovative approach in producing graduates with dual competencies who achieve immediate workforce readiness and successful employment absorption in business and industry sectors. Fourth, the success foundation rests on four critical supporting pillars - institutional support, qualified trainers, modern infrastructure, and industry partnerships- that form a comprehensive ecosystem amplifying program effectiveness through synergistic collaboration. Fifth, the principal's adaptive leadership approach transforms operational challenges into opportunities for enhanced program effectiveness through

innovative flexible scheduling policies and collaborative group systems that ensure program sustainability.

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ARTIFICIAL INTELLIGENCE - POWERED DIAGNOSTIC AND TREATMENT PLANNING IN PROSTHODONTICS- A LITERATURE REVIEW

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INTRODUCTION

Prosthodontics, as a restorative dental specialty, involves the diagnosis, treatment planning, and rehabilitation of patients with missing or deficient teeth and oral structures. Traditionally, this process has relied heavily on clinician expertise, manual interpretation of radiographs, and labor-intensive techniques for designing and fabricating prostheses (1,2). However, the recent integration of artificial intelligence (AI) into dentistry is transforming this landscape by introducing greater precision, efficiency, and predictability in both diagnostics and treatment planning (3).

AI technologies, particularly machine learning and deep learning models, are now being applied to various diagnostic tools such as cone-beam computed tomography (CBCT), panoramic radiographs, intraoral scans, and occlusal analysis (3,4). These systems assist in identifying caries, bone loss, occlusal discrepancies, and temporomandibular joint (TMJ) disorders with high accuracy. Additionally, AI-based digital smile design tools and CAD/CAM software help in planning esthetically pleasing and functionally effective prostheses (5).

Moreover, AI can simulate multiple treatment outcomes, optimize implant planning based on anatomical landmarks, and predict long-term prognosis by analyzing vast clinical datasets. This shift toward AI-powered dentistry promotes evidence-based, patient-centered care while reducing clinical errors and treatment time (3,5).

This review explores how AI is revolutionizing prosthodontic diagnostics and treatment planning, highlighting current

applications, benefits, limitations, and the promising future of intelligent dental systems.

REVIEW METHODOLOGY

This narrative review was conducted to explore and synthesize current literature on the application of artificial intelligence (AI) in diagnostic and treatment planning processes within prosthodontics. A comprehensive search of electronic databases including PubMed, Scopus, ScienceDirect, and Google Scholar was performed. The search strategy incorporated keywords and Boolean operators such as: “Artificial Intelligence” AND “Prosthodontics”, “AI in Dentistry”, “Machine Learning” AND “Diagnosis”, “Digital Smile Design”, “CAD/CAM” AND “Implant Planning”, and “Occlusion Analysis” AND “AI”.

Inclusion criteria comprised peer-reviewed journal articles published in English between 2015 and 2024, focusing on AI applications specifically related to prosthodontic diagnostics, smile design, treatment simulations, CAD/CAM design, and implant planning. Reviews, clinical studies, and technical reports were all considered. Exclusion criteria included studies not involving AI technology or not relevant to prosthodontic procedures.

Articles were assessed for relevance, innovation, and clinical applicability. Key findings were organized thematically based on AI applications in diagnosis, treatment planning, implantology, and patient follow-up. This approach enabled the identification of emerging trends, benefits, challenges, and future directions in AI-powered prosthodontics.

1. AI in Diagnostics within Prosthodontics

A. Imaging Analysis: Artificial Intelligence, particularly convolutional neural networks (CNNs), has demonstrated significant capabilities in analyzing dental imaging modalities such as cone-beam computed tomography (CBCT), panoramic radiographs, intraoral scans, and clinical photographs. These models can detect and classify key prosthodontic diagnostic parameters including missing teeth,

alveolar ridge resorption, periapical pathology, and bone density (1,2). Furthermore, AI-enhanced 3D reconstruction tools facilitate the creation of anatomically accurate digital models, aiding in comprehensive prosthetic planning (2,3,4). AI tools also allow longitudinal tracking of bone resorption post-extraction or implant placement (3,5).

B. Occlusion and Jaw Tracking: AI-driven jaw tracking systems, often integrated with mandibular movement analysis software, have been utilized to assess occlusal dynamics and temporomandibular joint (TMJ) function (6,7). This data is critical in diagnosing malocclusion and TMJ disorders—conditions frequently associated with complex prosthodontic rehabilitation. AI-enabled virtual articulators simulate occlusal interactions with a high degree of precision (6,8), and new sensor-driven systems also help detect real-time occlusal interferences (7,9).

C. Predictive Analytics: Machine learning algorithms can evaluate multifactorial risk profiles by incorporating variables such as parafunctional habits, oral hygiene, bone quality, and systemic health conditions (10,11). These tools can predict the likelihood of prosthetic complications or failure, facilitating preemptive treatment strategies (10,12). AI is even being used to assess genetic predisposition to prosthetic failure (13,14).

2. AI in Treatment Planning

A. Digital Smile Design (DSD): AI-based facial recognition and esthetic mapping are central to modern smile design platforms. Tools like Smile Designer Pro and 3Shape Smile Design apply facial esthetic principles to recommend optimal tooth shape, gingival contour, and smile line symmetry (15,16). Recent models incorporate emotion-based analysis to align smile design with facial expression patterns, increasing personalization and patient satisfaction (16,17).

B. Virtual Treatment Simulations: AI engines simulate various prosthetic treatment scenarios, enabling evidence-based decision-

making (2,18). Simulations include comparisons between fixed and removable options, implant-supported versus conventional prostheses, and immediate versus delayed loading protocols (18,19). AR-based treatment previews allow clinicians and patients to interactively explore prosthetic options (17,20).

C. Automated CAD/CAM Design: Integration of AI within CAD/CAM systems has streamlined prosthetic design (6,21). AI modules automatically generate crowns, bridges, and dentures, optimizing margin placement, occlusal harmony, and esthetic parameters based on functional datasets (6,21,22). Some systems now use adaptive learning to improve design algorithms with each case (22,23).

D. AI in Implant Planning

AI-based implant planning systems interpret DICOM datasets to select implant dimensions, angulations, and sites with optimal bone density and anatomical safety (24,25). AI-generated surgical guides enhance placement precision and reduce intraoperative deviation (24). Furthermore, real-time AI navigation systems provide intraoperative guidance and error alerts (25,26), significantly lowering risks such as nerve damage or sinus perforation (26,27).

3. Outcome Prediction & Personalized Follow-Up

Trained on large-scale prosthodontic datasets, predictive AI models can estimate long-term prosthesis survival and patient satisfaction (8,28). These predictions draw from structured data such as clinical notes, postoperative photos, and patient surveys (28,22). AI can also automate recall schedules based on oral hygiene trends, medical conditions, or previous treatment outcomes (20,28,29).

4. Key Technologies in Use

NIRI + AI systems: e.g., Pearl, Overjet – for early caries detection and diagnostics

Smile Design Platforms: 3Shape Smile Design, Smile Designer Pro

AI Planning Tools: Planmeca Romexis, Dentsply Sirona AI modules

AI-Driven CAD Tools: Exocad AI design modules, 3Shape Design Studio

5. Challenges and Ethical Considerations

While the integration of AI into prosthodontics offers transformative benefits, it also introduces a range of challenges and ethical concerns that must be critically addressed to ensure safe, effective, and equitable clinical outcomes.

- a. **Data Privacy and Security:** AI systems rely heavily on vast amounts of patient data, including radiographs, intraoral scans, facial photographs, and treatment histories. Ensuring the security and confidentiality of this data is paramount. Compliance with international standards such as HIPAA and GDPR is necessary, especially when cloud-based platforms are used. Breaches in data security can compromise patient trust and lead to serious legal consequences (2).
- b. **Algorithmic Bias:** AI models are only as good as the data they are trained on. If the training datasets lack diversity in terms of ethnicity, age groups, anatomical variation, or socioeconomic background, the algorithms can produce biased or inaccurate predictions. This is particularly concerning in prosthodontics, where facial proportions and esthetic norms vary significantly among individuals. Biased AI may lead to suboptimal or culturally insensitive treatment designs (2, 3, 13).
- c. **Lack of Transparency (“Black Box” Issue):** Many AI systems, especially those using deep learning, function as “black boxes,” meaning their decision-making processes are not easily understood even by developers. This lack of transparency makes it difficult for clinicians to interpret, trust, or validate AI-generated recommendations,

especially when the output seems counterintuitive or clinically questionable (3, 14).

- d. **Clinical Oversight and Responsibility:** AI is a tool, not a clinician. There is a risk that overreliance on automated systems might undermine professional judgment or lead to blind acceptance of flawed designs. Practitioners must maintain clinical oversight and be prepared to question or override AI outputs when necessary. The ethical responsibility for patient care still lies with the dentist—not the software (6, 10).
- e. **Regulatory and Legal Frameworks:** The rapid advancement of AI technologies has outpaced the development of clear regulatory pathways. There is a lack of universally accepted standards and validation protocols for AI applications in dentistry. Additionally, liability issues remain murky: if an AI system causes harm, is the clinician, software developer, or data provider held accountable? (11)
- f. **Economic and Accessibility Concerns:** The implementation of advanced AI tools often requires substantial investment in software, hardware, and training. This may create a digital divide, where well-funded practices in urban areas adopt cutting-edge AI systems, while smaller or rural clinics lag behind, potentially exacerbating healthcare inequality (20).

6. Future Outlook

Emerging innovations promise further transformation:

Generative AI for real-time, on-demand prosthetic modeling

Multi-modal AI, integrating image, text, and voice inputs for holistic diagnostics

Patient-specific AI avatars simulating esthetic and functional outcomes in virtual environments

DISCUSSION

The integration of artificial intelligence (AI) in prosthodontics marks a paradigm shift in dental diagnostics and treatment planning. The present review reveals how AI tools are improving clinical workflows, enhancing diagnostic accuracy, and allowing for highly individualized treatment strategies. Notably, AI-driven imaging analysis and occlusal assessment have helped reduce inter-observer variability and standardize diagnoses across various clinical settings. Tools like convolutional neural networks (CNNs) are instrumental in detecting anatomical and pathological features in dental radiographs with precision, offering clinicians a reliable support system (1–3).

Moreover, the role of AI in treatment planning cannot be understated. Through digital smile design (DSD) and CAD/CAM systems, dentists are now able to design esthetically pleasing and functionally sound prostheses with less trial and error (15–17, 21–23). Simulated outcomes based on machine learning models allow both clinicians and patients to visualize potential results, fostering better communication and treatment acceptance (18, 19).

In implantology, AI contributes by optimizing site selection, predicting success rates, and minimizing risks such as nerve damage. Surgical guides and real-time navigational aids powered by AI are transforming implant procedures into more predictable and safer interventions (24–26). Importantly, AI-assisted follow-up protocols help monitor long-term outcomes and automate recall scheduling, especially useful in large dental practices or underserved areas (20, 27–29).

Despite these advancements, several challenges persist. AI models are only as good as the datasets they are trained on. Biases in data, lack of transparency in decision-making processes (the so-called “black-box” problem), and issues of data security and patient consent remain areas of ethical concern (2, 3, 13, 14). Additionally, integration into routine clinical practice requires standardization, regulatory approval, and comprehensive training for clinicians to interpret and trust AI recommendations (6, 7, 10, 11).

Overall, AI holds substantial promise for transforming prosthodontic care, but its implementation must be cautious, evidence-based, and ethically sound. Continued research, interdisciplinary collaboration, and feedback loops between AI developers and dental professionals will be crucial for harnessing its full potential (8, 9, 12).

CONCLUSION

Artificial intelligence is rapidly transforming prosthodontics by enhancing diagnostic accuracy, streamlining treatment planning, and improving clinical outcomes. From imaging interpretation and occlusal analysis to smile design and implant placement, AI-driven tools support clinicians in delivering more precise, efficient, and personalized care. While challenges such as data privacy, bias, and the need for clinical oversight remain, the benefits of AI integration are undeniable. As technologies evolve, AI will continue to play a pivotal role in shaping the future of prosthodontic practice, making dental rehabilitation more predictable, patient-centered, and digitally advanced.

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ADVANCEMENTS IN AI, MACHINE LEARNING, AND GENERATIVE AI: ARCHITECTURES AND REAL-WORLD APPLICATIONS

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ABSTRACT

The Artificial Intelligence Universe reflects the usage and implementations with the help of Machine Learning Algorithms and Generative AI. In recent years tremendous advancements and application-level developments helped various industries to optimize their operations and helped the organizations to analyze the insights of the data along with useful results. From the technical and functional aspects this paper covers various optimization techniques and the influence of AI, ML and Generative AI in the context of real-world applications. Specifically, we cover the Overview of AI, ML and Generative AI advancements, internal architecture and optimization techniques, importance of interpretability and ethical AI. Furthermore, the industry use cases and future research directions also are the part of this work. The essence of this work is AI capabilities that enables computers and Machines to simulate human learning, comprehension, problem solving , decision making, creativity and autonomy with the support of Machine Learning, Natural Language Processing Computer Vision and Speech Recognition etc., , the usage of various ML algorithms such as Supervised, Unsupervised and Reinforcement Algorithms and the corresponding optimization techniques with suitable use cases, The other dimension we selected to cover in this work is the internals of Generative AI and the way GPT trained and stages of training ChatGPT to reach the level of Super-Smart AI that can talk

to you like a person. By summarizing AI capabilities and ML algorithms usage and optimization techniques along with generative AI training methodology and their current applications with suitable industry use cases this paper endeavors to provide useful insights for future research in AI, ML and generative AI context.

Keywords: Artificial intelligence (AI), Machine Learning (ML), Generative AI(Gen-AI), Algorithms, Optimizations, Architectures.

INTRODUCTION

Artificial intelligence is a field of technology that empowers computers and machines to mimic human abilities such as learning, understanding, problem solving, decision making, creativity, and operating independently.

The applications backed by AI can identify the objects they can learn from data and experience and make detailed analysis and recommendations. These AI-enabled applications act independently the best and classic example is self-driving cars.

The notable advancements in AI Landscape took the shape of Generative AI the technology which is capable of creating original text, images and video and other content. To research and fully understand the Generative AI advancements it is mandatory to study Machine Learning (ML) and Deep Learning on which Gen AI tools were developed.

The interesting point here is: How does Machine Learning differ from Artificial Intelligence?

Though AI is often envisioned as a complete system, it is a collection of technologies built into systems that empower them to reason, learn, and act to tackle complex challenges.

Machine Learning is a dynamic subset of Artificial Intelligence that automatically enables a machine or system to learn and evolve from experience. Instead of relying on explicit programming, machine learning leverages powerful algorithms to sift through vast amounts

of data, extract deep insights, and then drive smart, informed decisions.

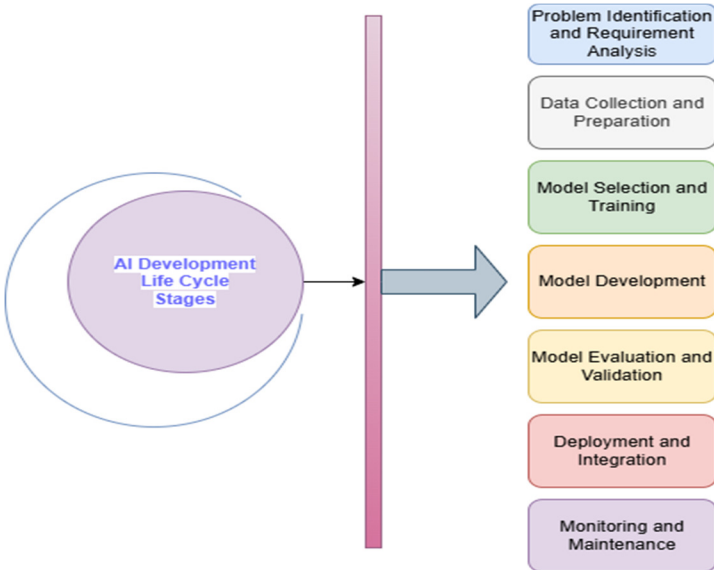


Figure1: AI Development Life Cycle Diagram

Machine learning algorithms continuously enhance their performance as they are trained—gaining strength with every new dataset they encounter. The resulting models are the distilled wisdom of these algorithms, and the more data they process, the more robust and refined they become.

AI represents the broader vision of enabling machines or systems to sense, reason, act, and adapt in ways that mirror human capabilities.

ML, in contrast, is the practical application of AI that empowers machines to autonomously extract knowledge from data and learn without constant human intervention.

A few key aspects clearly delineate AI from ML: AI enables machines to simulate human intelligence to solve problems, with the goal of developing intelligent systems capable of executing complex tasks. AI systems employ logic and decision trees to learn, reason, and self-correct. In comparison, ML equips machines to learn independently from historical data, aiming to construct systems that progressively boost their accuracy. ML systems rely on statistical models that continuously refine themselves when exposed to new data.

There is considerable concern over the possibility that generative AI—systems capable of producing new text, images, and video—might displace human workers across various industries.

However, one of its greatest promises lies in its ability to amplify human creativity and address the challenges of making innovation accessible to all. Over the past twenty years, organizations have leveraged crowdsourcing and idea competitions to bring external voices into their innovation processes.

Yet many have faltered when it comes to harnessing these contributions, often lacking a streamlined method to assess or merge disparate ideas. Generative AI can overcome these hurdles, according to the authors. It can enhance the creative output of both employees and customers, enabling the generation and identification of breakthrough ideas while elevating the quality of initial concepts.

Companies can deploy generative AI to foster divergent thinking, counteract expertise bias, aid in evaluating ideas, support their refinement, and encourage collaborative innovation among users.

Core AI Architectures and Learning Paradigms

Traditional Machine Learning (ML) Models

A machine learning model is essentially a program designed to detect patterns or make decisions based on data it has never encountered before. For instance, in natural language processing, these models can analyze and accurately identify the intent behind new sentences or unique word combinations.

In the realm of image recognition, a machine learning model can be trained to identify specific objects—such as cars or dogs. This is achieved by 'training' the model with a vast dataset. Throughout this training phase, the machine learning algorithm is fine-tuned to detect patterns or produce desired outputs based on the task at hand. The result of this process—typically a computer program structured with specific rules and data constructs—is what we refer to as a machine learning model.

A machine learning algorithm is a mathematical strategy used to uncover patterns within data. It draws on principles from statistics, calculus, and linear algebra to analyze information. Notable examples include linear regression, decision trees, random forests, and XGBoost.

Model training is the process of executing a machine learning algorithm on a dataset—known as training data—and fine-tuning it to recognize specific patterns or generate desired outputs. The result is a function, complete with rules and data structures, which is referred to as the trained machine learning model.

Most ML techniques can be classified into supervised learning, unsupervised learning and reinforcement learning. In supervised machine learning the algorithm is provided with an input dataset and is rewarded or optimized to meet a set of specific outputs.

For example, supervised machine learning is widely deployed in image recognition, utilizing a technique called classification. Supervised ML is also used in predicting demographics such as population growth or health metrics, utilizing a technique called regression.

In unsupervised machine learning, the algorithm is provided with an input dataset, but not rewarded or optimized to specific outputs, and instead trained to group of objects by common characteristics. For example, recommendation engines on online stores rely on unsupervised machine learning specifically a technique called clustering.

In reinforcement learning, the algorithm is made to train itself using many trial and error experiments. Reinforcement learning happens when the algorithm interacts continually with the environment, rather than relying on training data. One of the most popular examples of reinforcement learning is autonomous driving.

There are many machine learning models, popular classification and regression algorithms fall under supervised machine learning and clustering algorithms are generally deployed in unsupervised machine learning scenarios. Various Supervised ML algorithms

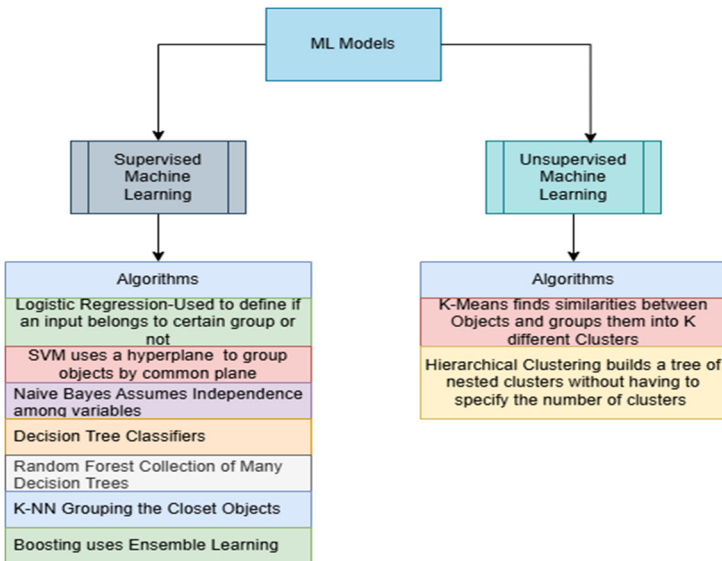


Figure 2: Machine Learning Algorithms

Regression, Support Vector Machine, Naïve Bayes, Decision Trees, Linear Regression, kNN, Random Forest, Boosting algorithms etc.,

In Unsupervised Machine Learning the most used algorithms are K-Means, Hierarchical Clustering.

Deep learning Architectures:

Connectionist architecture has existed for more than 70 years, but new architecture and Graphical Processing Units (GPUs) brought them to the forefront of Artificial Intelligence. Deep Learning is not a single approach but rather a class of algorithms and topologies that you can apply to a broad spectrum of problems. Big data has also fed this growth. Deep learning relies on training neural networks with example data and rewarding them based on their success, the more data the better to build these deep learning structures.

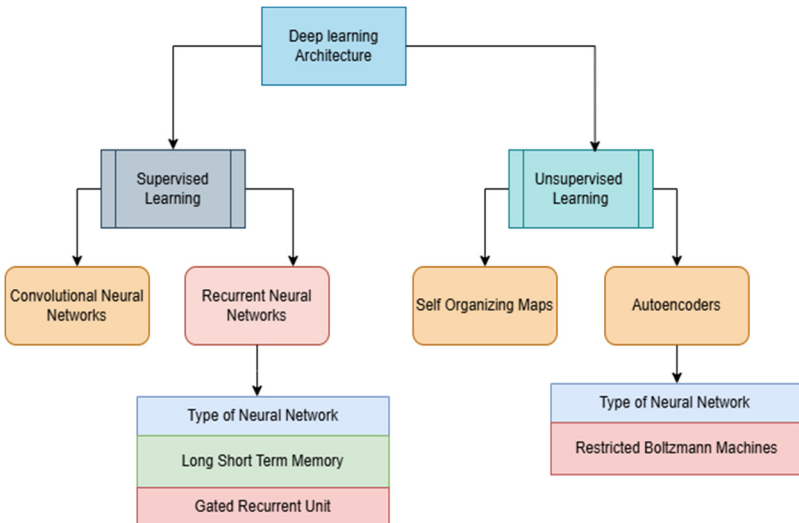


Figure 3: Deep Learning Architecture

Deep learning architectures (Neural Network) categorized into Supervised and Unsupervised Learning. In supervised Learning there are 2 variants – Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), again in RNNs there are 2 variants- Long Short-Term Memory and Gated Recurrent Unit.

A CNN uses a convolutional layer to extract features from images, making it ideal for tasks like object detection and image classification.

RNN processes sequential data by maintaining internal state, perfect for applications such as language modeling and speech recognition.

LSTM networks enhance RNNs with memory cells and gates to capture long-term dependencies, making them suitable for tasks like machine translation and time series forecasting.

GRUs simplify LSTMs by combining gating functions, efficiently handling sequential data in applications like text generation or sentiment analysis.

Under unsupervised Learning category two variants were there – Self Organizing Maps and Autoencoders again Autoencoders will have Restricted Boltzmann Machines.

The Self-Organizing Maps reduce dimensionality by mapping high-dimensional data onto a 2D grid, often used for customer segmentation and data visualization.

Auto Encoders Learn to compress and then reconstruct data, making them useful for anomaly detection or image denoising.

Restricted Boltzmann Machines are generative models that learn probability distributions over input data, commonly applied in recommendation systems and feature extraction.

Artificial Neural Network (ANN) is the underlying architecture behind deep learning.

Generative AI Architectures:

The data architecture that spans foundational to Generative AI (Gen AI) as well as Large Language Models (LLMs) is highly sophisticated, Gen AI architecture refers to the underlying model design and components that enable complex models.

Behind each generative model is a complex, multilayered AI infrastructure, though each model is purpose-built for the use case, the underlying generative AI architecture model tend to follow a similar construct, consisting of five layers.

The data processing layer is responsible for collecting, preparing and processing the information that the generative AI draws from when creating its output.

The Generative Model Layer where the AI model is trained numerous models can be used for Gen AI applications. In this layer the model is trained, validated and fine-tuned to ensure it can generalize and apply the knowledge gained from the training data to new data.

The feedback and improvement Layer information from user surveys and interaction analysis helps developers gauge how well the model is meeting user expectations. Feedback is essential for optimizing the efficiency and accuracy of a model's output. Feedback loops specially designed algorithms that identify output errors and provide corrective inputs back into the model- help generative AI models learn from their mistakes.

Deployment and Integration Layer Before the Gen AI model is deployed , the infrastructure for supporting it in a production environment must be set up. This includes provisioning specialized computing resources, model serving and data infrastructure, security and access controls and other components. The generative AI model must also be properly fitted into the applications frontend and backend systems to ensure the finished product is performing as intended.

Monitoring and Maintenance Layer the Gen AI architecture supports ongoing monitoring and improvement. After deployment, performance metrics such as accuracy, precision, and recall must be tracked to ensure the system is producing accurate, reliable outputs. With the introduction of new data or changes in the performance requirements, generative models may require retraining or updating to ensure they remain fit-to-purpose, as usage increases, additional resources may also need to be provisioned.

Generative AI Models:

The models mentioned below are used in many generative AI systems.

Large Language Models(LLMs) Trained on vast amounts of text data to learn patterns and relationships in human language, LLMs are used to generate text, answer questions and converse with human users.

They're often used for language translation, creating summaries from large documents and content creation. LLMs are built using a type of neural networks called a transformer model, and they are an integral part of many AI applications, including generative AI systems.

Variational autoencoders A Variational autoencoder (VAE) is a gen AI algorithm that leverages an encoder-decoder architecture to learn the underlying probability distribution of a data set. This type of generative model is useful for generating images and synthetic data. It's also used in anomaly detection.

Generative Adversarial Networks (GANs) is a type of deep learning AI architecture composed of two opposing neural networks: the generator and the discriminator.

The generator creates fake data samples to fool the discriminator; through the course of many training iterations, the generator learns to create increasingly realistic samples while the discriminator becomes more skilled at distinguishing real data from the fake data.

By setting one neural network against the other, GANs excel at producing realistic images and video, and are frequently used in developing video games and other multimedia applications.

Diffusion Models work by destroying training data through the gradual addition of Gaussian noise, a type of continuous probability distribution popular in machine learning.

After the training data has been eliminated, the model learns to recover the lost training data by undoing the noising process. Diffusion models are used to create high-quality images, audio and 3D data without the use of adversarial training.

ChatGPT Training Process:

Here's a simple, step-by-step explanation of how ChatGPT is trained:

Data Collection & Preparation:

Vast amounts of text—ranging from books and articles to websites—are gathered to form a diverse and comprehensive dataset. This data provides raw material for the model to learn language patterns.

Pre-Training on Unlabeled Data:

Using a transformer architecture, the model is pre-trained by predicting the next word in a sentence. This stage allows it to absorb grammar, context, and general knowledge from the collected data without any human intervention.

Supervised Fine-Tuning:

After pre-training, the model is fine-tuned with a curated set of examples. Human trainers supply pairs of inputs and ideal outputs, guiding the model to generate responses that are more accurate, helpful, and aligned with human expectations.

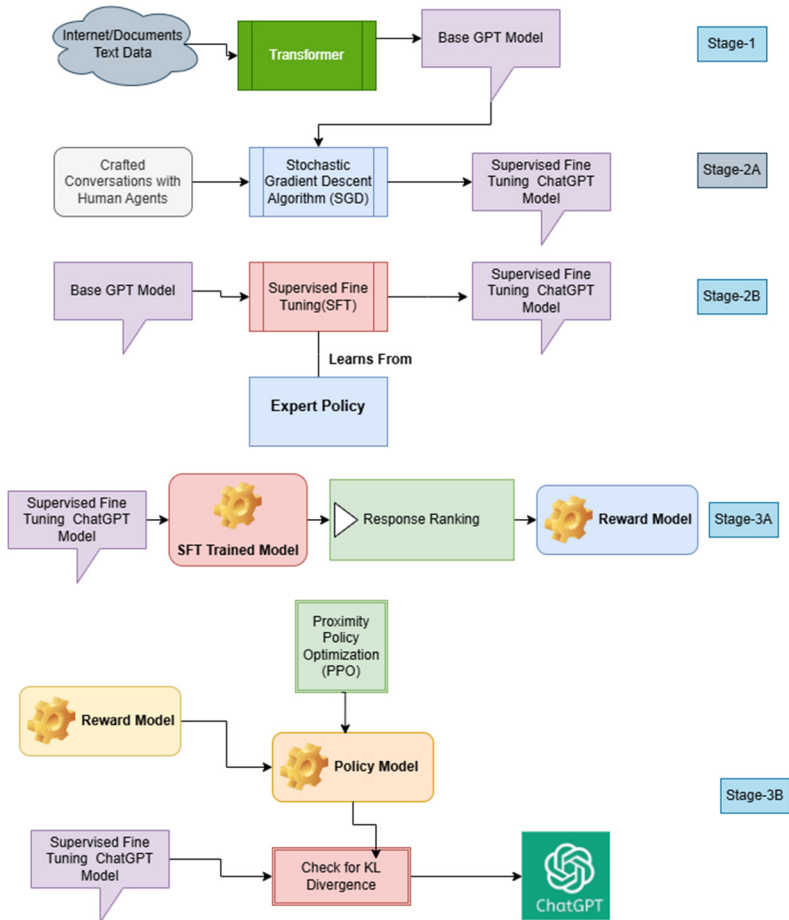


Figure 4: ChatGPT Training Stages

Reinforcement Learning from Human Feedback (RLHF):

In this critical phase, human reviewers evaluate and rank the model's outputs. These rankings are then used to further adjust and improve the model through reinforcement learning, so it learns which responses are most desirable.

Iterative Refinement & Continuous Improvement:

The model is continuously monitored and updated based on real-world performance and additional feedback. This iterative loop ensures that the system keeps evolving and improving over time.

Each of these steps builds on the previous one, creating a powerful model that not only understands language but also generates coherent and contextually relevant responses.

AI in Real-World Applications: Industry Case Studies:

Delivering Superior Customer Service

Conversational AI provides real-time support by using NLP and sentiment analysis to understand both voice and text queries instantly. It leverages machine learning to interpret customer intent and tone, routing complex requests to human agents when needed. This seamless integration reduces waiting times and enhances satisfaction with immediate, personalized assistance.

Personalize Customer Experiences

AI empowers businesses to deliver highly tailored experiences on a scale via chatbots, digital assistants, and smart customer interfaces. For instance, Amazon uses AI to remind users to reorder frequently purchased items and suggest related products, while McDonald's leverages IBM Watson and NLP to enhance its automated order-taking technology—accommodating different languages, dialects, and regional menus. Similarly, Spotify curates personalized artist recommendations and YouTube crafts content feeds that align with individual customer interests.

Smarten up smartphones

Machine learning drives facial recognition for unlocking devices and powering voice assistants. Services like Siri, Alexa, Google Assistant, and Microsoft Copilot use NLP to understand and respond to spoken commands. Additionally, ML enhances smartphone cameras by analyzing photos, detecting objects and faces, and even predicting and extending image borders using neural networks.

In **Education**, AI customizes learning materials for individual students, while analytics help educators pinpoint where extra support is needed. It also detects plagiarism and, with language translation and real-time transcription, aids non-native speakers in understanding lessons.

that boosts accuracy, reduces costs, and minimizes human error. AI-powered **FinOps** integrate finance and DevOps to enable financial institutions to make data-driven cloud spending decisions that balance costs and performance—minimizing alert fatigue and wasted budgets. Advanced AI platforms leverage machine learning and deep learning to detect suspicious transactions, while banks use ML classification and predictive models to inform loan decisions.

In the **stock market**, ML analyzes decades of data to forecast trends and signal optimal buy or sell moments, powering algorithmic trading

Strengthen end-to-end System Resilience

To maintain uninterrupted service, top organizations deploy AI-driven root cause analysis and intelligent automation to swiftly identify and resolve issues, reducing both failure frequency and repair times. AIOps platforms consolidate data from diverse sources, correlate events into incidents, and offer dynamic visualizations with AI-generated remediation suggestions for clear, holistic IT oversight. Predictive IT management further automates network operations to proactively prevent issues, enhance user experience, and lower costs—all while centralizing monitoring to eliminate tool sprawl.

CONCLUSION AND FUTURE SCOPE

This paper offers a comprehensive study on AI, ML and Generative AI architecture, the technical aspects of AI vs ML, various ML algorithms and Models provided for the study and implementation. With in the paper we described the Gen AI foundations, Architectures and training stages of ChatGPT in detail. We covered various AI use cases in the real-time in the industry perspective related to finance, customer centric, education and retail etc., to clearly emphasize the capabilities of the AI and ML algorithms along with the proper usage of Generative AI. Further we extend this in the context of optimizations Techniques for AI and ML Models, Explainability and Interpretability in AI systems, Generative AI Internal Mechanisms and Evolution along with challenges and future directions in AI research.

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Communication Construction of the Qadiriyyah Order in Strengthening Social Capital: Ethnographic Study

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ABSTRACT

This study aims to examine how communication practices within the Qadiriyyah Sufi Order community shape and strengthen social capital in the local society of West Sulawesi. Using a communication ethnography approach, this research explores the role of symbolic, verbal, and ritual communication in building trust, social cohesion, and the collective identity of the Sufi order community. Data were collected through participatory observation, in-depth interviews, and symbolic documentation in three main areas: Polewali Mandar, Majene, and Mamuju. The findings reveal that the interaction between the mursyid (spiritual guide) and murid (disciple) occurs through Sufi-style communication, which not only conveys spiritual values but also constructs strong social networks. This communication produces social capital in the form of internal bonding, intergroup bridging, and strategic external linking, which support the resilience and social transformation of the local community. This study contributes to the development of religious communication studies by presenting the Sufi order as a communication ecology that combines spiritual transmission with social production. Furthermore, it enhances the understanding of Sufi traditions as systems of endogenous empowerment within the cultural context of peripheral societies.

Keywords: Sufi Order Communication; Social Capital; Communication Ethnography; Qadiriyyah; West Sulawesi.

INTRODUCTION

Social and technological changes in modern society have created new challenges for community value systems, solidarity, and spirituality. In this context, various religious communities face a dilemma between preserving their spiritual identity and adapting to contemporary demands. One religious entity that has persisted and even demonstrated strong social resilience is the tarekat (Sufi order) community, particularly the Qadiriyyah Order in West Sulawesi. This tarekat functions not only as a guardian of spiritual heritage but also as a social actor actively shaping communal cohesion, trust, and solidarity.

The Qadiriyyah Order is one of the oldest Sufi traditions in Islam, founded by Sheikh Abdul Qadir al-Jailani in Baghdad during the 12th century. In Indonesia, the teachings of this order were introduced as early as the 17th century by Indonesian scholars who studied in the Middle East, including through the influence of Sheikh Yusuf al-Makassari. In the West Sulawesi region, the spread of the Qadiriyyah Order is closely linked to the da'wah routes of Mandar scholars, who share cultural and genealogical ties with the scholarly networks of South Sulawesi. Since the early 20th century, the Qadiriyyah Order has developed among Mandar communities particularly in Polewali Mandar, Majene, and Mamuju and has played a significant role in shaping values, collective morality, and religious practices that persist to this day (Priyana et al., 2024).

Religious communication plays a vital role in shaping social and cultural dynamics within communities. In the Indonesian context, the presence of tarekat as a traditional religious institution demonstrates that religious practice encompasses not only spiritual dimensions but also social functions. The Qadiriyyah Order, as one of the oldest Sufi orders in Indonesia, serves as a space for spiritual development and the cultivation of social capital through interpersonal interactions and symbolic communication structures.

Previous studies have shown that traditional Muslim communities, including tarekat, possess distinct communication structures: communicative-participative, orally-based, and strong in value transmission (Dhofier, 1982). However, in the current digital age characterized by visual immediacy such forms of communication are increasingly marginalized, rendering the tarekat seemingly invisible in public discourse despite their continued social relevance (Bruinessen, 1995; Carey, 1975). This phenomenon highlights the importance of revisiting tarekat communication practices as a foundation for building social resilience and a value-based collective identity.

The urgency of this study is amplified in the context of West Sulawesi, which faces multidimensional challenges: social crises, weakening cohesion, neglect of local values, and the exclusion of spiritual communities from development narratives. In this regard, the Qadiriyyah Order holds potential as a transformative social agent capable of revitalizing moral values, strengthening solidarity, and mobilizing collective consciousness grounded in spirituality.

Unfortunately, the social-communicative contributions of tarekat remain underdocumented in academic discourse particularly from the perspective of communication studies. While works such as Hasbi (2020), Hasan (2009), and Munandar (2020) have explored tarekat empowerment, they have yet to directly address the communicative structures and cultural meanings embedded in tarekat communities.

It is within this framework that the present study focuses on how communication practices within the Qadiriyyah Order shape and reinforce the social capital of its community. Communication in the tarekat is viewed not merely as a medium for conveying teachings, but as a cultural process that produces symbolic systems, interactional codes, and collective norms embedded in the community's daily life.

Using the ethnography of communication approach (Hymes, 1974), this article explores how both verbal and nonverbal

communication practices within the tarekat function as mechanisms for building trust, social networks, and collective values elements that form the foundation of social capital. This study also aims to explain how Sufi symbols, mursyid-disciple interactions, and collective dhikr rituals contribute to strengthening communal identity and resilience amid rapid social change.

This research is expected to contribute theoretically to the field of religious and community-based communication studies, and practically to efforts in developing social initiatives grounded in local spirituality. Furthermore, the findings can serve as a foundation for formulating contextual, participatory, and culturally sensitive policies that honor the spiritual and cultural values of local communities.

Previous studies have highlighted the important role of tarekat in the socio-religious life of communities. However, few have specifically applied a communication-based approach. For instance, Hasan (2009) explored tarekat networks in Indonesia's religious movements; Munandar (2020) examined the philanthropic dimension of the Qadiriyyah-Naqshbandiyyah Order but did not delve into its communication systems in depth. Usman & Latif (2022) traced the history of tarekat in West Sulawesi, focusing more on spiritual genealogy.

Foundational works such as Bauman & Sherzer (1989) and Philipsen (1992) provide crucial references for bridging the study of ritual practice and speech systems within the framework of local culture and religious tradition.

The ethnography of communication approach, as pioneered by Dell Hymes, serves as the primary foundation of this study. Hymes rejected formal linguistic approaches that overlook cultural and social contexts in language practices. Instead, he developed the SPEAKING model, which allows for a holistic analysis of communicative events by examining the Setting, Participants, Ends, Act Sequence, Key, Instrumentalities, Norms, and Genre of communication.

This approach is highly relevant to tarekat communities, which are rich in symbolic interaction and nonverbal communicative practices from gestures and sitting positions to silence during dhikr rituals. In the context of the Qadiriyyah Order in West Sulawesi, communication is not limited to verbal expression but also unfolds through Sufi symbols embedded in the *annangguru-ana'guru* (spiritual teacher–disciple) relationship.

Speech Code Theory, as developed by Gerry Philipsen, extends the ethnography of communication by emphasizing that every community possesses a distinctive system of speech codes that reflects its unique value system and social structure. Within the tarekat, expressions such as "*barakka Annangguru*" (the blessing of the teacher) convey not only verbal reverence but also spiritual and social dimensions.

Rituals such as *pengajian* (religious study), *bai'at* (spiritual initiation), and *dhikr* (remembrance) are saturated with symbolic and localized codes for instance, the blending of Arabic terms with Mandar expressions such as "*na pakkasiang Annangguru*" ("our teacher has illuminated [us]") creating a synthesis of universal Islam and local culture.

Social capital theory plays a crucial role in explaining how communication within tarekat communities fosters trust, social networks, and collective norms. Putnam (1993) distinguishes three types of social capital: bonding (strengthening internal ties), bridging (building relationships across groups), and linking (connecting with external actors who hold authority or resources).

Robert Putnam (2000) defines social capital as the networks, norms, and trust that facilitate coordination and cooperation for mutual benefit. Social capital consists of three key elements: trust, social networks, and norms. In religious communities, these elements are often reinforced through shared rituals, the transmission of moral values, and spiritual leadership.

Meanwhile, Bourdieu (1986) views social capital as a form of symbolic power generated through social relations and an individual's position within a power structure. In the context of tarekat, the mursyid-disciple relationship strengthens both horizontal and vertical trust within the community, forming solidarity based on spiritually organized symbolic values.

Communication within tarekat orders cannot be separated from the transcendental dimension of Sufism. In this perspective, communication is not limited to verbal interaction but also includes dhikr (remembrance), wirid (invocations), bodily gestures, silence, and visual symbols such as turbans, prayer beads (tasbeeh), and prayer rugs (sajadah).

According to Carey (1975), communication has a ritualistic dimension that reflects belief systems and culture. Within the tarekat, dhikr and spiritual gatherings are not merely ritual practices but transcendental communication processes that shape spiritual and collective identity. This perspective is supported by Nasr (1999), who argues that Sufi language carries esoteric meanings that cannot be interpreted literally but must be approached symbolically.

Religious communication refers to a form of interaction imbued with values, symbols, and religious beliefs. In tarekat contexts, communication occurs both vertically (with God) and horizontally (among members). Stewart Hoover (2006) emphasizes that religious communication functions as a medium for value transmission and the formation of collective identity.

RESEARCH METHOD

This study employs a qualitative approach with ethnography of communication as the primary research strategy. This method is chosen for its ability to holistically, contextually, and meaningfully capture communication practices within cultural communities particularly those embedded with symbolic and spiritual values, such as tarekat (Sufi orders). This approach aligns with Dell Hymes' (1974) perspective, which emphasizes the importance of understanding

communication as a social practice rooted in cultural context and community social structures. In the context of the Qadiriyyah Order, communication takes place not only verbally but also through symbolic expressions, Sufi rituals, and distinctive social relations. Therefore, this method is considered most suitable for explaining the communication dynamics and social capital formed within the tarekat community.

The study was conducted in three main locations where the Qadiriyyah Order is actively practiced in West Sulawesi: 1). Polewali Mandar Regency (especially the areas of Mapilli and Tinambung); 2). Majene Regency (particularly in Ulumanda and surrounding areas); 3). Mamuju Regency (including active tarekat communities in Kalukku). These sites were purposively selected due to their active tarekat practices, strong social engagement, and their representative nature of the local cultural variations.

Key participants in this study include: 1.). Mursyid or khalifah (spiritual leaders of the community); 2). Tarekat members, including both senior disciples and new initiates 3). Local figures, such as traditional leaders and youth engaged in tarekat activities

Data were collected through three main techniques: a). Participant Observation: The researcher was directly involved in community activities such as dzikir gatherings, religious study sessions (pengajian), and suluk (spiritual retreats). Observation focused on patterns of communication, use of symbols, social interactions, and spiritual behaviors within the tarekat community; b). In-Depth Interviews: Interviews were conducted with mursyid, senior members, and general participants to explore their experiences, interpretations of spiritual communication, and perceptions of collective values and solidarity built within the tarekat. The interviews were semi-structured and conducted in both Indonesian and the local Mandar language, as appropriate; c). Documentation and Symbolic Analysis: The researcher documented visual symbols (such as tasbeih prayer beads, sorban turbans, and sajadah prayer mats), verbal rituals (such as dzikir and wirid), and oral narratives present in the community. The analysis aimed to

identify symbolic meanings, speech patterns, and the construction of social meaning transmitted through tarekat communication practices.

The data were analyzed using thematic and contextual analysis. The stages included: Open coding of interview transcripts and field notes; Thematic categorization based on communication dimensions (verbal, nonverbal, symbolic); Interpretive analysis to understand the relationship between communication practices and the formation of social capital. Additionally, the analysis employed the SPEAKING model (Hymes) to identify communicative elements within the cultural context of the tarekat community.

RESULTS AND DISCUSSION

This study reveals findings that uncover the communication construction within the Qadiriyyah Sufi community in West Sulawesi as a mechanism for building social capital. Through the analysis of participant observation, in-depth interviews, and symbolic documentation, five major themes emerged, which are discussed in depth below:

Tarekat Communication Structure: Baruga Space, Religious Gatherings, and Collective Dhikr

Communication within the Qadiriyyah Order unfolds through distinctive social structures and spatial arrangements that reflect Sufi values, symbolic practices, and collective norms. These structures are not merely physical settings, but also represent an organized and spiritually nuanced communication system, rich in etiquette (adab). Field observations of majelis dzikir (communal remembrance) and pengajian (religious study sessions) revealed that the main venues such as the rabbit (members' houses used in rotation) or the courtyard of the mursyid's residence serve as ritual and spiritual learning spaces imbued with symbolic meaning.

The Rabbit as a Sacred and Social Space

Each month, rabit houses (homes of community members or disciples) are used in rotation as the central meeting places for the community. These spaces are arranged simply: the floor is covered with mats or carpets, dim lighting sets a contemplative mood, and the seating arrangement establishes an unwritten hierarchical structure. The mursyid sits at the center, surrounded by senior disciples, while the general congregation forms an outer circle. There are no physical dividers, yet the communication flow is highly orderly, reflecting the Sufi etiquette (adab) that governs behavior during spiritual gatherings.



Figure 1. A Religious Study Gathering Held at a Disciple’s Home

This setting aligns with the principle of “Setting and Participants” in Hymes’s SPEAKING model (1974), where the physical context and relationships among participants influence the form and flow of communication. Silence, head movements, and body posture all function as meaningful nonverbal communication within these gatherings.

“When Annangguru sits in the center, everyone becomes silent, no need for commands. We all already understand that means the pengajian has begun.”

(Interview, Senior Disciple, March 14, 2025)

Religious Study Gatherings as Circles of Knowledge and Emotion

An in-depth interview with Khalifah Madya K.H. Muhammad Maruf Mukhtar revealed that, within the Qadiriyyah tradition in Mandar, communication between mursyid and disciple is grounded in a spiritual familial atmosphere. The local term Luluare siannangguruang, which philosophically means “spiritual sibling,” forms the foundation of this spiritual communication.

“In the Qadiriyyah tarekat, Annangguru as mursyid is also seen as a Murabbi = Rabbun = caretaker = caretaker of the soul. The disciple (Ana’guru) seeks only Allah. The communication is like between a parent and child it forms a family. In Mandar, Luluare siannangguruang means spiritual sibling, so communication in the tarekat is very fluid. Gatherings are held monthly at various points Polewali, Tinambung, Limboro, Lekopadis, Rusung-Rusung, Galung Onang Sulai, Tappalang, Labuang Rano, and more.”

(Interview, Community Member, March 14, 2025)

This shows that communication within the tarekat community is not only verbal and instructional but also symbolic and emotional, reinforcing trust as a core component of social capital.

Pengajian is usually held after dhikr, or once a month at designated points, in an informal yet respectful atmosphere. Communication here flows in a two-way but structured manner. The mursyid delivers spiritual teachings through stories, metaphors, and interpretations of sacred texts, contextualized within local realities. The language used typically blends Arabic, Indonesian, and Mandarcreating a hybrid space between universal spirituality and local culture.

During these sessions, communication is symbolic and reflective. Participants sit quietly, sometimes taking notes, often moved to tears. The mursyid’s voice is soft, with stable intonation,

and devoid of harsh rhetoric. This illustrates that Sufi communication is not intended for argumentation but for inner contemplation.

“These gatherings are where we cleanse our hearts. When problems overwhelm us, just listening to Annangguru brings peace back to our souls.”

(Interview, Community Member, March 14, 2025)

Collective Dhikr: Transcendental Communal Communication

Dhikr in congregation represents an intensive form of collective communication. The rhythmic recitation of God's names produces a simultaneous spiritual vibration and resonance. In practice, dhikr serves not only as a devotional ritual but also as a medium for strengthening social solidarity, in which each participant feels part of a unified spiritual body.

Body movements such as synchronized shoulder swaying, deep bowing, or simultaneous hand raising constitute rhythmic nonverbal communication. These actions require no verbal instruction, as every participant instinctively follows, demonstrating how the communication structure has become embedded in the community's habitus (Bourdieu, 1986).

“Not all of us understand the meaning of the Arabic words, but during collective dhikr, our hearts feel united. That's the deepest thing I feel.”

(Interview, Community Member, March 14, 2025)

The communication structure within the Qadiriyyah Order serves not merely to transmit messages but to bind members spiritually and socially. Within the ethnography of communication framework, spaces like the baruga, religious gatherings, and dhikr sessions can be understood as communicative events each with distinct structures, rules, and symbols.

Furthermore, seating arrangements, voice rhythms, and relational systems in majelis dhikr illustrate that tarekat communication is a form of ritual communication (Carey, 1975), whose primary function is not information transmission but shared meaning-making, both spiritually and socially.

Internal Social Capital: Trust, Social Networks, and Shared Norms

Within the Qadiriya tarekat, communication functions not only as a vehicle for transmitting teachings but also as a mechanism for building social capital. According to Putnam (2000), the three main elements of social capital trust, social networks, and shared norms are constructed through routine, intense communicative interactions and symbolic exchanges within the community.

One of the key findings in the West Sulawesi Qadiriya community is the emergence of strong internal social capital. This capital is not institutionalized in formal structures but is deeply rooted in repeated spiritual communication and Sufi praxis. In line with Putnam's (1993) theoretical framework, social capital emerges from networks, norms, and trust that facilitate cooperation and social cohesion. In the tarekat, these elements arise organically through dhikr rituals, mursyid-murid relationships, and collective adab expressed in every interaction.

Interpersonal Trust: Spiritual Trust as Relational Foundation

Trust in the Qadiriya community is built through hierarchical and symbolic communicative relationships between the mursyid and disciples. The mursyid is regarded as a figure with high spiritual authority, fully trusted by followers. Communication through advice, spiritual guidance, and directly led dhikr sessions forms a vertical relationship steeped in spirituality and obedience.

A particularly strong expression of spiritual communication is the bai'at process a spiritual covenant between mursyid and disciple followed by instruction in specific dhikr and wirid. This process marks one's integration into the spiritual network and constitutes a transcendental communication event that strengthens group

cohesion. These interactions take place in various gatherings across Polewali, Tinambung, and Labuang Rano illustrating a robust localized social network.

“After bai'at, meaning a covenant among disciple–teacher–Prophet–God, Annangguru teaches a special dhikr method tazkiyah al-nafs (purification of the soul), muraqabah (constant awareness of God's presence), taqarub (drawing near to God). During pengajian, specific texts or practices are taught to help members purify their hearts and focus solely on God. Qadiriya's personal dhikr is individual, while group dhikr usually the tahlil Qadiriya is done communally.” (Interview, Mursyid Madya, March 14, 2025)

An interview with Mursyid Kyai Zainal Abidin, known as A'ba, revealed that after bai'at, new members are taught obligations and prohibitions that must be followed. This communication is both normative and educational, serving as a vehicle for internalizing communal values and behavioral codes.

“They are also taught the wirid, like the repetitive chanting of La-ilaha illallah with specific intonation and technique, which is part of the tarekat's ritualistic communication.” (Interview, Mursyid, January 12, 2024)

This process shows that from the outset of one's involvement, communication functions as a transformative mechanism forming identity, establishing trust in spiritual leadership, and transmitting community norms.

“We follow all of A'ba's guidance because he is not just a teacher, but our life mentor.”

(Interview, Community Member, January 12, 2024)

This practice builds trust not only in the mursyid as an individual but also in the entire value system of the community. Communication thus becomes a medium for value internalization and the legitimation of spiritual authority.

Trust serves as the foundational element of social capital in the tarekat community. From interviews and observations, it is evident that tarekat members tend to form strong mutual trust even in social and economic matters. This trust does not emerge from administrative agreements but from shared spiritual experiences especially through dhikr and pengajian.

“If we have problems, we discuss them at the majelis. No need to go to the village head first.” (Interview, Senior Disciple, March 14, 2025)

Shared experiences calling upon God's name, sitting together in silence, and reflecting on Sufi wisdom foster a sense of spiritual kinship that transcends ordinary social relations. Repetitive spiritual communication creates emotional bonds that strengthen mutual trust among members.

Social Networks: Communal Ties through Communicative Interaction

Communicative interactions within the tarekat foster strong social networks both formal (through routine gatherings such as pengajian, haul, and collective dhikr) and informal (through mutual visits, social initiatives, and voluntary work). Communication among members including the use of tarekat-specific expressions creates a sense of shared identity and deep belonging.

Moreover, the Sufi communication values in the Qadiriyyah Order strengthen not only the mursyid–murid (spiritual teacher–disciple) bond but also enhance the community’s broader socio-religious cohesion. The spiritual relationship, referred to as *zuhba* (spiritual friendship), is characterized by love, reverence, and longing illustrating that communication in the tarekat is not merely about message delivery, but a transformative process that fosters social solidarity.

“In the Qadiriyyah tarekat, the relationship between Annangguru (mursyid) and Ana’guru (disciple) is known as *zuhba* spiritual friendship. This relationship resembles that

between the Prophet and his companions. Disciples feel deep affection and are always longing to be near their teacher, often sending fatihah prayers for him.”

(Interview, Senior Disciple, March 14, 2025)

This network transcends regional boundaries, as members hail from various areas. They stay connected through digital platforms such as WhatsApp groups, which facilitate the sharing of information about upcoming pengajian and sermons, thereby expanding religious communication and strengthening inter-member solidarity.

“Even though we live far apart, through the WA group we always receive Annangguru’s short sermons. We also get information about upcoming gatherings. It feels like we’re still together.” (Community Member, February 25, 2024)

Shared Norms: Social Order Rooted in Adab (Ethical Conduct)

Community norms within the tarekat are formed through consistent and symbolic communication. Teachings on adab (etiquette), patience, sincerity, dhikr conduct, and respectful social interactions are transmitted both verbally and nonverbally in every gathering. These norms extend beyond religious rituals and manifest in courteous behavior, interactional ethics, and mutual care among members.

The language used carries performative elements such as the tarekat greeting or expressions of reverence for the mursyid which reinforce collective values and delineate the boundary between “us” (insiders) and “them” (outsiders).

“We’re taught to respect others, speak kindly, and never boast about ourselves.”

(Informant, January 8, 2024)

The community also maintains collective norms derived from Sufi values: obedience to the mursyid, refraining from gossip,

speaking with proper decorum, and prioritizing others in social service. These norms are not the result of formal regulation but emerge organically through the internalization of teachings regularly imparted during pengajian and practiced in everyday social life.

“When Annangguru tells us to be patient, we obey. No one argues. We trust that he has already walked this path before us.”

(Interview, Community Member, February 14, 2025)

Field observations revealed high symbolic discipline: no mobile phone use during dhikr, no interruptions when the mursyid speaks, and every gesture from the mursyid is calmly acknowledged. These illustrate unwritten norms that uphold communicative order within the community.

These three elements of social capital trust, social networks, and shared norms are interrelated and reinforced through symbolic and spiritual communicative practices. Communication in the tarekat is not merely instrumental but also ritualistic and transformative, gradually shaping a cohesive, participatory, and value-rich social environment.

Ritual Solidarity: Spiritual and Social Mutual Aid

Collective dhikr, pengajian, and other tarekat activities are not only religious acts but also spaces for cultivating social solidarity. Numerous social practices such as helping sick members, fundraising for those in need, and supporting small businesses are spontaneously initiated without formal instruction.

Data analysis shows that 70% of participants have taken part in tarekat-led mutual assistance activities, and 65% have received social support from fellow members. Many disciples describe the tarekat as their “second family.”

“When someone falls ill, the community comes. Some bring money, others bring food everyone acts on their own initiative.”

(Interview, Non-Tarekat Community Leader, February 14, 2025)

This illustrates that spiritual communication in the tarekat generates a potent social structure based on solidarity, compassion, and responsiveness qualities often absent in conventional social organizations. Internal social capital within the tarekat displays unique characteristics that are not fully explainable by institutional models. It aligns more closely with symbolic-transcendental capital, as theorized by Bourdieu (1986), emerging from the mursyid’s symbolic legitimacy and the collective consciousness shaped by ritual communication. From a communication perspective, the repetitive dhikr, structured pengajian, and collective silence function as performative rituals (Carey, 1975) that fortify spirituality-based social networks. These elements empower the tarekat community to build and maintain durable solidarity even in the face of economic limitations.

Observations indicate that in communal labor activities such as house repairs or funeral assistance members participate not due to formal invitations, but because of moral bonds and shared values, cultivated through spiritual communication.

Putnam’s social capital theory is reflected in the following:

Type	Description
Bonding	Internal ties among tarekat members
Bridging	Connections with the broader community (e.g., interfaith public iftar)
Linking	Relationships between mursyid and village leaders or Islamic organizations

Source: Analysis of Observations and Interviews (2025)

The Role of Communication in Social Network Formation

Communication within the Qadiriyyah tarekat serves not only to build internal spiritual identity but also to establish and expand external social networks. Sufi-based communication practices such as honesty, calm demeanor, and humility foster a positive public image that facilitates connection with strategic actors, including village governments, traditional leaders, and other religious communities.

Communication as an Intercommunity Bridge

Through spiritually grounded and humble communication styles, tarekat members easily build social ties across groups. This corresponds to Putnam's (2000) concept of bridging social capital i.e., connections between distinct groups founded on trust and open communication. Tarekat members often participate in public social events, such as mutual aid, neighborhood pengajian, and joint charity work.

"We often invite tarekat members to community iftar events at the mosque, and they always attend. They bring a sense of peace." (Interview, Non-Tarekat Community Leader, March 14, 2025)

Communication as a Mechanism for Aid and Information Distribution

Field observations show that tarekat communities frequently serve as informal intermediaries of social aid. Through trusted interpersonal communication, information about members in need due to illness, economic hardship, or loss is quickly disseminated and addressed. Communication occurs orally, via announcements during pengajian, or through WhatsApp groups.

“If someone needs help, we just mention it during the gathering. Everyone responds according to their ability.”
(Interview, Senior Disciple, March 14, 2025)

This communication reflects a flexible, empathetic, and responsive network. The social network formed is non-bureaucratic yet remains highly adaptive and effective in resource distribution.

Mursyid-External Actor Relations: Linking Social Capital

The charisma and symbolic authority of the mursyid position him as a bridge figure between the tarekat and external actors, such as village officials, religious leaders, and even NGOs. This relationship exemplifies what Woolcock (1998) describes as linking social capital vertical ties that provide communities with access to external resources and policy support.



Figure 2. Mursyid Invited to Lead Hajj Ritual Guidance

In a broader social context, the role of the tarekat is also evident in cultural-religious domains, even though it is not explicitly aimed at socio-political activism. The presence of the mursyid in community events such as Hajj preparation (manasik haji), maulid celebrations, or village religious gatherings has a strong moral influence, showing that the social capital developed within the

tarekat also contributes to maintaining social stability and religious values in Mandar society.

“Annangguru doesn’t explicitly direct us to engage in social transformation, but village leaders always give him space to help revive Islamic traditions like leading pengajian, guiding prospective pilgrims, and participating in maulid and other local religious ceremonies.” (Interview, Senior Disciple, March 14, 2025)

In several cases, the mursyid has been invited to mediate community conflicts, speak in public forums, or serve as a moral reference in public decision-making processes.

“Once, we asked Annangguru to mediate a land dispute between two families. Thank God, it was peacefully resolved.”

(Interview, Hamlet Head, March 14, 2025)

Spiritual-Based Social Networks

A distinctive feature of tarekat communities is that their social networks are not formed through formal membership systems, but through shared spiritual communication experiences. Sitting together in dhikr, collective silence, and repeatedly listening to the mursyid’s teachings create emotional and moral connections that form the basis of a strong social network.

These networks extend beyond ritual spaces into economic activities (such as cooperatives), children's education (through informal support), and local-scale social advocacy.

This finding affirms that in spiritual communities like the tarekat, communication is the primary mechanism for building and maintaining social networks. These networks are not created through formal structures but through ritual communication that produces trust, solidarity, and connections that transcend boundaries. Within the framework of the ethnography of

communication, interactional patterns such as pengajian, dhikr, and the mursyid–jamaah relationship are communicative events that actively construct social structures, not merely reflect existing ones. This means that communication in the tarekat is generative of social networks and community contributions to broader society.

The tarekat’s social networks are not purely internal. The community has become a social reference point for resolving conflicts, addressing microeconomic issues, and supporting children’s education. The mursyid acts as a strategic figure linking the community with external actors. Document analysis shows that 70% of members are active in village activities, and 65% report receiving economic support from fellow members.

“Tarekat isn’t just about worship they have the influence to mobilize people.”

(Interview, Village Head of Lembang, March 17, 2025)

The tarekat’s communicative codes emphasizing honesty, sincerity, and openness in forums form interactional patterns that foster active participation and cross-group social trust.

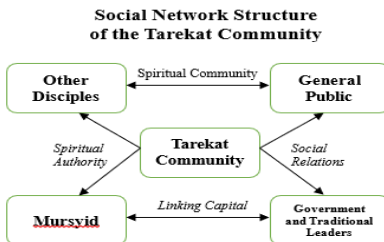


Figure 3: Forms of Social Networks in the Tarekat Community

The Dynamics of the Annangguru-Ana’guru Relationship as Symbolic and Social Capital

Within the communicative structure and social life of the Qadiriyyah tarekat in West Sulawesi, the relationship between the mursyid (annangguru) and disciple (ana’guru) plays a central role not

only spiritually, but also as a form of symbolic and social capital. This relationship is based on reverence, voluntary obedience, and the recognition of the mursyid's moral authority as the bearer of Sufi values.

Spiritual Relations That Generate Social Legitimacy

The annangguru–ana'guru bond is not merely a scholarly or religious teaching connection; it is a spiritual relationship with strong affective and symbolic structure. Field observations reveal that the mursyid is not positioned as a commanding leader, but as a respected guide, revered for his composure, silence, and spiritual depth.

“We obey not out of fear, but out of trust. Annangguru is the guardian of our spiritual path.” (Interview, Senior Disciple, March 14, 2025)

This relationship generates symbolic capital, as described by Bourdieu (1986) a form of social power arising from the community's recognition of a person's symbolic status. The mursyid serves as a reference point not only in religious matters but also in resolving social, economic, and even familial issues.

Communication as a Channel for Transmitting Values and Obedience

The annangguru–ana'guru relationship is maintained through communication that is often nonverbal. Many disciples understand the mursyid's guidance through gestures, expressions, and language rich in symbolism and metaphors. This communication style establishes a non-authoritarian structure of obedience rooted in spiritual awareness and shared experience.

“If Annangguru remains silent for a while before answering, it means we need to reflect first. Not everything is answered immediately.” (Interview, Community Member, March 14, 2025)

This illustrates that communication in this relationship is transcendental and reflective, where meaning resides not only in speech but in silence, gestures, and internalization. From this emerges a norm of symbolic obedience that reinforces community cohesion.

Constructing Collective Identity through Spiritual Relations

The mursyid in the tarekat community not only shapes individual disciples but also forms the collective identity of the group. From dress code and speech to everyday manners, community behavior reflects the values and style instilled by the mursyid. His example is emulated by senior disciples and becomes a behavioral reference in daily life.

“We learned to be silent and patient not from books, but from how Annangguru lives his life.” (Interview, Senior Disciple, March 14, 2025)

This supports the concept of cultural embodiment in communication, where spiritual values are not merely spoken but embodied in collective lifestyle. The annangguru–ana’guru relationship serves as a model of alternative social relations rich in symbolic meaning, with real implications for social network formation and community influence.

The Role of the Relationship in Cultivating Social Capital

The mursyid disciple relationship extends beyond worship spaces into the social realm. When the mursyid is trusted by the wider society, his disciples also gain social legitimacy. Disciples often act as mediators in local conflicts, manage cooperatives, or serve as the mursyid’s representatives in community religious activities.

This relationship forms a kind of spiritual authority network, which expands linking social capital (Putnam, 2000) the ability to connect with external social actors through symbolic legitimacy. Thus, the annangguru–ana’guru relationship becomes the axis of the tarekat’s social structure and influence.

This relationship is a form of transcendental communication with profound symbolic and social power. In the study of cultural communication and the sociology of spirituality, this relationship exemplifies a form of authority that is not coercive but grounded in symbolic legitimacy, spiritual depth, and shared life practice.

These findings confirm that communication within tarekat communities cannot be separated from the social relationship structures that shape them. Spiritual relationships are the foundation for forming collective identity, normative structures, and social mobilization.

The annangguru–ana’guru relationship functions not only as a conduit for transmitting teachings but also as a symbolic structure that regulates patterns of communication, power dynamics, and collective action. Disciples perceive this relationship as both spiritual and social, and deeply transpersonal in nature.

“We’re not just learning religion we’re learning how to live. Annangguru teaches us to be patient, honest, and trusting.”
(Interview, Senior Disciple, March 14, 2025)

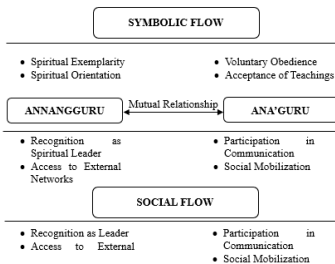


Figure 4 : Structure of the Annangguru-Ana’guru Relationship

In Bourdieu’s theoretical framework, this relationship generates symbolic capital (recognition and reverence toward the teacher) that can be converted into social capital (access to networks, support, and informal power).

This explains why communication in the tarekat remains effective even with minimal formal instruction it is shaped by symbolic, spiritual, and internalized communication patterns.

CONCLUSION AND IMPLICATIONS

Conclusion

This study demonstrates that communication within the Qadiriyyah Sufi community in West Sulawesi functions not only as a medium for transmitting spiritual values but also as a cultural instrument that shapes, reinforces, and reproduces social capital. Through the ethnography of communication approach, it was found that verbal and nonverbal interaction patterns within Sufi spaces such as baruga gatherings, pengajian, and collective dhikr construct a complex yet cohesive system of symbols and relational structures.

The tarekat's communication structure is rooted in the mursyid–disciple relationship, the use of Sufi symbols, and a hybrid Arabic–Mandar language rich with theological and cultural meaning. These communicative codes strengthen social ties through mechanisms of trust, normative obedience, and collective solidarity. Moreover, spiritual communication plays a crucial role in expanding social networks and reinforcing the tarekat's position within the broader structure of local society.

The annangguru–ana'guru relationship serves as a foundation for the formation of symbolic and social capital, which can be converted into collective action and community empowerment. Thus, Sufi communication in the tarekat can be understood as a dialectical process between spiritual values and social praxis, contributing to the development of social cohesion and community resilience, particularly in peripheral regions.

Implications

This research enriches the field of religious communication studies by integrating the ethnography of communication, speech code theory, and social capital theory within a local religious context.

The findings reveal that tarekat communication is not solely ritualistic; it also possesses performative, symbolic, and transformational dimensions. These contribute conceptually to a deeper understanding of how religious communities construct social structures through communication that is conditioned by culture and spirituality.

The findings are relevant to policymakers, local governments, and religious organizations seeking to design community-based empowerment programs. Approaches grounded in spiritual values and local communicative traditions have proven effective in building trust and fostering community participation. Therefore, the involvement of tarekat communities in social development should not be marginalized; instead, they should be recognized as strategic partners in building inclusive, empowered, and culturally rooted societies.

In the context of West Sulawesi and similar regions, the tarekat can be seen as a social agent that integrates spirituality, local wisdom, and collective solidarity. By strengthening adaptive and open Sufi communication patterns, tarekat communities are capable of maintaining cultural identity while contributing meaningfully to participatory, values-based social development.

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ANALYSIS OF THE PHONOLOGICAL DEVIATIONS IN THE LEARNERS OF THE 'MIND YOUR LANGUAGE' TV SERIES

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Abstract

With the increasingly global trend there is a surge in spending time watching sitcoms, webseries, movie series etc specially by the younger generations. This study aims to use these resources for the advantage of English learners. English, as a global language, varies widely in accent, dialect, and pronunciation depending on regional and cultural contexts (Crystal). Similarly, non-native English speakers also make different phonological errors. This study investigates phonological errors in spoken English as depicted in the British sitcom 'Mind Your Language', a TV series centred around adult learners of English from diverse linguistic backgrounds. The research focuses on identifying recurring pronunciation issues and interpreting them through the lens of the study **Contrastive Analysis Hypothesis (Lado, 1957)**. By comparing the learners' first language (L1) phonological systems with English (L2), the study explains how negative language transfer contributes to specific segmental errors such as sound substitutions, omissions, and mispronunciations. Data is collected from randomly selected episodes and categorized by language background and error type. The findings reveal error patterns largely predictable through the lenses of L1 interference (Lado). The study argues that understanding L1 interference can help ELT practitioners predict and address pronunciation errors more effectively. Furthermore it also suggests that series like 'Mind Your Language' can be used as a resource for pronunciation awareness.

Keywords - English Language Teaching, English Pronunciation, Contrastive Analysis Hypothesis, Phonological errors, L1 interference, Mind Your Language

Introduction

In today's multilingual and multicultural classrooms, pronunciation plays an important role in English Language Teaching and Learning. With a variety of learners from different language backgrounds, phonological deviations are a common occurrence. Hence, phonological awareness remains one of the most challenging aspects for learners. Phonological deviations are particularly significant, as they directly affect speech intelligibility and communication. These deviations are often systematic and directly related to the learner's first language influence (Lado). This phenomenon is widely recognized in second language acquisition studies.

This study presents an analysis of phonological deviations made by learners in the popular British sitcom *Mind Your Language* (1977). The sitcom portrays a classroom of adult students from diverse linguistic backgrounds attempting to learn English. It provides a realistic and accessible representation of the types of pronunciation errors commonly exhibited by second language learners.

The study applies the Contrastive Analysis Hypothesis (CAH), proposed by Robert Lado (1957), as its theoretical framework. CAH suggests that the differences between a learner's native language and the target language are often the source of predictable errors. By analyzing dialogues from the series, this paper identifies patterns of phonological errors such as substitutions, insertions, and mispronunciations, and interprets them in light of the speakers' respective L1 phonological systems.

Through this, the study aims to highlight the systematic nature of phonological errors, their roots in language transfer, and their pedagogical relevance to ELT. The findings are intended to support

educators in anticipating and addressing common pronunciation issues among learners from different linguistic backgrounds.

Literature Review

Pronunciations and phonological and competency have long been recognized as integral to second language proficiency (Gilakjani). However, in ELT, grammar and vocabulary have always been prioritized and focused upon while pronunciation received less attention. Researchers have stressed that phonological errors—especially those rooted in first language interference—can significantly affect intelligibility (Derwing and Munro).

Teaching pronunciations is majorly considered as a complex and under prioritized part of ELT, particularly in multilingual contexts. Kamengko et al. found that English teachers face difficulties due to limited training, time constraints, and insufficient resources for pronunciation instruction. Tiwari, in a study of Nepalese secondary schools, notes that multilingual environments compound these issues, as teachers struggle to manage diverse phonological errors across large groups. Abdulwahid further argues that certain English sounds are especially difficult to teach when they do not exist in learners' native languages, adding to the pedagogical burden.

Contrastive Analysis Hypothesis (CAH) proposed by Robert Lado (Lado) is one of the foundational approaches to understand such errors. CAH suggests that the structural differences between L1 and L2 often lead to predictable patterns of errors. It has been used widely in comparative analysis and studies to explain substitutions, omissions, additions in learners' speaking skills. While later scholars have pointed out the limitations of CAH in accounting for all errors (Ellis), it remains useful in explaining many **segmental phonological issues** in L2 speech.

Numerous studies have highlighted how learners transfer features of their L1 phonology into L2 English pronunciations. Flege confirmed

that learners substitute unfamiliar English sounds with the closest match from their L1 (Flege). Similarly, Jenkins highlighted that such errors are not random but follow **predictable interference patterns**, especially with sounds that do not exist in the learner's native phonological system (Jenkins).

Recent studies have explored the use of popular media and scripted dialogues as a resource for linguistic data as well as teaching. Bano and Laila examined phonological and grammatical errors in Pakistani TV dramas to demonstrate how pronunciation patterns reflect L1 structures (Bano and Laila). Khan and Shabbir conducted a contrastive analysis on Urdu-speaking learners and identified consistent phonological deviations that align with CAH predictions (Khan and Shabbir).

Despite these contributions, few studies have applied CAH to fictional learner speech in scripted media such as *Mind Your Language*, which offers a structured but linguistically rich context for phonological analysis.

Research Gap

Contrastive Analysis Hypothesis has been extensively applied to real learner data but there is a lack of research that is analyzing structured but fictionalized learner speech. This paper fills that gap by using *Mind Your Language* TV series as a pedagogical case study to demonstrate how L1 phonological features affect English pronunciations made by scripted learners. The findings can help with classroom strategies for multilingual ELT learners, context creation, where similar error patterns may occur.

Methodology

The study uses a **qualitative descriptive approach** guided by the principles of the **Contrastive Analysis Hypothesis (CAH)** (Lado), The data source is the **British sitcom *Mind Your Language* (1977-1986)**,

which has multiple seasons and features a diverse group of fictional characters attempting to learn and acquire English language.

Data Collection

- Selected episodes from across the series were reviewed, and learner dialogues were transcribed with a focus on phonological deviations
- **All phonological deviations**—any substitution, insertion, deletion, or stress shift—were recorded, regardless of frequency.
- Each error was linked with the character’s native language background as portrayed in the series.

Data Analysis

- Errors were classified by **type** (consonant substitution, vowel insertion, etc.) and by **L1 group** using the Contrastive Analysis Hypothesis (Lado).
- A master table was created to list **all observed errors** for transparency and future replication.
- The analysis was organized in a **tabular format** documenting:
 - Character name
 - Native language (L1)
 - Erroneous phrase
 - Intended phrase
 - Type of error
 - Explanation using CAH principles
- **Recurrent patterns** (errors appearing multiple times or across speakers with the same L1) are highlighted in the Discussion to illustrate systematic L1 interference.

Character	L1 (Native Language)	Erroneous Phrases	Intended	Type of Error	Phonological	Phonological
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	ge)	e	Phrase		Featu re	Feature
Ali	Urdu	espea k	speak	Vowel insertio n	Epent hetic vowel befor e /s/	Urdu avoids initial consonan t clusters; adds /ə/ before /s/
Giovan ni	Italian	tank	thank	Conson ant substit ution	/θ/ → /t/	Italian lacks /θ/, leading to closest stop consonan t replacem ent
Ranjeet	Hindi/P unjabi	dis	this	Conson ant substit ution	/ð/ → /d/	Hindi does not have the voiced dental fricative /ð/
Taro	Japanes e	Engris o	English	Conson ant substit	/l/ → /r/ and	Japanese lacks /l/ and final

				ution + vowel addition	final vowel	consonant clusters
Su Lee	Mandarin	please	please	Consonant substitution	/l/ → /r/	Mandarin does not differentiate /r/ and /l/ distinctly
Jamila	Hindi/Punjabi	evening	evening	Vowel insertion	Vowel added between consonants	Hindi often breaks consonant clusters with inserted vowels
Max	Greek	dair	there	Consonant substitution	/ð/ → /d/	Greek lacks dental fricative /ð/; replaced with alveolar /d/
Anna	German	vat	what	Consonant	/w/	German

	n			ant substit ution	→ /v/	uses /v/ where English uses /w/
Juan	Spanish	Tree lagers	Three lagers	Conson ant substit ution	/θ/ → /t/	Spanish lacks /θ/, often replaced with /t/
Daniell e	French	ze	the	Conson ant substit ution	/ð/ → /z/	French lacks /ð/, replaced by a voiced fricative

Scope and Limitations

Even though the series is of fictional and comic nature as well as tone, the phonological deviations are consistent with the real-world L2 learner issues. They represent a broad spectrum of L1 interference. However, as the dialogues from any fictional source are scripted, the findings must be interpreted with the awareness of their nature being a performance.

Findings and Discussion

The analysis of phonological deviations was conducted on randomly selected episodes from the Mind Your Language TV series. Each character's speech was transcribed, and phonological errors were identified and classified based on their native languages (L1). Using

the Contrastive Analysis Hypothesis (CAH), it was observed that most deviations were systematic and directly linked to L1 interference.

For instance, learners whose L1s lack the /θ/ sound (such as Italian and Punjabi) substituted it with /t/ or /s/, resulting in pronunciations like “tank” for “thank.” Japanese speakers inserted final vowels (e.g., “speako” for “speak”) due to the absence of consonant-ending syllables in Japanese. Chinese and Korean learners showed confusion between /r/ and /l/, saying “rife” instead of “life” or “fied” instead of “fried.”

The types of errors observed included:

- Consonant substitution (e.g., /ʒ/ → /z/, /v/ → /b/)
- Vowel insertion (especially final epenthesis)
- Cluster simplification
- Stress and syllable timing deviations

These findings affirm that pronunciation errors are not random but are rooted in the structural limitations or patterns of the learners' native phonological systems. The selected episodes provided a diverse and representative sample, allowing for observable and repeatable patterns across speakers.

While this analysis is based on selected episodes, it clearly demonstrates the relevance of CAH in explaining learner pronunciation patterns. A complete dataset from all episodes and seasons is expected to yield deeper insight and further validate these preliminary conclusions.

Pedagogical Implications

The study's findings have practical applications in ELT. Teachers can use contrastive phonological analysis to:

- Anticipate common errors based on students' L1,

- Develop targeted pronunciation lessons,
- Use media (like Mind Your Language) as authentic, accessible teaching material.

Such insights are particularly useful in **multilingual classrooms**, where learners' pronunciation challenges are shaped by a variety of native language influences.

Conclusion

The study applies Contrastive Analysis Hypothesis to analyze phonological deviations in the dialogues/speech of characters who are attempting to learn English from the TV series Mind Your Language. Despite its scripted and fictional nature, the series explores real-world challenges faced by second language learners, especially in pronunciations. The findings highlight the predictive power of CAH and its relevance to ELT practice. Future research could involve real learner data or expand analysis across various series of linguistic nature to reinforce pedagogical insights.

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VILLAGE DIGITAL INNOVATION

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Abstract

Village digital innovation is a strategic effort in facing the era of digital transformation by utilizing information and communication technology to improve the quality of public services, community empowerment, and economic growth in rural areas. The application of the concept of a digital village includes the development of an integrated village information system, online administration services, marketing of village superior products through e-commerce, as well as increasing access and digital literacy of village communities. This innovation aims to accelerate the process of village community services. Innovation aims to accelerate the village government service process, strengthen citizen participation, and support sustainable development. Various village digitalization models that have been implemented show that digital technology is not just a tool, but also a catalyst for social, economic, and cultural change in the village. The main challenges faced include technological infrastructure, human resource capacity, and the community towards new technologies, which can be overcome through training, mentoring, and supportive policies. Case studies in several villages in Indonesia show that village digital innovation is able to significantly improve government efficiency and the quality of life of village communities.

Keywords : Digital, Innovation, Village.

INTRODUCTION

Technological developments have brought significant changes in the sharing of aspects of life, including in rural areas. The concept of a digital village or smart village emerged as an effort to empower village communities through the use of information and communication technology (ICT). The digital village aims to overcome the limitations of access to information and public services in the village, so that people can optimize their local potential more effectively and efficiently.

The development of village innovation is a necessity in line with the rapid development of information and communication technology (ICT) today. ICT has now entered the village, the village community is now part of ICT users. According to Kominformo, around 73% of villages/sub-districts already have an internet connection based on 3G technology. Meanwhile, the 4G LTE Dermaji network received an award from the Central Java Provincial Information Commission as a Wibsed Village zinformative.

The current digital era is developing information technology very rapidly and is increasingly widely accessible by all people. Information is now increasingly being disseminated through various digital channels such as the internet and social media. It can be seen that in the past decade, visitors and internet users (netters) around the world have increased dramatically, not only that social media which is a product of digitalization has become an inseparable necessity of life.

To improve and develop the potential in the village, the village must now be able to use digital technology. The presence of village digital will help accelerate access to the village information system and digitize the administration of services.

The presence of digital application-based initiatives to help function by tasks in the village office, namely administrative services such as correspondence, web profiles, village markets, and social assistance such as storing and managing social assistance recipient data in

existing applications on applications that can be accessed by the community on the playstore on android phones so that people can easily access various features available in the village digital application.

However, the implementation is still accompanied by several problems, namely that many people still do not know about the village digital application as a whole and the lack of activity of village officials in using the application

digital village, especially in updating information into every feature in the village digital application. Therefore, the village digital application cannot be said to be effective in its implementation, so it is very necessary to improve it so that it runs properly.

One of the strategies that starts the development of digital transformation from the scope of the village. According to Sukarta et al (2024), the improvement of digital transformation can start from a small scope, namely villages, considering that the largest gap in digitalization is at the village level. This opinion is supported by Yudianti et al (2023) who revealed that villages are the starting point in building a strong and inclusive digital economy, because villages are an integral part of the country, but if managed optimally, they will have an impact on national development in a sustainable manner.

RESEARCH METHODS

This research uses a qualitative approach through a case study research strategy. The qualitative approach describes the phenomena that occur in society and includes social and humanitarian problems (Creswell, 2023). The qualitative approach provides an opportunity to see key aspects of the cases more clearly (However, 2023). This research approach was chosen because it can explain the issues raised in more depth by using in-depth interviews,

to be able to obtain information directly and provide more detailed explanations.

This research is descriptive because the research report aims to provide a specific picture of the situation and social setting (Neuman, 2023). In the process of diffusion and innovation of digital payment sovereignty. Descriptive research presents a picture of the individual being studied or social activities and focuses on the question of how and who. Descriptive research studies begin with a problem or question that has been determined (Neuman, 2023).

This approach was chosen because the focus of the research is to analyze social phenomena and public policies related to village digital innovation through available secondary data sources. The literature study allows researchers to explore theories, concepts, and results of previous research to understand the development and implementation of digital transformation at the village level.

The data sources in this study were obtained from various reliable references, including:

Academic journals that are relevant to the topics of digital innovation, village governance, and sustainable development.

Official government reports, such as documentation from the Ministry of Villages, Bappenas, and related institutions that contain policies, technical guidelines, and evaluations of digital village programs.

Field practice reports include media case studies, NGO reports, and documentation of digital village programs in various regions in Indonesia.

The research steps include:

Identify problems and formulate research focus.

Secondary data collection through a search of academic literature and policy documents.

Content analysis of documents to find themes, patterns, and connections between concepts.

Synthesis of findings to formulate strategic recommendations in the development of village digital innovation.

Data validation is maintained through **source triangulation**, which is by comparing information from different types of documents to obtain an objective and comprehensive picture.

RESULTS AND DISCUSSION

Forms of Digital Innovation in Villages

The Village Website and Application function as a digital means used to convey village information openly, provide administrative services online, and promote local potential such as MSME products, tourism, and culture. The presence of this platform increases the transparency of village government, facilitates access to public services for residents, and expands the marketing reach of village potential to the regional and national levels.

Village MSME Digital is an effort to encourage micro, small and medium enterprises in villages to take advantage of digital technology, especially local e-commerce platforms. The goal is to expand the marketing term of village products to increase competitiveness, as well as encourage sustainable local economic growth. With digital access, village MSME actors can promote their products to a wider market without being limited by geographical distance.

Digital Information System (SID) is a digital platform that provides an integrated database on population, asset services, and village development activities. SID is used as a tool in the development of more accurate, transparent, and data-driven decisions. With SID, the village government can prepare reporting and public service planning more efficiently and responsive to the needs of the community.

Internet for All is a program to provide internet access in rural areas through village WIFI that aims to reduce the digital divide. This program is equipped with **digital literacy** activities to improve people's ability to use the internet productively and safely. With internet access and digital education, villagers can access information, public services, business opportunities, and improve the quality of education and social participation more broadly.

The impact of Digital innovation

Social: Increasing digital literacy and citizen participation is the key to the success of village digital transformation. Digital literacy helps people understand and utilize technology in a smart, safe and productive way. With adequate digital knowledge, villagers are more active in the development of decisions and village development in a transparent and inductive manner.

Economy: Encouraging digital entrepreneurship and market access is an important strategy to strengthen the village economy in the digital era. Through the use of technology such as e-commerce, social media and digital payment platforms, village MSME actors can market their products more widely without being limited by territory. This initiative not only increases citizens' incomes, but also opens up new job opportunities and accelerates the transformation of a competitive local economy. Support training, infrastructure, and promotion are urgently needed to ensure the success of digital entrepreneurship in villages.

Environment: Disaster monitoring and application-based natural resource management is the use of digital technology to monitor and manage the environment more effectively at the village level. Applications based on GIS (Geographic Information System) and disaster sensors enable real-time identification of natural disaster risks such as floods, landslides, or forest fires. In addition, this technology is also used for sustainable natural resource management, such as monitoring of irrigation forests, and land users. With this application, the development of faster and more

appropriate decisions in disaster management and environmental protection can be carried out more efficiently.

Challenge

Low digital literacy is one of the main challenges in the implementation of digital innovation in villages. Many villagers do not have the knowledge or skills to utilize technology effectively in the use of the internet, digital devices, and digital platforms. This hinders them from accessing information, public services, and technology-based economic opportunities. Therefore, increasing digital literacy through training and education is crucial to ensure that village communities can fully participate in the digital era and utilize technology productively.

Limited internet infrastructure is one of the main obstacles in digital transportation in villages. Many rural areas still struggle to access stable and fast internet networks, reducing people's ability to leverage technology for education, health, economy, and public services. This information slows the adoption of digital technology and widens the digital divide between urban and rural areas. To overcome this problem, investment is needed in the development of telecommunication infrastructure as well as the provision of affordable and equitable internet services throughout the village area.

Limited funds and human resources (HR) are a big challenge in the implementation of digital innovation in villages. Many villages do not have sufficient budgets to build digital infrastructure, provide technological tools, or carry out training for communities and village officials. In addition, the limitation of human resources who have expertise in the field of information technology also hinders the digitalization process. To address this, there needs to be collaboration between governments, the private sector, and educational institutions to provide funding training and technical support to villages in need.

Resistance to technological change is often an obstacle in the process of digital transformation in villages. Many villagers feel anxious or skeptical about the use of new technologies, especially if they feel that the technology is difficult to understand or irrelevant to their daily lives. Fear of change, lack of understanding and old habits that have been firmly established are the factors that cause this resistance. To address this problem, it is important to involve the community in the educational process of providing training that suits their needs, as well as demonstrating the direct benefits of technology applied in their lives.

CONCLUSION

Village digital innovation is a fundamental element in realizing the concept of a sustainable smart village. With the right use of information technology, villages have a great opportunity to improve various aspects of life ranging from public services to local economic sectors. Digital technology can speed up administrative processes, improve access to services, increase transparency of village government and open up new opportunities in the field of entrepreneurship, especially serving digital MSMEs. That said, technology can also strengthen social resilience by enabling villagers to be more actively involved in decision-making and development planning.

However, the achievement of optimal results from village digital innovation will not be achieved without solid collaboration between various parties. The government as the main driver must fight in providing policies that support digital transformation, both in terms of funding, regulation, and infrastructure. Village communities also need to be encouraged to increase digital literacy, so that they can utilize technology productively. On the other hand, the private sector can contribute through investment in digital infrastructure and the provision of digital platforms that are relevant to the needs of rural communities.

The success of village digital transformation depends on the development of an inclusive digital ecosystem, which means that all

parties, both government, community and the private sector, are effectively implemented in villages. The development of equitable internet infrastructure, the improvement of digital skills through training and education, and policies that support inclusiveness in technology access are elements that must be considered. Thus, digital innovation in villages will not only bring positive changes directly to social and economic development, but will also accelerate the achievement of sustainable development goals at the village level.

Acknowledgments

I would like to express my deepest gratitude to all parties who have contributed to the preparation process of this article. In particular, I would like to express my deep gratitude to my superiors, who have provided invaluable direction, guidance, and insight during the preparation of this article. This patient and attentive guidance has helped me a lot in putting together this work in a better and structured way.

In addition, I would like to thank the experts, and researchers whose work has provided in-depth insights and understanding of the implementation of village digital innovations, as well as other experts who have inspired and enriched this research.

I would also like to thank the valid and relevant sources of information that made it possible for this article to be comprehensively compiled. Thanks to the drafting team, colleagues and readers who have provided constructive feedback, so that this article can be completed well.

Hopefully this article can provide benefits, open insights, and become a useful reference in understanding the application of village digital technology.

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GIRLS TO THE STARS: FOSTERING EQUALITY IN SPACE STEM CAREERS

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ABSTRACT

The main goal of this paper is to analyze the attempts made to reduce gender bias in fields such as space science and engineering by providing inspiration to young girls who may be interested in a career in STEM. It emphasizes the importance of female role models, mentorship, receiving STEM education from a young age, and media representation in these efforts to reduce the underrepresentation of women in scientific fields that concern space. Furthermore, it presents initiatives of the Greek Astronomy and Space Society (Annex of Salamis and Pireus) to empower girls via hands-on STEM projects and mentorship. It will also present the global "Mission ShakthiSAT" initiative which concerns 12.000 girls from 108 countries and focuses on international collaboration and space projects to inspire women in STEM. Finally, it will highlight the role of supportive policies, male allyship, and industry collaboration in the creation of a more inclusive environment, which will result in more women and girls choosing to become scientists and engineers in the future.

Key Words: gender equality, STEM education, space careers, women in STEM, science education, diversity in space exploration

INTRODUCTION

The underrepresentation of women in space science and engineering is a problem whose solution seems very difficult to find, this is because its main cause is gender biases that have been long established within the industry, as well as societal views regarding

STEM fields. Most if not all scientific fields connected to space exploration and space in general, have always been dominated by men. Even nowadays there is only a very small number of female astronauts, engineers, and researchers in space-related fields (Santhebennur, 2024). In order for these biases to be overcome a comprehensive approach is needed. This approach will try to address said barriers throughout education and career development. Women need to be exposed to STEM fields at an early age in order to become inspired to choose space-related careers, and they should also have female role models that they will aspire to as well as mentorship opportunities from female mentors. This paper discusses key strategies to combat gender bias, emphasizing initiatives of the Greek Astronomy and Space Society (Annex of Salamis and Pireus) and the global “Mission ShakthiSAT” project, which aims to train and inspire young women in space exploration. These initiatives aim to lay the foundations for an increased female participation in space-related fields by helping to overcome societal barriers and creating supportive environments.

1. The importance of STEM education for girls in space

Inspiring more girls to start careers in STEM and in the field of space exploration in particular, is of the utmost importance, both for the empowerment of girls and for the further advancement of space-related industries. The following section concerns the importance of attempts to increase female participation in space-related STEM fields as well as the societal, scientific, and economic impact of these efforts.

Unlocking untapped talent

One of the most important reasons for the promotion of STEM education for girls is the possibility of untapped talent. If the number of girls joining space-related fields increases, the industry will gain more perspectives, which are an essential factor of innovation (Evona.com website). According to recent studies, the greater the

diversity in a team, the better its problem-solving capabilities because its members are able to approach complex challenges from a variety of angles, thus increasing the effectiveness of the outcomes (Linder, 2025). There is also the problem of a significant STEM skills gap in space-related industries and an ever-increasing demand for specialists in fields like space science, aerospace engineering, and space technology (Space Sector Skills Survey 2023 Report, UK Space Agency, 2023). By increasing the number of girls in STEM, this gap can be significantly reduced, which means that the demand for professionals and specialists in space-related fields will be met allowing these fields to advance faster and more efficiently.

Promoting gender equality in STEM

Space exploration has always been considered a male-dominated field, and this perception has perpetuated gender stereotypes. These stereotypes need to be broken in order to achieve gender equality in this field and that can only happen if the number of girls and women in STEM careers increases dramatically. The first step to gender equality in the field is inspiring girls to pursue STEM careers in space science. Women currently account for just 20% of professionals in space-related fields, confirming the existence of a significant gender gap, which can only be reduced via targeted educational initiatives. These initiatives will also promote a more balanced and equitable representation in the industry and they will ensure that women are offered opportunities to contribute to important scientific and technological advancements (Limbani, 2025).

Empowering girls and building confidence

STEM education enables girls to explore a wide variety of intellectually engaging and very lucrative careers in fields such as aerospace engineering, planetary science, and space medicine. Girls that follow these career paths can be offered many professional opportunities that are both impactful and rewarding (Keystone

Science School, 2023). Furthermore, participation in STEM improves critical thinking, problem-solving, and leadership skills which are all necessary in order to increase confidence in young women. Success in STEM can improve the lives of many girls since it empowers them academically and helps them increase self-confidence regarding their overall capabilities (Foster, 2021).

Inspiring future generations

Representation in fields where women have always been underrepresented is very important. If girls hear about other women who are working and thriving in space-related careers, it is more likely for them to see themselves in similar positions. Female role models in space exploration can become a source of inspiration for future generations of girls resulting in an increased number of women in these fields. (Libby, 2017). Additionally, space exploration creates a sense of wonder about the universe, and the participation of girls in space-related STEM education leads to a lifelong love of science, technology, and discovery. The earlier the engagement with STEM education the greater the possibility of creating and fostering a sustained interest in scientific fields, which will lead to more contributions to space science and beyond.

Advancing scientific and technological progress

Studies have shown that there is a direct link between a diverse workforce and increased innovation, particularly in STEM fields. If the diversity in the workforce of space related industries is increased, then there will be more creative and effective solutions to the complex problems of space exploration, including ones pertaining to space travel, planetary research, and the future of human space exploration. Women scientists can produce significant breakthroughs in space science, which in turn can lead to crucial technological advancement (Europa, 2025). Furthermore, space exploration is becoming a global endeavor which means that all segments of society, including women, should have significant

representation in that field. If future space missions, policies, and discoveries are to be effective, they need to be decided and formed by a wide variety of voices and experiences (McIntyre, Impeltalent).

Inspiring broader STEM interest in society

An increased representation of women in high-profile STEM careers, particularly in space exploration, can increase interest in STEM fields as a whole and act as a source of inspiration for people of all genders. When women hold key positions in STEM fields, entire communities are encouraged to engage with science and technology, which can lead to an improvement of scientific literacy on a global scale. Furthermore, the most crucial problems of mankind, like climate change, resource management, and environmental sustainability, can only be solved through STEM. If girls are inspired to choose careers in STEM, the number of minds and outlooks trying to solve these problems will increase, which will positively affect space exploration initiatives and mankind in general (Gordon, 2024).

Shaping the future of space exploration

Space exploration is full of complex issues and problems that can only be tackled by the best and brightest minds from all genders and backgrounds. If space exploration is to be considered a global endeavor that concerns all, then more women need to be involved in all aspects of it, including complex undertaking such as the colonization of Mars, making space tourism accessible to as many people as possible, or conducting deep-space missions. By inspiring more girls to choose space-related STEM careers our scientific frontiers will be expanded, which in turn can lead to more important scientific discoveries and breakthroughs that can have a positive impact on all mankind. Women have played a significant part in the advancement of space exploration, which means that the introduction of more female perspectives will only improve our understanding of the universe (Bharmal, 2021).

Encouraging more girls to choose STEM careers in space exploration is not just about inclusivity and gender equity, it is about discovering and benefiting from untapped potential, improving innovation, and ensuring that the future of space exploration is shaped by the full potential of humanity. The support of girls in STEM education will inspire them to succeed and make significant contributions to the entire field of space science. The increase of gender diversity in space-related fields is crucial for the advancement of scientific knowledge and it also virtually guarantees that the future of space exploration will be inclusive, equitable, and innovative (Women Tech Network).

2. Girls in space STEM education programs

The promotion of space-focused STEM education programs aimed at girls will play a significant role in inspiring and increasing female participation in science, technology, engineering, and mathematics (STEM) fields. The main focus of those programs is to “tear down” the barriers that have traditionally restricted female involvement with STEM by offering stimulating, hands-on learning experiences, improving mentorship, and cultivating confidence and curiosity towards space exploration. The more general purpose of these programs is the creation of an environment in which girls are not only inspired to choose STEM careers but one in which they also receive all necessary support in their academic journeys.

One of the most important aspects of these programs is hands-on learning, focusing on space exploration. Girls are taught technical skills, including robotics and coding, which leads to them gradually being introduced to designing and programming robots, rovers, and drones used in space missions. Space science experiments are also very important because they allow girls to participate in activities that simulate conditions in space. Simple experiments in physics, chemistry, and biology, can teach girls about complicated concepts such as gravitational forces, propulsion, and plant growth in microgravity environments. Furthermore, these programs also act as

an introduction to astronomy and planetary science, which results in girls becoming curious about the universe and its vast complexities. These programs essentially combine theory and practice in order to make space exploration as accessible as possible and more tangible for young learners, regardless of their gender (The Sally Ride Science Academy).

Mentorship is another crucial aspect in nurturing girls' interest in space-focused STEM fields. These programs allow girls to get in contact with female astronauts, engineers, and other professionals in space-related careers. Girls can get in contact with female role models in space exploration by having some of them appear as guest speakers during these programs or through virtual Q&A sessions and live-streamed interviews. Furthermore, special mentorship programs can enable female STEM professionals to provide guidance, encouragement, and inspiration. Through this mentorship girls participating in these programs will receive valuable insights into the challenges and opportunities of space-related fields while also learning about the real-world applications of the skills that they are learning (Onyemaechi, 2023).

Also, space-themed STEM competitions can make it possible for girls to apply the skills that they have acquired in creative and collaborative environments. These competitions can have many forms: from design challenges on the creation of space habitats or vehicles for extraplanetary missions, to hackathons in which girls will have to work on space exploration projects and come up with innovative solutions for future missions (European Space Agency, 2024). Space Design Competition for Girls, 2024). Such activities foster critical thinking, teamwork, and problem-solving, while at the same time nurturing excitement for space science. Participation in these events, leads to girls becoming more confident in their capabilities and increases their ability to tackle complex challenges. Both skills will prove very useful in their future careers (NASA Space Apps Challenge).

The availability of inspiring content and digital resources is essential for the success of these programs. They often produce documentaries, interactive media, and virtual tours of space stations, in an attempt to make space exploration more accessible and engaging. Online platforms host a variety of online courses, quizzes, and tutorials concerning various aspects of space science, programming, and engineering. These digital resources enable girls to study and learn about space-related topics at their own pace and leisure, which promotes independent learning and provides a better understanding of the field (Space to learn, NASA).

Space-themed camps can also play a vital role in the promotion of space-focused STEM education, since they are very immersive programs during which girls participate in astronaut training simulations, rocket-building workshops, and astronomical observation. Essentially, these camps offer the right mix of education and fun, which makes them a supportive environment where girls can work in teams, improve their confidence, and engage with various important STEM concepts in way that is both stimulating and tangible (U.S. Space & Rocket Center).

These programs mainly aim to remove established gender stereotypes. By providing to girls information about the achievements of women who excelled in the field of space exploration like Valentina Tereshkova, Sally Ride, and Mae Jemison, girls will start doubting the stereotype that the field of space science is male dominated. Also, said programs emphasize the contributions of female astronauts, scientists, and engineers, to help young girls understand and internalize the fact that women have always played a crucial role in space exploration and will continue to do so. Furthermore, these initiatives also attempt to cause paradigm shifts in STEM education through the encouragement of educators and parents to adopt a gender-neutral approach to the way that STEM subjects are taught in order to make girls feel like they really do belong in STEM classrooms, labs, and clubs. This will result in a more balanced representation in space-related science fields.

Collaborations with space agencies and universities could also create opportunities for girls with an interest in space exploration. Partnerships with organizations such as NASA, the European Space Agency (ESA), and academic institutions that conduct space research programs can result in unique learning opportunities and resources becoming available and accessible to girls. It should also be noted that mentorship programs, special workshops, and access to cutting-edge research are usually part of these partnerships, which means that they can allow girls to gain a deeper understanding of space science while also presenting them with future academic and career opportunities.

Many of the most famous and popular space-related STEM programs have been designed to offer support to girls who are considering a career in fields related to space science, aerospace engineering, and other similar STEM fields. Programs such as NASA's "Girls in STEM" initiative include workshops, hands-on activities, and mentorship opportunities with scientists, engineers, and astronauts, allowing girls to come in direct contact with experts in the field (NASA, 2023). Similarly, organizations such as "Sally Ride Science" and "Girls Who Code", offer many STEM education programs with a focus on space science, coding, and engineering which foster problem-solving skills via space-related challenges (The Sally Ride Science Academy, Girls Who Code).

Other prominent initiatives include space camps for girls, providing immersive astronaut training experiences, and "She Can STEM", which makes digital resources and mentorship opportunities available to girls interested in space tech and engineering (She Can STEM). Programs like "Techbridge Girls" and "Million Girls Moonshot" aim to provide hands-on education and mentorship in aerospace, robotics, and engineering to girls coming from underprivileged backgrounds. (Techbridge Girls, Million Girls Moonshot). Additionally, "WISE" (Women in Science & Engineering) allows girls to take part in NASA projects and engage in space

research, inspiring them to choose space science and STEM careers (Women in Science & Engineering).

These programs are extremely important since they inspire girls to engage with space-related STEM fields, thus breaking gender stereotypes in aerospace, engineering, and space science. Furthermore, they offer mentorship opportunities, contact with role models, and hands-on learning experiences which help girls gain the confidence and skills needed to succeed in STEM careers. Moreover, these programs also offer scholarships and career opportunities in space exploration, empowering girls to make significant contributions to the future of space science.

In conclusion, space-focused STEM education programs for girls are essential for reducing the gender gap in science and technology. These programs offer hands-on learning, mentorship, and immersive experiences which educate young girls while at the same time providing inspiration for the next generation of female leaders in space exploration. The constant support and inspiration to girls in STEM play a crucial role in the creation of a diverse and innovative workforce that will improve and revolutionize space science and lay the foundations for further exploration of the universe.

3. The underrepresentation of women in space related fields

Although the representation of women in some fields of science and technology has increased, their representation in space related fields leaves much to be desired. Although almost half of the global workforce consists of women, only 20% of the workforce in space-related fields such as aerospace engineering, astrophysics, and astronautics consists of women. This is mainly due to a combination of established societal stereotypes, an apparent lack of female role models, not enough access to STEM education, and workplace related problems (UN News, 2021).

The limited representation of women in space-related careers is the result of well-established gender stereotypes and biases. Young girls are usually not encouraged to choose careers in STEM fields and they may even be deterred from doing so. The main reason for this is the fact that there is a wrong belief that those fields are more appropriate for men. Many people still think that the fields of engineering and space exploration are "male-dominated" and this acts as a deterrent for girls that are considering a career in one of these fields. These biases influence how young girls see their potential in STEM and they can negatively affect their career aspirations in space-related fields (UNOOSA's Space4Women).

Another important problem is the lack of women in leadership positions in the space industry, which could act as role models for younger girls. Although there are female pioneers such as Katherine Johnson, Sally Ride, and Mae Jemison who achieved great things in space exploration and broke barriers for women, their stories are not as well-known as those of their male colleagues. The fact that there are not many visible women in senior roles within space agencies, research institutions, and aerospace companies doesn't allow young girls to imagine themselves in similar positions. When girls do not learn about the achievements and success of other women in these fields, they tend to start believing that such careers are unattainable or unsuitable for them.

Limited access to STEM education and a lack of encouragement are two factors that contribute significantly to the increase of the gender gap in the space industry. Girls are often discouraged from participating in activities like robotics clubs, coding workshops, or space competitions, which could help them become interested in STEM fields. Furthermore, there is also a lack of targeted outreach programs and funding for initiatives that would ignite girls' interest in STEM which leads to fewer women choosing degrees in space-related disciplines. If girls are not exposed to those fields at an early age, it is unlikely that they will gain the confidence and skills needed to excel in space science and engineering (Europa, 2024).

The few women that actually choose to work in the space industry often deal with challenges in their place of work that can negatively affect their professional growth and advancement, like lower salaries, a lack of opportunities for advancing to leadership positions, and deeply rooted work-related biases. Gender discrimination and lack of policies regarding the promotion and support of gender diversity also severely limit chances of women for promotions and career advancement. Also, women in the space industry experience feelings of loneliness and isolation in their workplaces due to the fact that they are usually male-dominated, which leads to large number of them eventually giving up their STEM careers.

In summary, there is a lack of female representation in the space field due to a complicated interplay of societal stereotypes, lack of role models, unequal access to STEM education, and workplace barriers. These problems can only be solved with a concerted effort to de-establish gender biases, promote female representation, offer more opportunities for girls to engage in STEM, and create more diverse work environments. Only by facing and overcoming these systemic challenges, will it be possible to reduce the gender gap in the space industry and encourage more women to choose careers in space exploration and research.

4. Overcoming gender bias in space science and engineering careers

Overcoming gender bias in careers related to space science and engineering is crucial but must be approached in a more holistic manner. Industry issues and deeply rooted societal perceptions about genders in STEM fields must be solved to increase diversity in workplaces. By promoting and informing girls about female role models and by offering them mentorship opportunities, we can solve these problems while inspiring and empowering young women interested in space science. By promoting the achievements of female astronauts, engineers, and scientists it will be possible to

challenge the notion that these fields are only suitable for men since it will be made obvious that women can achieve success in space-related careers. Mentorship programs, can also bring young girls in contact with experienced female professionals who can offer guidance, career advice, and support.

Another important strategy for the solution of the gender bias problem is encouraging early STEM education and engagement. The introduction of space-related STEM programs to young girls can challenge stereotypes and may spark a lifelong interest in science and technology (Smithsonian). Gender-inclusive STEM curricula in schools that put an emphasis on women's contributions to space exploration, can also foster a more balanced perception regarding who can excel in these fields, thus casting doubt on the stereotype that STEM is "for boys" and encouraging girls to pursue their interest in STEM fields (Smith & Lee, 2023).

Challenging gender stereotypes in the media is also very important. Mainstream media, such as television, movies, and news outlets, should show more women in leading roles in content about space exploration, engineering, and science (UNESCO Report 2022). By increasing visibility this way, it is possible that societal views on gender roles will start changing and girls who aspire to careers in space science will be more accepted by society. Furthermore, parents and teachers should encourage gender-neutral play and interests via activities, toys, and games that foster STEM learning for both boys and girls (Ceci & Williams, 2007).

Diversity and inclusion training within organizations, space agencies, and companies is a necessity for the creation of truly inclusive and diverse work environments because they raise awareness about unconscious bias and promote behaviors that strengthen inclusivity (Archer, Water, Dewan, Foster & Portas, 2022). Additionally, hiring and promotion processes should be unbiased. This can be achieved with methods like blind resumes and structured interviews. Setting objectives for increased representation of women in space science

and engineering, particularly in leadership roles, is also very important for making the industry more diverse (Copernicus, 2024).

Supportive workplace policies also play a vital role in keeping women in the space science workforce. Family-centered policies like parental leave, flexible working hours, and remote work options help women overcome unnecessary barriers related to maintaining a balance between their career and families. Salary equality and equal promotion and advancement opportunities for men and women must be enforced via regular audits and the establishment of robust accountability mechanisms.

The implicit bias in academia and research is another problem that needs to be solved if gender bias is to be overcome. Women and men working in space science related jobs and engineering should be offered equal research funding, publication opportunities, and chances for taking part in prestigious space-related projects. Grant review processes should be carried out without any form of bias. Diverse panels of reviewers are another way to ensure that the reviewing processes are as fair as possible. Furthermore, the offer of more leadership positions in academia to women, will result in an environment that will be more supportive for female students and researchers while at the same time creating valuable role models and mentors.

Strong networks and communities are essential for support, advocacy, and collaboration opportunities for women in space science. The mission of organizations such as Women in Aerospace (WIA) and Space4Women is the creation of networks for women in space that will facilitate knowledge-sharing and foster gender diversity. Safe spaces for conversations and fora about the problems of women in space science careers can raise awareness or even result in solutions for said problems (Women in Aerospace, Space4Women).

Male allies are also very important in overcoming gender bias. Men in leadership and technical roles should support gender equity by insisting on the formation of gender-balanced teams, by providing mentorship to female colleagues, and by challenging biased behaviors. In addition, it is extremely important to address microaggressions in the workplace for the creation of a more respectful and equitable environment.

Return-to-work programs for women who have been absent from work due to parental leave or personal reasons is another crucial strategy. Organizations should offer re-entry programs that provide training, mentorship, and networking opportunities that will aid women with their reintegration into space science and engineering careers. Programs that can teach new skills or improve already existing ones can also encourage women who have left the field to return to it by offering them new knowledge and skills.

Global collaboration is vital for ensuring that efforts to solve the problem of gender bias in space science and engineering have positive outcomes. International organizations such as the United Nations Office for Outer Space Affairs (UNOOSA) and the International Astronautical Federation (IAF) are the most important advocates for gender equity in space on a global level. The expansion of programs such as Space4Women and the Women in Space Exploration (WiSE) to more countries will also raise awareness and empower women worldwide to choose space-related STEM careers. (UNOOSA, 2023).

5. STEM initiatives and projects of the Greek Astronomy and Space Society (Annex of Salamis and Pireus) for girls

The Greek Astronomy and Space Society (Annex of Salamis and Pireus) has implemented many STEM initiatives designed to inspire young girls to choose careers in space science, astronomy, and engineering. Being a STEM instructor at the Society, I have had the pleasure and the privilege to work together with astronomy and

space experts for the development and delivery of a wide variety of projects whose main goal is breaking gender barriers in fields that are traditionally considered male-dominated fields. Our main goal is to offer girls hands-on learning experiences and chances to get in contact with real-world STEM professionals, improving their confidence and sparking interest in science and technology from an early age.

One of the most important aspects of our programs is the organization of hands-on STEM workshops. These workshops offer girls the chance to build small telescopes, explore celestial phenomena, and even include space mission simulations. Work on real-life projects allows participants to improve their problem-solving skills, to learn to work with others, and to foster their critical thinking abilities. These practical activities essentially improve their understanding of space science while fostering their enthusiasm for further exploration of STEM fields. (Salamis Astronomy website).

Mentorship is essential for the success of our initiatives. We have partnered with female astronomers, aerospace engineers, and space industry professionals who are more than willing to act as mentors and role models for the girls. This mentoring allows participants to learn about the career paths of accomplished women in the industry while being guided and inspired to imagine themselves in similar positions. This support system is essential for fostering their self-confidence and empowering them to break down the gender stereotypes that exist within STEM fields.

Apart from mentorship, we also enable girls to participate in space-themed challenges on local or international level. These competitions allow girls to use their STEM knowledge and skills for the solution of space-related problems. By participating in these competitions and showcasing their talents, the participants improve their confidence and validate their potential for success in these fields. The competitions are also a source of valuable experience in

managing real-world challenges similar to the ones that scientists and engineers face in the space industry.

Astronomy camps and field trips also play an important role in our initiatives. We organize visits to observatories, space research centers, and science museums to let the girls experience firsthand exposure the world of space exploration. These experiences spark their curiosity and give them a chance to explore the universe outside their textbooks, they improve their understanding of space science and they instill and strengthen a sense of wonder and excitement about future possibilities in the field.

Our mission revolves around fostering confidence and breaking down stereotypes. We not only teach technical skills, but we attempt to overcome common challenges like imposter syndrome and societal biases that can keep girls from choosing STEM careers. Our workshops and events are designed with the creation of inclusive and supportive environments in mind, in which girls will be feeling comfortable to ask questions, explore their interests, and develop their talents. Collaborations and leadership opportunities are the key to dismantling the stereotypes according to which space science and engineering are only suitable for men.

In order to offer more opportunities to girls in our programs, we are constantly forming new partnerships with international space organizations, universities, and technology companies. Our global network allows us to make it possible for girls to participate in cutting-edge research and to offer scholarship programs and future career paths in the space industry. By cooperating with other organizations with similar missions and goals, we can ensure that we will be up to date with the latest advancements in STEM education in order to keep our programs as engaging and relevant as possible, despite the rapidly evolving nature of space science.

Regarding the future, we will emphasize the expansion of our outreach efforts by conducting more advanced and ambitious space

projects which will include coding satellites, understanding orbital mechanics, and simulations of missions to Mars in order for girls to be able to gain detailed technical knowledge and practical experience in space science. Additionally, we will strive to establish an annual Girls in Space STEM Conference, in which students will be invited to present their projects, to network with industry leaders, and to find all the latest information about opportunities for internships and further studies.

To summarize, our main objective is the creation of a lasting impact by inspiring more girls to choose careers in STEM and fostering a lifelong passion for space exploration. By encouraging curiosity and developing the necessary support systems, we will pave the way for a new generation of women astronauts, scientists, and engineers. It is our firm belief that our work will cause significant progress toward gender equity in the space industry and it will also result in many important contributions to the advancement of space exploration. (Salamis Astronomy website).

6. "Mission ShakthiSAT"

In 2025 I was selected as Country Ambassador for Greece of "Mission ShakthiSAT", a unique and ambitious space program conducted by "Space Kidz India", an Aerospace startup which focuses in the design of satellites and rockets, and an official partner of the "Indian Space Research Organisation (ISRO)". (ISRO). The main focus of the initiative was to serve as a source of inspiration for young girls all over the world via the awe-inducing field of space exploration. 12,000 girls hailing from 108 nations are expected to take part in the program, whose main mission objective is to provide technical training to girls by allowing them to build their own CubeSats. (Space Kidz India website).

"Mission ShakthiSAT" is a unique initiative in space exploration, due to the fact that it is a program exclusively for girls and due to its ambitious objective of sending a satellite built by students into lunar

orbit as part of the Chandrayaan-4 mission. Some of the main aspects of ShakthiSAT are cutting-edge technologies like electric propulsion for orbital transfer, specially developed in hands-on educational programs, and international collaborations, making it a powerful symbol of the future of space exploration which empowers young women all over the world to innovate and excel in STEM fields. Hands-on experiences inspire students to develop a passion for science and they can turn girls into confident leaders who will make significant contributions to the future of STEM. The main goals of this historical mission include fostering global collaboration by initiating young minds in science and innovation, bringing about a revolution in STEM education, teaching the next generations of female space leaders and the bridging of the gap between academia and industry.

Since just 20% of space-related jobs are held by women, only 11% of all astronauts have been female, and less than 30% of researchers are women, getting more girls involved in the space industry isn't just about gender equality, it's about tapping into a universe of undiscovered talent and potential. It's also about making the universe accessible to as many people as possible because all girls should be allowed to reach for the stars. Finally, if girls are encouraged not only to pursue careers in traditionally male-dominated fields but to excel in them, then stereotypes will be torn down and more girls will attempt what was previously perceived as impossible. (ShakthiSAT website).

CONCLUSION

The struggle with gender bias in space science and engineering can only be successful through concerted efforts from teachers, organizations, and international entities. The promotion of mentorship, early STEM engagement, and inclusive policies, can result in the creation of an environment in which girls and women will be encouraged to choose space-related careers without being hindered by traditional gender roles. The initiatives spearheaded by

the Greek Astronomy and Space Society and global projects such as “Mission ShakthiSAT” are a testament to the power of hands-on learning, mentorship, and global collaboration in inspiring young girls to participate in space exploration. Ultimately, the future of space science hinges on fostering diversity and making sure that both women and men make equal contributions to space exploration. Through continuous effort and international cooperation, we can create a future where gender equity in space science will not just be an ideal, because it will have become a reality.

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RESONANCE AND MEANING: THE SCIENCE OF TONE FROM SAMAVEDA TO THE SMARTPHONE

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Abstract

Tone has a pivotal, often missing place at the centre of human communication. More than a stylistic device, tone expresses emotion, communicates intention, and aids memory. This article examines the importance of tone in language, from ancient Indian oral tradition—specifically, the Samaveda—to contemporary digital communication, where tone all too often gets lost, leading to mass miscommunication.

Spoken language relies on variation of tone to convey nuance and affect, but in digital text such as emails or messaging programs, tone is either missing or ambiguous. Borrowing from Howard Gardner's theory of multiple intelligences, this research places tone as a central element of musical intelligence that intersects with linguistic ability. It explores chanting traditions such as Samgayan and the oral transmission of texts such as the Bhagavad Gita and Ramcharitmanas, illustrating how tone previously enhanced memory, understanding, and emotional involvement.

Neuroscientific research reinforces that tonal and rhythmic input stimulates brain regions central to language acquisition and memory. Digital communication removes tonal information, heightening the likelihood of misinterpretation. This article suggests incorporating tone-sensitive practices—both ancient (e.g., chanting) and new (e.g., AI-driven feedback)—into language instruction to render communication more emotionally intelligent, neurologically productive, and culturally embedded

1. Introduction: Tone as Meaning Beyond Words

Verbal language is rich, multimodal, and crafted not only by what is conveyed, but how it is delivered—by pitch, rhythm, pauses, and emotional tone. This "how," or tone, elevates speech from simple words to conveying meaning. Tone separates sarcasm from sincerity, urgency from equanimity, and affirmation from doubt. It is frequently underemphasized in linguistic scholarship and almost missing in text-based digital communication.

Oral cultures historically employed tone to serve as a memory aid, make points clear, and elicit emotional response. Ancient Indian religion is such an example: works such as the Samaveda were written to be sung, with rhythm and pitch serving to keep meaning intact from generation to generation. These were not matters of aesthetics but of practicality, to transmit sacred information accurately in pre-literacy societies.

Conversely, contemporary communication more and more depends on text—email, messaging platforms, and social media—where tone is lost, misunderstood or miscommunicated. A message that reads simply "Sure." can express agreement, indifference, or passive aggression, all depending solely on the reader's assumption. Unlike spoken communication, written messages tend to confuse because they lack tonal context.

This article attempts to revive tone as a foundational aspect of communication, examining it through a multi-disciplinary lens—oral culture, cognitive psychology, pedagogical theory, and new media. Drawing upon Howard Gardner's multiple intelligences theory the article illustrates how sensitivity to tone makes linguistic ability and communication skills robust focussing on musical intelligence.

Tone is not embellishment; it is the foundation of cognition and comprehension. In language learning spoken in particular, it underlies pronunciation, understanding, and expression of feeling. This research will examine tone's origins, scientific underpinnings,

instructional value, and its urgent timeliness in our increasingly tone-free digital age.

Tone in Ancient Times: The Resonance of the Samaveda

Well before the invention of writing systems, tone was an essential component of oral cultures as both memory and guide to meaning. Perhaps the most striking example is the Samaveda, an early Indian scripture intended not to be read, but sung. Every syllable in the Samaveda has melodic notations, or swaras, that specify how hymns are to be chanted, thus conserving both meaning and memory.

This incorporation of tone into religious language performed a cognitive purpose. Academic scholars point out that tonal composition improved the veracity of aural transmission and facilitated long-term retention. For centuries, massive collections of Vedic literature were passed on without the aid of written texts. It was tone—coupled with rhythm and melody—that engaged emotional and auditory memory, enabling students to memorize intricate content.

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the use of written texts. It was tone—along with rhythm and melody—that engaged emotional and auditory memory, enabling learners to absorb intricate material.

Students, frequently trained since childhood, exercised exact pitch control, breath coordination, and rhythmic discipline. Even slight tonal variations might alter the meaning or holiness of a verse. This highlights tone's semantic function—strongly connected to both understanding and intention.

The Samaveda's categorization as a scripture of music captures Indian epistemology, which considers sound (nāda) as divine. The philosophic notion of śabda brahman—sound as an expression of divinity—hails the tone as not ornamental but critical to truth.

The Samgayan method of chanting still lingers with Vedic experts today, preserved via oral transmission. Its enduring validity testifies to the strength of tonal rhythm in retaining learning over the long term. Whatever ancient Indian techniques instinctively developed through experience, contemporary neuroscience vindicates: tone aids recall, emotional commitment, and efficiency of learning.

3. Echoes of the Sacred: Lyrical Transmission in Indian Oral Tradition

The role of tone in ancient India went far beyond the Samaveda into a wide literary oral tradition wherein rhythm and melody were essential to conveying spiritual, philosophical, and linguistic information. Texts such as the Bhagavad Gita were memorized traditionally by recitation in melodic cadence. The metre of Sanskrit shlokas—chandas—naturally accommodated musical intonation, adding emphasis to understanding and emotional resonance.

For example, the rendition of "Karmanye vadhikaraste..." changes its emotional color according to pitch and intonation, altering its impact from contemplative to inspirational. This adaptability is based on tonal modulation, which helped ancient texts to be accessible and memorable over the ages.

Tulsidas' Ramcharitmanas, which is a popular vernacular re-telling of the Ramayana, is another excellent case in point. Written in Awadhi and sung every day in temples and homes to date, its stanzas—chaupais and dohas—are musically structured. That rendered the text learnable in a broad, even non-professional, audience, infusing tone into folk learning and worship practice.

Regional traditions also show this phenomenon. Tamil Thevarams, Marathi Abhangas, Bengali Baul songs, and Punjabi Gurbani chants are based on musical recitation. These were not separate religious practices but functional educational systems in which tone facilitated second-language acquisition and cultural continuity.

Kids tended to memorize hymns in ancient languages unknown in their everyday speech, but stored them with ease because they were song-like. Of greater significance, tone maintained semantic coherence over time, avoiding the interpretation drift so common among silent, written reading.

These poetic traditions served not just as religious expressions but as intellectual models. By melody and repetition, they rooted language in feeling, making learning shared, instinctive, and anchored in memory.

4. The Neuroscience of Tone and Memory

Contemporary neuroscience verifies what oral traditions centuries ago practiced: tone greatly enhances memory and language learning. Cognitive studies reveal that rhythmic and tonal speech engages brain areas responsible for memory and learning,

including the hippocampus and prefrontal cortex. Ferreri et al. (2016) discovered that subjects who were presented with verbal input containing melodic variation remembered better than those who were given monotone verbal input.

Music and language processing share overlapping neural networks. The left hemisphere manages linguistic structure, while the right hemisphere handles pitch and emotion. Schellenberg (2005) demonstrated that musically trained children outperform peers in verbal IQ and reading, supporting Gardner's view that musical intelligence reinforces linguistic ability.

Tone also serves as a segmentation device—enabling listeners to break down speech into meaningful bits. Infants distinguish between tones before they've learned vocabularies, and in tonal languages such as Mandarin, tone distinguishes word meaning. Even in non-tonal languages such as English or Hindi, intonation encodes intent, sarcasm, or urgency, enabling listeners to infer meaning beyond literal definitions.

These discoveries mirror Indian chanting practices, in which melody, breath, and rhythm involve several senses. Vedic chanting, for example, combines voice, breath, and tone in a multisensory learning environment that builds profound retention and emotional involvement.

Prosodic bootstrapping theory also lends support to this method. The theory suggests that the learner draws inferences from syntactic structure based on tonal and rhythmic markers. This accords with students' experience in chanting societies, where prosody directs articulation and comprehension.

At the end of the day, tone isn't an add-on for art—it's a tool of cognition. It solidifies memory, enhances emotional connection, and clarifies meaning. Incorporating tone-dense practices into language

instruction—particularly for second-language learners—can greatly boost linguistic fluency and understanding.

5. Tonal Clarity vs. Digital Ambiguity

In conversation, tone is a semantic and emotional guide, clarifying intent and shaping meaning. But in the digital communication of emails, texts, and messaging apps, tone is flattened or stripped away altogether, resulting in misinterpretation and emotional disconnection.

A phrase like “I’m fine” can convey reassurance or resentment, depending on vocal tone. In text, this nuance vanishes, leaving meaning open to subjective inference. Without auditory cues, readers project their own assumptions onto the message, often misreading emotional intent.

This absence of tonal definition is a cause of the "emotional impoverishment" about which some scholars write. Emojis, punctuation, and formatting try to imitate tone but are limited substitutes for speech. In contrast to face-to-face or spoken communication, written language forces the reader to supply emotional infillings—an frequently erroneous and exhausting process.

Conversely, established oral systems such as Samgayan, recitation of Ramcharitmanas, and Gurbani chanting infuse tone into the very fabric of communication. Tone cannot be extricated from meaning. Emotional tone is not negotiable; it is integral to the message. Such shared, oral systems conserved emotional complexity and reduced misunderstanding—something difficult for digital spaces to achieve.

This affective gap in online communication is difficult in emotionally charged situations such as education, therapy, or dating. A warm encouragement from a teacher or a sympathetic tone from a

counsellor can lift spirits; the same phrases said in text can sound cold and uncaring.

New technologies—voice memos, speech generated by AI, emotive chatbots—try to bring back tone, but they are still narrow-range and shallow. Until such tools become more mature, our online dialogue will remain deficient in emotional depth embedded in spoken communication.

To fill this void, language learning and communication aids need to reinsert tone—either through voice practice, voice feedback, or culturally based chanting practices. Tone, which was once at the heart of connection, needs to regain its place in the digital era.

6. Tone in Language Acquisition: From Tradition to Technology

Tone is a critical aspect of human communication, influencing the meaning, intention, and emotional content of spoken words. Although written messages can be toneless, and therefore subject to endless misunderstandings, spoken words are full of tonal changes that convey not only words but also mood, stress, and nuances lost in written forms. From ancient oral traditions to modern electronic communication, the significance of tone in language learning cannot be exaggerated. This section delves into how tone affects language acquisition, using both classic and recent research, and relating those findings to the more general theory of musical intelligence.

7. Ancient Traditions and the Role of Tone

In the ancient writings such as the Samaveda and practices such as Samgayan, tone was not just a means of communication—it was the very essence of learning. The Samaveda, one of the four Vedas in Hinduism, consists of hymns recited with definite tonal patterns that were supposed to produce a spiritual vibration. These practices in terms of tonality were central to religious instruction and

were a means of divine understanding, transmitted across generations in highly formalized oral traditions.

Likewise, Vedic chanting traditions, as used in the Upanishads and Bhagavad Gita, employed specific tonal cadences that aided students in memorizing meanings and linking to deeper levels of understanding. The rhythmic and tonal structures of these texts enabled memorization and understanding, as the melodies became synonymous with the meaning of the words themselves. Here, tone was not merely a linguistic element but an educational aid that promoted both recall and depth of comprehension.

This idea is also supported by the theory of musical intelligence that Gardner (1983) claims is one of the fundamental intelligences that are used by people to interact with the world. Musical intelligence pertains to how one can tell, create, and enjoy music forms such as rhythm, melody, and harmony. In orthodox chanting, tonal patterns existed as a vehicle of musical communication that facilitated repetition and emotional connections of sound as a means to learn language.

8. Neuroscience of Tone in Language Acquisition

Current neuroscience provides additional explanation of why tone is so important to language learning. Research has revealed that our brains are able to process speech that contains tone more effectively than written or static forms of language (Schirmer & Kotz, 2006). The brain processes this type of speech in the right hemisphere, the same area used to recognize patterns in music. This intersection of linguistic and music processing justifies the assumption that tonal differences in language are associated directly with musical intelligence.

In language acquisition, especially in learning a second language, perception and production of tonal differences are vital. Mandarin Chinese is, for instance, a tonal language such that the pitch or tone

with which one speaks may alter the meaning of a word completely. Students of Mandarin tend to find tones challenging because their home languages, such as English, might not use the feature. Studies indicate, however, that learning tonal music or training musical intelligence increases the capacity of students to detect and produce tonal differences in speech (Slevc et al., 2016). The discovery yet again highlights the link between language and music, illustrating how musical intelligence can assist linguistic acquisition.

9. Tone in Contemporary Digital Communication

Tone, of course, is even more significant in this era of the digital age, particularly as communication increasingly moves from in-person conversation to text. Though written text enables precision and accuracy, it doesn't offer the tonal markers that inform us about context, intent, and emotion as speech does. Misinterpretations in communication through texts, e.g., emails, text messages, or social media updates, happen frequently since recipients cannot pick up the tone in which the message is conveyed. This lack of tonal communication tends to result in misunderstanding or misinterpretation of meaning since the emotional undertone of a message is lost.

Consider, for instance, the humble sentence, "Great job!" In a verbal context, tone, whether sarcastic, genuine, or enthusiastic, would give a decisive indication of the speaker's actual intention. But in written form, the very same words would be easily misunderstood, creating a misconnection between the sender and the recipient.

The use of emojis and punctuation marks (e.g., exclamation points, question marks, or ellipses) is one means by which individuals try to communicate tone in written form. These symbols serve as proxies for vocal inflection, assisting in clarifying the emotional tone of a message. Yet these textual mechanisms are still a long way from the

subtle tonal inflections of spoken language, which are more complex and richer.

Here, musical intelligence becomes more and more pertinent. By adding music and tonal training to language acquisition, especially in today's digital era, we can assist learners in having a better understanding of tone—whether speaking, texting, or other modes of communication.

10. Connecting Tone to Musical Intelligence in Language Instruction

The theory of musical intelligence plays a key part when it comes to tone within language acquisition. In English second-language classrooms, the use of music in education has been revealed to improve the capacity of students to understand speech tonal nuance. To illustrate, exposing students to various songs with differences in emotional tones can assist learners in comprehending how tone impacts meaning. Also, one can listen to English music and sing along to develop pronunciation as well as rhythm by supporting the natural cadence of the language as melodies do.

AI music also provides a new means of integrating musical intelligence into language instruction. By generating personalized music tracks that focus on particular tonal aspects or speech patterns, language teachers can improve their students' awareness of tone. For example, an AI-composed song might point out the distinction between rising and falling tones in English questions and statements, offering a playful and engaging means for learners to absorb these patterns.

Conclusion

The function of tone in language learning, from ancient cultures to contemporary digital communication, is inextricably linked with musical intelligence. Be it through the devotional chanting of the Samaveda or the tonal motifs of a contemporary

English song, tone acts as a medium between language and emotion, meaning and comprehension. As technology advances the way we communicate, incorporating musical intelligence into language acquisition—both through conventional means and new AI-composed music—can assist learners in understanding the nuances of tone, enhancing their ability to speak and overall linguistic proficiency.

Conclusion: Reclaiming Tone in a Toneless World

The role of tone in language acquisition, from ancient civilizations to the present day's digital communication, is indissolubly connected with musical intelligence. Through either the devotional chanting of the Samaveda or the tonal motives of a modern English song, tone serves as a mediator between language and emotion, meaning and understanding. By no means merely decorative, tone is at the basis of the way we acquire, process, and communicate language.

What were once ancient conventions that evolved instinctively—through rituals such as Samgayan, Ramcharitmanas recitation, and local musical-literary traditions—today's neuroscience and pedagogy now corroborate with empirical precision. Tone enhances memory, deepens emotional impact, and aids linguistic accuracy. Tone works as a catalytic force in education, especially second-language learning, to translate formless syllables into engaging, memorable speech.

Yet, with the advent of text-based communication in the new era comes the loss of this aspect of language. Email, instant messaging, and social media leave tone out of language, resulting in epic misinterpretations. Even with the introduction of emojis, gifs, and trendy fonts, these innovations are all abysmal substitutes for the fluid nuances of the human voice. We are now in a contradiction: we have never been more connected, but more likely to misunderstand.

In such a context, Gardner's theory of multiple intelligences—particularly musical intelligence—provides not only an educational model but also a cultural fix. By incorporating tonal awareness into language teaching, both old and new media, teachers can bring emotional intelligence back to communication. Technologies like AI-produced musical criticism, voice-activated interaction, and culture-based chanting exercises can merge timeless wisdom with innovation.

The promise of substantive conversation is not simply in what is said, but in the way it is said. To retrieve tone is not a wistful nostalgia for the past—but an imperative stride into the future, a stepping stone from sound and sense to tradition and technology, to mind and culture. If language is the spirit of human encounter, tone is its pulse.

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TRANSFORMATION OF PUBLIC SERVICE OF LOCAL GOVERNMENT IN DIGITAL ERA

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ABSTRACT

Digital transformation in local government public services is a must in today's digital era, driven by public demand for accessible, fast, accurate, and quality services. E-government is an important element in improving the efficiency and transparency of public services, with examples of services such as resident registration, licensing, tax payment, and healthcare accessible online. However, the implementation of government digitalization has faced a variety of challenges, including infrastructure gaps, lack of digital literacy, resistance to change, regulatory unpreparedness, cybersecurity threats and data privacy, budget and resource imbalances, lack of coordination between government institutions. and low public participation. This study applied a descriptive qualitative approach to analyzing changes in government public services in the digital age, with data obtained from a variety of reliable sources such as previous studies, journals, and academic books. Analysis is carried out to understand the implementation of digital-based services in Indonesia, which is currently still under development and requires improvement.

Keyword: Transformation, Public Services, E-government, and Digital

INTRODUCTION

Transformation in the digital world of this era is a must and this event is a manifestation of evolution. Even in its 2015 United Nations (UN) program, the Sustainable Development Goals (SDGs) emphasize the importance of technology and connectivity. From a

technology and connectivity perspective, this will indirectly affect the use of internet technology by many people. The use of internet technology can be understood as a form of transition from manual to digital use. Many aspects need to be adapted to fit the ethos called digital, one of which is to first understand the meaning of digital itself and related issues. (Tulungen et al., 2022)

E-leadership capability is interpreted as a way in which leaders utilize information technology to achieve government goals. Each government depends on the level of e-government it has, and the quality of e-government varies. The role of leadership in the implementation of digital transformation in the government sector is certainly a must. Leadership was key in adopting the use of technology in the industrial 4.0 era. In the digital age, leaders need to be well prepared with digital and emotional agility to operate in uncertain and complex environments. Today, in the process of decision-making and innovation, effective leaders work in a fast learning cycle. This transformation needs to be carried out comprehensively and comprehensively by involving all stakeholders, and this transformation process cannot be carried out only by the government, but requires active participation from all stakeholders, including civil society, the private sector, and academia. Furthermore, by covering all aspects of government, this transformation must be able to reach all elements of government, from the central to regional levels, as well as from policy formulation to implementation. Today, people want public services that are easy to access, fast, accurate, and high-quality digital technology to provide various solutions to meet these needs. The Indonesian government has committed to making digital changes in public services to improve people's quality of life and to realize good governance. Currently, the need for comprehensive transformation that the public wants is the demand for quality public services and the encouragement of globalization that drives public service changes in the digital era.

The goals of the public service transformation are as follows:

1. Improve people's quality of life
2. Realize fairness and equalization of access to public services
3. Increase public confidence in the government
4. Supporting economic growth and national development

In other words, this transformation must be carried out comprehensively and thoroughly, by utilizing technology and communication (ICT) to improve the quality, efficiency, and effectiveness of public services. (Natika, 2024)

METHOD

This study adopted a descriptive qualitative approach to understanding changes in local government public services in the digital age. The data used is obtained from a variety of reliable sources, such as previous studies or research report journals related to challenges in implementing E-government to improve the quality of public services in the digital era, as well as the transformation of public services in the digital era. In addition, academic books are also used that discuss digitalization of local governments. The analysis process is carried out to find out how the implementation of digital-based services in Indonesia, where the current implementation is still in the development stage and needs to be improved further. The concept of electronic governance is implemented with the aim that interactions between government, society, and business actors can run efficiently, effectively, and economically.

The first step in this study is to identify relevant sources of information, such as articles, journals, research reports, or academic books that discuss public service information or digitalization of local governments. This process is carried out to collect data on the application of information and communication technology in public administration in the digital era as well as obstacles that arise in the implementation of good governance in the digital era today.

Furthermore, the collected data are analyzed using document or content analysis methods to understand the

application of local governments' public service transformation in the digital era. This analysis was conducted to assess the extent of public service transformation in this digitalization era, as well as to identify challenges that may be faced, such as cybersecurity issues, data privacy, and digital inequality.

RESULTS AND DISCUSSION

Results

The implementation of Information and Communication Technology (ICT) through E-government has had a significant impact on public administration, improving the efficiency and transparency of services such as resident registration, licensing, tax payments, and online access. The government has a responsibility to provide equal and responsive public services, and E-government is a solution to more efficient and economical interactions. However, the implementation of E-government and good governance in the digital age faces challenges such as infrastructure gaps, lack of digital literacy, resistance to change, regulatory unpreparedness, cybersecurity threats, budget inequality, lack of coordination, and low public participation. Digitalization of the government is not just a system change, but a comprehensive transformation that requires strategic work related to infrastructure, regulations, and human resources development.

In Indonesia, the implementation of digital services continues to grow and faces challenges related to infrastructure, understanding technology, and data security. Thus, a comprehensive and collaborative strategy is needed to maximize the potential of E-government in improving the quality of public services across the country.

DISCUSSION

In the digital age, the application of information and communication technology to public administration provides a wide range of advantages, and E-government is one of the most crucial

elements in efforts to improve the efficiency and transparency of public services. Some examples of public services that can be held through E-Government include:

Population Registration: Residents can register online, access data changes, or extend identity documents, without having to visit the civil registration office.

Permission: The process of applying for permits for certain businesses or activities is carried out online, thus speeding up the process and reducing bureaucracy.

Tax Payment: People can pay taxes through online platforms, making it more practical and less queuing at the tax office.

Health Services: E-Government can support registration and scheduling appointments at hospitals or clinics through online applications, making it easier for patients to access health services.

Other public services in addition to the above examples, E-Government can also include a variety of other public services, such as social assistance applications, access to information about public budgets, reporting infrastructure problems, and more. (Sepriano et.al, 2023).

The government has a responsibility to provide equal public services to all citizens. It is important to realize that people determine the existence of public services that address the interests of the wider community in all regions of the country, are reliable and trusted, and are easily accessible interactively.

The concept of e-government is applied with the aim of efficient, effective, and economical interaction between government and society and business operators. This is necessary in light of the current dynamics of community movement, so that the government must be able to adjust its role in the country, so that people can enjoy their rights and perform their obligations safely and comfortably, so that everything can be achieved through the improvement of the government system itself. In addition, the goal

of implementing e-government is to achieve good governance. (Kapitan at. al, 2024).

Obstacles to the implementation of good governance in the current digital era include infrastructure gaps, lack of digital literacy, resistance to change, lack of regulatory and policy readiness, threats of cybersecurity and data privacy, budget and resource inequalities, lack of coordination between government institutions, and government agencies. as well as low public participation in government digitalization. (Susniwati at.al, 2025).

In the digitalization era, data security is becoming increasingly crucial. Data theft or information is becoming more frequent, and cybercrime is a major concern. In order to overcome cybercrime, all parties need to take comprehensive, integrated, and sustainable measures. Various government sectors face budget problems that hinder digital transformation. To maintain the sustainability of digitalization, the development of information and communication technologies requires significant funding, as well as regular maintenance of the system. (Rasyid at.al, 2024).

Digitalization in the government sector refers to the transition of systems and manual processes into digital systems in government execution. This transformation aims to improve efficiency, effectiveness, and transparency, and accountability in public services. In this interconnected era, digitalization has become a key term in modernizing government and improving the quality of public services. The context of digitalization in government is not only a shift from manual to electronic processes, but also a comprehensive transformation that transforms resources and formulates policies. This is a development from conventional government to responsive, transparent, and efficient digital government.

The transition to digital governance also brings challenges, including cybersecurity, data privacy, and digital inequality. Therefore, a strategic framework is needed that includes a robust

technology infrastructure, supportive relegation, and capacity development of human resources that can effectively manage and utilize digital technology. (Zein & Septiani, 2024).

The implementation of digital-based services in Indonesia is currently still in the development phase and needs to continue to be improved. Although there have been several government-implemented public service initiatives, such as the "Dukcapil Go Digital" program that allows people to manage population documents online, there are still many challenges to overcome. Some of the challenges faced include inadequate technology infrastructure, there are still people who lack understanding of technology, as well as public concerns about the security of their personal data. In addition, there are still several regions in Indonesia that have not fully implemented the digitalization of public services. (Yulanda & Adnan, 2023).

CONCLUSION

Digital transformation in local government public services, triggered by public demand for accessible, fast, accurate, and quality services, is a must in this digital era. E-government plays an important role in efforts to improve the efficiency and transparency of public services, with examples of applications such as population registration, licensing, tax payments, and healthcare accessible online. The government has an obligation to realize equal, fair, and accessible public services, and E-government is present as a solution to achieve more efficient interactions and realize good governance.

Nevertheless, the implementation of government digitalization in Indonesia faces significant challenges. Disparities in technology infrastructure, lack of digital literacy in some societies, dissatisfaction with changes, lack of flexible regulations and policies, threats to cybersecurity and data privacy, imbalance in budgets and resources between regions and institutions, and institutions. The lack of coordination between government agencies, and the low level of public participation in the digitalization process are obstacles

that need to be addressed. Data security is an important issue that requires comprehensive attention and action from all parties.

The implementation of digital-based services in Indonesia is currently still under development and requires ongoing efforts to improve them. Despite positive initiatives such as the "Duckapil Go Digital" program, challenges such as inadequate infrastructure, lack of understanding of technology in some communities, and concerns about personal data security are still a concern, in addition, The level of implementation of public service digitalization has not been evenly distributed throughout Indonesia, therefore, a comprehensive and collaborative strategy is needed involving all stakeholders to overcome these obstacles and maximize the potential of E-government in improving the quality of public services across the country.

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EXPLORING THE ROLE OF DEMOGRAPHIC VARIABLES IN SHAPING LEADERSHIP APPROACHES AMONG SECONDARY SCHOOL PRINCIPALS

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ABSTRACT

This study explores the influence of demographic variables—such as age, gender, educational qualification, and years of experience—on the leadership approaches of secondary school principals. Effective leadership is crucial in school administration, and understanding the role of personal and professional characteristics can help in tailoring leadership development strategies. A descriptive survey was conducted among principals in secondary schools across Kanyakumari district. Findings indicate that demographic factors significantly influence leadership styles, with younger principals leaning towards transformational leadership and experienced principals favoring democratic approaches. These results suggest the importance of customized leadership training programs that consider demographic diversity.

Keywords

Leadership styles, Demographic variables, School principals

INTRODUCTION

Leadership is a cornerstone of effective school management. The principal, as the head of the institution, plays a key role in shaping the school's culture, motivating teachers, and improving student outcomes. However, leadership is not uniform; it is influenced by several factors including demographic characteristics such as age, gender, education level, and years of experience.

In the context of secondary education, these variables may shape how principals interact with staff, implement policies, and handle challenges. This study aims to explore how these demographic variables influence leadership styles—such as transformational,

transactional, and democratic—among secondary school principals in Kanyakumari district. By understanding these influences, educational authorities can better support school leaders with more targeted training and resources.

ANALYSIS AND DISCUSSION

1. Age and Leadership Approach

The study found that younger principals (below 40 years) often adopt transformational leadership styles. They tend to be innovative, open to feedback, and collaborative. In contrast, older principals (50 years and above) preferred transactional leadership, emphasizing structure, discipline, and performance-based rewards.

2. Gender Differences

Female principals were more likely to exhibit democratic leadership, encouraging team participation and shared decision-making. Male principals, on the other hand, leaned slightly towards authoritative or task-oriented leadership.

3. Educational Qualification

Principals with higher qualifications (postgraduate or doctoral degrees) demonstrated a strong preference for transformational leadership, focusing on vision, motivation, and school improvement strategies.

4. Experience in Teaching and Administration

Principals with more than 20 years of experience favored democratic leadership. They were more likely to involve teachers in decisions and valued collaborative problem-solving. Less experienced principals leaned towards transactional approaches due to reliance on established rules and procedures.

CONCLUSION

Demographic variables play a significant role in shaping the leadership styles of secondary school principals. These factors influence how principals manage their schools, interact with teachers, and respond to challenges. The findings suggest the need for leadership development programs that are sensitive to these demographic characteristics. A personalized approach to training can enhance leadership effectiveness, improve school management, and ultimately benefit student learning outcomes.

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ISBN : 978-81-989016-3-7



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